

GRADENEXT · SMART LAB

Simulation Experiment Book

Grade 7 · Unit D · Matter and Energy in Ecosystems

California Integrated Science, Grade 7 · Domain: Biology / Ecology · 6 topics · 30 experiments

NGSS performance expectations anchored: MS-LS2-1, MS-LS2-2, MS-LS2-3, MS-LS2-4, MS-LS2-5

Specification standard: v1.0 · Assembled 04 September 2026

Build specifications for interactive simulation experiments · one per syllabus subtopic · for developers, artists and curriculum designers

How to read this book. Every subtopic in the GradeNext California Science syllabus has exactly one simulation experiment specification. Each specification is a *build order*: it tells a developer what to code, an artist what to model, and a curriculum designer what the student is meant to discover. Nothing here is a worksheet; every experiment is a running, manipulable scene with a control panel, more than one scenario, and something measurable coming out.

Anatomy of a specification

Section	What it is for	Who reads it first
Header fields	Experiment name, render mode, simulation engine, interaction level, session length, NGSS anchor	Product, curriculum
Theme & scene	What the student sees on load: setting, camera, palette, lighting, foreground/background, where the panel sits	Art, UX
Objects & components	Every entity the engine must instantiate, including invisible ones (fields, colliders, probes). Build notes give geometry, appearance, counts, behaviour. The <i>Editable</i> column says which objects the student can drag, place, swap, resize or connect directly	3D/2D art, engine
How it works — the model	The rules the engine runs: equations, rate laws, state machines, datasets; what updates per tick; what is deliberately simplified and why	Engine, science review
Control panel	Every control with widget type, range or options, default, unit and visible effect. At least one control is <i>structural</i> (changes which objects exist or how they are arranged)	Engine, UI
Scenarios	Named presets that reconfigure the same simulation to ask a different question. Presets reference control names from the panel	Curriculum, QA
Student activities	Numbered actions the student performs and what they record	Curriculum
Outputs & measurement	Readouts, graphs, tables, badges the sim produces and when they update	Engine, UI, assessment
What the student should realise	The single idea, the misconception it displaces, and the sentence the student should be able to say	Curriculum, assessment

Glossary of tags

Render mode · *2D Canvas* — vector/sprite scene on a flat canvas, the mode of the existing collision and projectile sims · *3D Scene* — camera, lights, meshes and materials · *2.5D Layered* — parallax layers or cutaways with depth ordering but no free camera · *Hybrid 2D+3D* — a 3D viewport alongside a 2D graph, diagram or map panel · *Data Dashboard* — datasets, filters, plot canvases and claim-building tools as the primary scene.

Simulation engine · *Rigid-body* (masses, forces, collisions) · *Particle system* (many small agents with simple rules) · *Field/vector* (scalar or vector fields sampled in space) · *Molecular* (atoms, bonds, valence, CPK colouring) · *Agent-based* (organisms or entities with behaviours) · *Fluid/thermal* (heat flow, buoyancy, convection) · *Ray/wave* (light and sound propagation) · *Data-driven model* (real or realistic datasets driving the scene) · *State machine* (staged processes with transitions) · *Procedural geology* (layers, uplift, erosion, plate motion).

Interaction level · *Explore* (look, zoom, toggle overlays) · *Manipulate* (drag, place, swap objects) · *Investigate* (control variables, measure, record) · *Design* (build, test, score, iterate) · *Argue-from-data* (select evidence, build and score claims).

Object class · *Actor* (moves or acts) · *Structure* (static or slowly changing body) · *Particle* (instanced small entity) · *Field* (invisible spatial quantity) · *Instrument* (measures something) · *Overlay* (visual layer on top of the scene) · *Environment* (backdrop, terrain, container) · *UI-Probe* (cursor-following tool).

Control widgets · Slider · Stepper · Dropdown · Toggle · Radio · Colour · Dial · Drag-handle · Multi-select · Numeric field · Timeline scrubber.

Engineering conventions

- **Experiment ID** is G<grade>-<subtopic> (for example G6-B2.1). It is stable and unique across the whole programme.
- **Scenario presets** use control=value pairs separated by semicolons. Control names in presets are shortened versions of the panel labels; the JSON companion resolves them.
- **Every control carries a unit.** Where a control is unitless the unit column reads "—".
- **The JSON companion** for this unit (json/G<grade>_Unit<letter>.json) contains every field of every specification in structured form, with numeric ranges parsed into min/max and option lists into options[], ready for ingestion by the Smart Lab content pipeline.
- **Ages 11–14.** All simulated hazards are clearly virtual. No procedure here is a real-lab instruction.

Unit map

ID	Subtopic	Experiment	Render	Engine	Level	Session
D1.1	Limiting resources	Which Shortage Bites First: Tomales Point	Hybrid 2D+3D	Agent-based + Data-driven model	Investigate	18–25 min
D1.2	Carrying capacity	The Island Fills Up: Foxes on Santa Cruz	Hybrid 2D+3D	Agent-based	Investigate	18–25 min
D1.3	Reading real population data	Three California Censuses, One Argument	Data Dashboard	Data-driven model	Argue-from-data	18–25 min
D1.4	Predicting the response to scarcity	Call It First: The Scarcity Forecast Desk	Hybrid 2D+3D	Agent-based + Data-driven model	Argue-from-data	18–25 min
D1.5	Competition for the same limiting resource	One Reef, One Ration: Urchin Against Abalone	3D Scene	Agent-based	Investigate	18–25 min
D2.1	Producers, consumers and decomposers	The Floor That Eats: A Sierra Nutrient Ledger	2.5D Layered	Agent-based + Fluid/thermal	Investigate	18–25 min
D2.2	From food chain to food web	Draw the Web: Monterey Kelp Forest	Hybrid 2D+3D	Data-driven model + Agent-based	Design	18–25 min
D2.3	Trophic levels	Half a Level Up: Where Omnivores Sit	Hybrid 2D+3D	Data-driven model + Agent-based	Investigate	18–25 min
D2.4	Why energy decreases at each level	Follow the Joule: Where the Energy Goes	Hybrid 2D+3D	Particle system + Agent-based	Investigate	18–25 min
D2.5	Reading an energy pyramid	Build the Pyramid, Then Break It	Hybrid 2D+3D	Data-driven model	Argue-from-data	18–25 min
D3.1	Matter cycling between organisms and the environment	The Sealed Jar: Nothing Leaves, Nothing Arrives	Hybrid 2D+3D	Agent-based + Data-driven model	Investigate	18–25 min
D3.2	Decomposers closing the loop	Rot Chamber: Who Unlocks the Nutrients	3D Scene	Agent-based + Fluid/thermal	Investigate	18–25 min
D3.3	The carbon cycle, introduced	Carbon on Tour: Reservoirs and Fluxes	2.5D Layered	Data-driven model + Particle system	Argue-from-data	18–25 min
D3.4	The nitrogen cycle, introduced	Under the Alfalfa: Four Microbe Crews	3D Scene	Agent-based + Molecular	Investigate	18–25 min
D3.5	Modeling matter and energy together	The Loop and the One-Way Street	Hybrid 2D+3D	Data-driven model + Particle system	Argue-from-data	18–25 min
D4.1	Competition and predation	Chase, Lag and Crowd-Out	Hybrid 2D+3D	Agent-based + Data-driven model	Investigate	18–25 min
D4.2	Mutualism, commensalism and parasitism	The Fitness Ledger: Who Gains, Who Pays	3D Scene	Agent-based + Data-driven model	Investigate	18–25 min
D4.3	The same pattern in very different ecosystems	Three Places, One Pattern	2.5D Layered	Agent-based + State machine	Design	18–25 min
D4.4	Interactions that shift over time or condition	When Partners Turn: Finding the Tipping Point	Hybrid 2D+3D	State machine + Agent-based	Argue-from-data	18–25 min
D4.5	Interaction patterns and population change	Draw the Curve: From Interaction to Population	Data Dashboard	Data-driven model + State machine	Argue-from-data	18–25 min
D5.1	Physical disruptions	Fire, Flood and Slide: The Landscape Stress Test	Hybrid 2D+3D	Agent-based + Fluid/thermal	Investigate	18–25 min
D5.2	Biological disruptions	Uninvited: Invaders, Blights and a Missing Keystone	Hybrid 2D+3D	Agent-based	Investigate	18–25 min
D5.3	Constructing an argument from evidence	The Evidence Docket: Prove What Wrecked the River	Data Dashboard	Data-driven model	Argue-from-data	18–25 min

ID	Subtopic	Experiment	Render	Engine	Level	Session
D5.4	Succession after a disruption	The Comeback: Ninety Years on a Burnt Slope	2.5D Layered	State machine + Procedural geology	Investigate	18–25 min
D5.5	Short-term versus long-term change	One Year Later, Fifty Years Later	2.5D Layered	Data-driven model	Investigate	12–18 min
D6.1	Naming ecosystem services	The Invisible Payroll: What the Land Does for Free	Data Dashboard	Data-driven model	Investigate	12–18 min
D6.2	Threats to biodiversity	Pressure Stack: Layering the Threats on a Range Map	2.5D Layered	Field/vector + Data-driven model	Investigate	18–25 min
D6.3	Named solution categories	The Solutions Bench: Six Tools, One Watershed	Hybrid 2D+3D	Agent-based + Data-driven model	Design	18–25 min
D6.4	Evaluating competing solutions	Weights Change Winners: The Trade-off Bench	Data Dashboard	Data-driven model	Argue-from-data	18–25 min
D6.5	A biodiversity solution for a real ecosystem	Delta Commission: A Budget, a Room, and Thirty Years	Hybrid 2D+3D	Agent-based + Data-driven model	Design	18–25 min

D1 · Resource availability and populations · MS-LS2-1

D1.1 · Limiting resources

Experiment name: Which Shortage Bites First: Tamales Point

Render mode: Hybrid 2D+3D

Simulation engine: Agent-based + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-1

THEME & SCENE

A high three-quarter view over the Tamales Point headland at Point Reyes: yellow-green coastal prairie running out to a grey Pacific, fog lying in the swales, the elk fence a thin dark line across the neck of the peninsula. Roughly two hundred tule elk graze as individual animals, heads down, calves tucked beside cows, a bull bugling on the ridge. Two shallow stock ponds glint; a seep darkens one gully. Coyote brush thickets throw hard afternoon shade. On the left, five vertical gauge staves labelled Forage, Water, Minerals, Shade, Space fill and drain live. The control panel docks left beneath them; a hover probe reads any animal's private hunger and thirst.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Elk cow agent	Actor	Low-poly cervid ~2.0 u nose-to-tail, tan body, darker neck ruff; walks 0.6 u/s, grazes head-down 8 s bouts; carries private hunger, thirst, salt, heat and space needs; instanced ×20–600	Yes: drag
2	Elk bull agent	Actor	Same body 15 % larger, six-point antlers, mahogany mane; holds a rut territory disc in autumn; higher forage demand	Yes: drag
3	Elk calf agent	Actor	0.9 u, pale dappled coat, tracks its cow within 4 u; dies first when any need is unmet	No
4	Forage tile field	Field	8 m grid over 1,050 ha of coastal prairie; each tile holds standing biomass 0–4,000 kg/ha, drawn as grass height plus a green-to-straw colour ramp; regrows on a seasonal logistic curve	Yes: resize
5	Stock pond	Structure	Shallow ellipse basins 20–60 u across, volume 0–4,000 m ³ , water plane drops visibly as drunk and evaporated; cracked mud ring appears below 15 %	Yes: place
6	Seep spring	Structure	4 u dark wet patch feeding 0.5–6 m ³ /day regardless of season; supports only ~12 animals/day	Yes: place
7	Mineral lick patch	Structure	6 u scuffed pale-grey soil scar; supplies sodium; elk queue at it, one animal per 20 s	Yes: place
8	Coyote brush thicket	Structure	Clumped 3 u dark-olive shrub domes casting real shade; each thicket shelters ~8 animals from heat load	Yes: place, resize
9	Elk fence	Structure	2.4 m woven wire on posts across the peninsula neck, 4.8 km long; a hard collider blocking emigration; can be opened at a gate or removed	Yes: swap
10	Rut territory field	Field	Invisible repulsion discs radius 60 u around each bull in autumn; caps how many bulls the headland holds	No
11	Need-state bars	Overlay	Five 12 px micro-bars floating above each selected animal: forage, water, salt, heat, space; red below threshold	No
12	Limiting-stave gauge	Overlay	Five vertical staves, tallest = most abundant; the shortest stave glows amber and is labelled "limiting now" every tick	No
13	Scarcity spotlight	Overlay	Desaturates the whole scene except the currently limiting resource, which keeps full colour	No
14	Fog drip layer	Environment	Volumetric low cloud rolling in from the west on summer mornings; adds 0.2–1.4 mm/day condensation to forage and ponds	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Census drone	Instrument	Small quadcopter running a fixed transect every simulated month; produces the count the graph plots, with $\pm 8\%$ sighting error	Yes: drag
16	Carcass marker	Overlay	Small bone-white pin left where an animal died, colour-tagged by cause of death	No

HOW IT WORKS — THE MODEL

Each elk is an agent with five independent need meters that drain at rates set by body size, season and heat. Every tick an animal moves to the nearest tile that satisfies its most urgent unmet need, consumes from that stock, and refills that meter. Stocks are finite and deplete: forage tiles lose the kilograms eaten and regrow at the productivity rate; pond volume falls by drinking plus evaporation; lick and shade have hard queue capacities. Survival is governed by the single worst-off meter, not by an average, so the population is capped by whichever resource runs out first, exactly as Liebig's law of the minimum states. Raising any other resource changes nothing measurable. The model must never show a generic "food" bar: the five stocks are separate objects with separate units, because the misconception being attacked is that animals need "enough resources" in general.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Forage productivity	Slider	200–3,000	1,600	kg/ha per season	Regrowth rate and peak height of every grass tile
Pond water volume	Slider	0–4,000	1,200	m ³	Starting drinking water; pond surface redraws
Summer fog days	Slider	0–60	30	days/season	Fog drip added to forage and ponds; slows evaporation
Herd starting size	Stepper	20–600	300	head	Number of elk agents instantiated
Mineral lick patches	Stepper	0–6	2	count	How many sodium sources exist on the headland
Shade thicket cover	Slider	0–40	15	% of area	Number and size of coyote brush domes; heat relief capacity
Fence state	Dropdown	Closed (historic) / Gate open / Fence removed	Closed (historic)	—	Whether elk may leave for Limantour range when a need fails
Season preset	Dropdown	Wet spring / Dry summer / Drought summer 2021	Dry summer	—	Loads real rainfall, temperature and forage figures
Scarcity spotlight	Toggle	On / Off	Off	—	Greys out every resource except the limiting one
Playback speed	Dial	0.5×–8×	2×	—	Simulated days per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Green year	forage=2,400; pond=3,000; fog=45; herd=300; season=Wet spring	With everything topped up, which stave is still the shortest, and how many elk does the headland hold?
S2	Drought summer 2021	season=Drought summer 2021; pond=250; forage=900; herd=445	Grass is still standing in places, yet elk are dying. Which stock actually ran out, and how do the carcass tags prove it?
S3	Grass gone, water fine	forage=350; pond=4,000; fog=50; herd=300	Water is unlimited here. Why does adding more of it not save a single animal?
S4	Open the fence	fence=Fence removed; forage=900; pond=250; herd=445	When animals may walk to the next range, does the shortage still limit the herd? What replaces death as the response?

STUDENT ACTIVITIES

1. Run S1 for two simulated years, then record which stave the gauge labels "limiting now" in each of the four seasons. Write the four answers in a row.

2. Set pond volume to 4,000 m³ and change nothing else in S3. Record herd size after one year, then compare it with the S3 result and explain the identical number.
3. Drag ten elk onto the map to push the herd past 450. Record how many days pass before the first carcass marker appears and which cause colour it carries.
4. Place two extra mineral licks and one extra pond in the drought preset. Record herd survival, then state which single addition would have worked better and why.
5. Switch the fence to Fence removed and re-run S2. Record emigration count and final herd size, and describe what limits the population once escape is possible.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Herd size	Line graph	head	Census-drone count per simulated month, with the $\pm 8\%$ sighting band shaded
Resource stocks	Line graph	mixed	Four traces: forage kg/ha, pond m ³ , lick queue length, shade slots free
Limiting resource log	Data table	—	One row per month naming the shortest stave and its value
Cause of death tally	Bar chart	count	Starvation, dehydration, heat, salt deficiency, old age
Individual need probe	Live numeric	%	The five meters of the animal under the cursor, updated each tick
Emigration count	Counter	head	Animals that left through an open gate or removed fence
Run comparison	Data table	mixed	Exportable CSV: settings in, final herd size and limiting resource out

WHAT THE STUDENT SHOULD REALISE

Students believe a population is limited by "not enough resources" as a general shortage, so adding any resource should help. Here four stocks can sit full while one empties, and the herd falls to the level the emptiest one allows. Pouring in water when grass is the problem changes nothing measurable. The student should be able to say: "A population is set by whichever single resource runs short first, and topping up the others does nothing."

D1.2 · Carrying capacity

Experiment name: The Island Fills Up: Foxes on Santa Cruz

Render mode: Hybrid 2D+3D

Simulation engine: Agent-based

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-1

THEME & SCENE

A slow orbit above Santa Cruz Island: chaparral ridges in grey-green, oak groves in the canyons, fog pooling on the north shore, the Santa Barbara Channel a hard blue rim. Cat-sized island foxes trot along ridgelines in ones and twos, each with a faint energy ring around it. Deer mice glitter as small pale sparks in the scrub; toyon and manzanita bushes carry countable red fruit clusters that empty as they are eaten. Dens show as dark holes under rock outcrops. The right-hand panel holds two graphs stacked: fox number against time, and growth per fox against fox number. Nothing on screen is labelled with a capacity figure.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Island fox agent	Actor	0.5 u body, rust-and-grey coat, black tail tip; trots 0.9 u/s along ridge paths; carries energy 0–100, age, sex and a den claim	Yes: drag
2	Fox pup	Actor	0.25 u, woolly grey; appears in litters of 1–5 in spring, stays within 6 u of the den for 60 days	No
3	Energy ring	Overlay	Thin arc around each fox, green above 60, amber 30–60, red below 30; fox dies at 0	No
4	Deer mouse stock	Field	Per-cell prey density 0–140 mice/ha drawn as pale sparks; regrows logistically toward its own ceiling; depleted where foxes hunt	Yes: resize
5	Toyon and manzanita shrub	Structure	2 u domed shrubs with 0–40 instanced fruit spheres, scarlet; fruit count drops as eaten and refills on a seasonal mast schedule	Yes: place
6	Insect patch	Field	Jerusalem cricket and beetle density under leaf litter, 0–60 units/ha; low-value, always-available food floor	No
7	Den site	Structure	Dark 0.6 u burrow mouth under a rock outcrop; a fixed number exist; only a fox holding one may breed	Yes: place
8	Chaparral terrain tile	Environment	250 km ² heightfield split into 8 ha cells with slope and vegetation type; sets travel cost and mouse regrowth	No
9	Golden eagle	Actor	1.8 u wingspan raptor circling at altitude; stoops on a fox every N minutes when present; historically the cause of the 1990s crash	Yes: place
10	Feral pig herd	Actor	1.2 u dark bodies rooting in oak groves; piglets subsidise eagles, raising eagle numbers without any fox involvement	Yes: place
11	Energy budget field	Field	Invisible per-agent ledger: intake minus travel cost minus basal cost per day; drives the energy ring	No
12	Radio-collar census	Instrument	Annual mark-recapture routine returning an estimate with a 90 % interval; this is the number graphed	No
13	Growth-per-fox plot	Overlay	Scatter of annual per-capita growth rate against that year's fox number; the descending line and its zero crossing are the point of the sim	No
14	Overshoot marker	Overlay	Red bracket drawn on the time graph wherever fox number exceeds the level the food stock can sustain	No
15	Carcass and cause tag	Overlay	Bone-white pin colour-coded starvation, predation, age, disease	No
16	Prey-stock gauge	Instrument	Island-wide totals for mice, fruit and insects as three thin horizontal bars above the graphs	No

HOW IT WORKS — THE MODEL

No carrying capacity is stored anywhere in the code. Each fox spends energy on basal metabolism and travel and gains it by eating whatever prey cell it reaches; prey stocks are real depletable quantities that regrow at their own rates. A fox above 70 energy that holds a den breeds; a fox at 0 dies. When there are few foxes, prey is thick, energy gain is easy, litters survive and numbers climb steeply. As foxes accumulate they eat the prey stock down, per-fox intake falls, litters fail and deaths rise until births equal deaths. The plateau the student sees is that balance point, and the growth-per-fox plot falls linearly to zero at exactly that number. The failure to avoid is drawing a smooth S-curve from an equation: the curve must be the emergent census of individuals, noisy, and capable of overshooting and crashing.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Starting foxes	Stepper	2–400	10	foxes	How many agents exist at year zero
Island area in use	Slider	10–250	250	km ²	Terrain cells available; fewer cells means less total prey
Mouse regrowth rate	Slider	0.1–1.2	0.6	per year	How fast the deer mouse stock refills after being hunted
Fruit crop	Dropdown	Failed / Poor / Normal / Mast year	Normal	—	Fruit spheres per shrub and their refill schedule
Den sites	Stepper	5–400	220	count	Breeding slots; a structural cap independent of food
Litter size	Slider	1–5	3	pups	Pups born per denning female per spring
Golden eagles	Stepper	0–8	0	birds	Predation pressure; each bird removes foxes at a set rate
Feral pigs	Toggle	Present / Removed	Removed	—	Whether an eagle food subsidy exists on the island
Drought severity	Slider	0–80	0	% cut to regrowth	Multiplier applied to all prey and fruit regrowth
Playback speed	Dial	0.5×–10×	3×	—	Simulated years per real minute

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Ten foxes, empty island	foxes=10; area=250; mouse regrowth=0.6; eagles=0	Sketch the shape of the population curve. At what number does it stop rising, and what stopped it?
S2	Dropped in at the top	foxes=400; area=250; mouse regrowth=0.6	Starting far above the plateau, what happens in the first three years, and why does the population fall below the plateau before settling?
S3	Half the mice	mouse regrowth=0.3; foxes=10; fruit=Normal	Predict the new plateau before running. By what fraction did it move, and did the shape of the curve change?
S4	Eagles and pigs, 1994	eagles=4; pigs=Present; foxes=1,400 scaled; drought=0	The island held foxes easily for decades. Why does the population now crash to double figures with food untouched?

STUDENT ACTIVITIES

- Run S1 for 40 simulated years. Record fox number at years 2, 5, 10, 20 and 40, then draw the curve from your five points before looking at the graph.
- Read the prey-stock gauge at year 2 and again at year 40. Record both, and state in one sentence what the foxes did to the mice.
- Halve the mouse regrowth rate and re-run from 10 foxes. Record the new plateau and calculate the ratio between the two plateaus.
- Set den sites to 30 with mouse regrowth back at 0.6. Record the plateau and explain why plenty of food no longer sets the ceiling.

5. Read the growth-per-fox plot and record the fox number at which the line crosses zero. Compare it with the plateau on the time graph.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Fox population	Line graph	foxes	Annual mark-recapture estimate with its 90 % interval band
Prey stocks	Line graph	units/ha	Three traces: deer mice, fruit, insects, island-wide
Growth per fox vs number	Line graph	per year vs foxes	Scatter plus fitted line; the zero crossing is the emergent capacity
Measured plateau	Live numeric	foxes	Mean of the last ten years, reported only after the run ends
Mean fox energy	Live numeric	0–100	Population average, updated yearly
Deaths by cause	Bar chart	count/year	Starvation, predation, age, disease
Overshoot flag	Pass-fail badge	—	Amber whenever numbers exceed what current prey can sustain
Run archive	Data table	mixed	Exportable CSV of settings and resulting plateau for every run

WHAT THE STUDENT SHOULD REALISE

Students think carrying capacity is a fixed property of a place, like a stadium's seat count. Here the plateau is never entered as a setting: it appears because foxes eat the prey stock down until births match deaths, and it moves whenever prey regrowth, fruit crops, den sites or drought change. The same island holds different numbers in different years. The student should be able to say: *"Carrying capacity is not a number the island has. It is where growth runs out, and it moves when resources move."*

D1.3 · Reading real population data

Experiment name: Three California Censuses, One Argument

Render mode: Data Dashboard

Simulation engine: Data-driven model

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS2-1

THEME & SCENE

A wide dark-slate analyst's desk. Left, three dataset cards stacked like field notebooks: an island fox card with a fox photograph and the Channel Islands in silhouette, a tule elk card showing the Tomales Point fence line, a sea otter card with a raft of otters in kelp. Centre, a large plot canvas, pale grid, one bold data series in warm amber with open circles for each survey year and a translucent uncertainty ribbon behind it. Right, a claim-builder tray of blank cards and an evidence basket. A small California map inset marks the survey area for whichever card is loaded. Annotation pins hang below the time axis waiting to be opened.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Dataset card — island fox	Instrument	Notebook tile holding per-island counts 1993–2016 for San Miguel, Santa Rosa and Santa Cruz; loads on drag to the canvas	Yes: drag
2	Dataset card — tule elk	Instrument	Tomales Point herd counts 1978–2023, annual, from the reintroduction of ten animals through the drought die-off	Yes: drag
3	Dataset card — sea otter	Instrument	California spring survey index 1985–2023, three-year running mean supplied alongside raw counts	Yes: drag
4	Plot canvas	Overlay	900×520 px axes; year on x, count or density on y; supports two series at once with paired axes	Yes: resize
5	Data point marker	Overlay	Open 7 px circle per survey year; hovering shows year, count, method and observer note	No
6	Uncertainty ribbon	Overlay	Translucent band around each series showing the published interval; collapses to a line when the toggle is off	No
7	Annotation pin	Overlay	Twelve draggable pins carrying real events: golden eagles arrive 1994, captive breeding begins 1999, last releases 2004, delisting 2016, sea star wasting 2013, marine heatwave 2014, drought 2020–21	Yes: drag, place
8	Smoothing engine	Field	Running-mean filter of width 1–5 years applied live; draws the smoothed line over the raw points	No
9	Model overlay curve	Overlay	Selectable dashed curve: none, exponential, logistic, or logistic with a step event; fitted parameters shown beside it	Yes: swap
10	Capacity guide line	Overlay	Horizontal dashed line the student drags to their estimate of the plateau; reports the value it sits at	Yes: drag
11	Residual strip	Overlay	Thin panel under the plot showing data minus model per year as up or down bars	No
12	Claim card	UI-Probe	Blank card with a dropdown claim stem and a slot for three evidence chips	Yes: place
13	Evidence chip	UI-Probe	Chip minted by clicking any data point, pin or measured value; carries its number and source	Yes: drag
14	Claim scorer	Instrument	Checks whether the chips in a claim actually support its stem; returns green, amber or red with a written reason	No
15	Method note tooltip	Overlay	Explains how each series was collected: mark-recapture grids, ground counts from the fence line, shore and aerial otter counts	No
16	California map inset	Environment	240 px orthographic state map with the active survey area highlighted and a scale bar	No

HOW IT WORKS — THE MODEL

Nothing here is invented. Each card loads a stored survey series with its published years, values, method and interval, and every drawn element is derived from those numbers. The smoothing engine recomputes a running mean of the chosen width each time the stepper moves, so the student can watch a jagged otter series turn into a trend and back again. The model overlay fits its curve to the visible window only, reporting growth rate and, for logistic, the plateau it inferred; residuals are the honest difference between fit and data. The claim scorer parses the selected stem, checks the chips against the series, and rejects claims whose evidence is off-topic, out of window, or contradicted. The sim must never redraw a data point to fit the model, and must never hide a survey year with a wide interval, because the skill being taught is reading data that argues with you.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Dataset	Dropdown	Island fox (San Miguel) / Island fox (Santa Cruz) / Tule elk (Tomaes Point) / Sea otter (California)	Island fox (San Miguel)	—	Loads a different real series, map inset and method notes
Second series	Dropdown	None / any of the four above	None	—	Adds a paired series on a second y-axis for comparison
Year window	Timeline scrubber	1978–2023	Full range	year	Which survey years the plot and any fit use
Smoothing window	Stepper	1–5	1	years	Width of the running mean drawn over the raw points
Uncertainty ribbon	Toggle	On / Off	On	—	Shows or hides the published interval around each point
Model overlay	Dropdown	None / Exponential / Logistic / Logistic + event	None	—	Which curve is fitted to the visible window
Capacity guide	Drag-handle	0–3,600	Off	animals	Student's own estimate of the plateau, reported live
Y-axis quantity	Radio	Total count / Density per km ²	Total count	—	Rescales using the surveyed area of the active dataset
Annotation pins	Toggle	On / Off	On	—	Reveals the dated real events under the time axis

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The fox that nearly went	Dataset=Island fox (San Miguel); window=1993–2016; pins=On; overlay=None	The series falls from hundreds to fifteen and then climbs again. Which two pins sit at the turning points, and what does each say?
S2	A herd against a fence	Dataset=Tule elk (Tomaes Point); window=1978–2023; overlay=Logistic; capacity guide=drag	Where does the herd first stop rising, and does the 2020–21 fall look like normal wobble or something new?
S3	Noise or trend	Dataset=Sea otter (California); smoothing=1 then 3 then 5; ribbon=On	At a one-year window the index goes up and down constantly. With a three-year mean, what single sentence describes 1985–2023?
S4	Two islands, one threat	Dataset=Island fox (San Miguel); second series=Island fox (Santa Cruz); window=1993–2010	The two islands crash at different times and different depths. What does that tell you about the cause?

STUDENT ACTIVITIES

1. Load the San Miguel fox series and record the lowest count, its year, and the two nearest annotation pins.

2. Set the smoothing window to 1, then 3, then 5 on the otter series. Record the 2000 value each time and state which width you would report and why.
3. Drag the capacity guide to where the elk herd levels off and record the number, then compare it with the plateau the logistic fit reports.
4. Build a claim card reading "This population was limited by a resource shortage" and fill it with three evidence chips. Record the scorer's colour and its written reason.
5. Compare the two fox islands on shared axes and record, for each island, the first year of decline and the year recovery began.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Population series	Line graph	animals	The loaded survey values with their intervals and method markers
Smoothed trend	Line graph	animals	Running mean of the chosen width, redrawn live
Fitted parameters	Live numeric	mixed	Growth rate per year and inferred plateau for the selected overlay
Residuals	Bar chart	animals	Data minus model per year, positive above the axis
Capacity guide value	Live numeric	animals	Where the student's dashed line currently sits
Claim score	Pass-fail badge	—	Green, amber or red on each claim card with the reason text
Evidence log	Data table	mixed	Every chip minted, with its value, year and source dataset
Export	Data table	mixed	CSV of the visible window, raw and smoothed, for use outside the sim

WHAT THE STUDENT SHOULD REALISE

Students treat a population graph as a picture of what happened, and read every wiggle as a real event. Working with three genuine California series, they find intervals wide enough to swallow a wiggle, a smoothing choice that changes the story, and a crash whose cause is a pin rather than a shortage. The student should be able to say:

"Before I say a population changed, I have to check whether the survey could tell the difference."

D1.4 · Predicting the response to scarcity

Experiment name: Call It First: The Scarcity Forecast Desk

Render mode: Hybrid 2D+3D

Simulation engine: Agent-based + Data-driven model

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS2-1

THEME & SCENE

A forecaster's desk split down the middle. On the right, a live 3D window into one California ecosystem: a Monterey kelp forest with otters cracking urchins in the canopy shadows, or the Santa Cruz Island scrub, or the Tomales Point headland. On the left, a broad prediction chart, pale on charcoal, with the population history drawn solid up to today and empty grid beyond it. A dashed forecast spline waits with six draggable control points, and a translucent tolerance ribbon follows it. Below sits a deck of twelve mechanism cards, face up, each naming one way a population can answer a shortage. A red event marker on the timeline shows when the scarcity will hit.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Ecosystem window	Environment	3D viewport, 620×620 px, rendering whichever system is loaded with live agents at 1 frame per simulated day	Yes: drag (camera)
2	Sea otter agent	Actor	1.3 u body, dark brown, floats on its back with an urchin on its chest; forages in dives, switches prey when urchins thin	No
3	Purple urchin agent	Actor	0.15 u spiny hemisphere, violet; grazes kelp holdfasts; survives long fasts by shrinking gonads	No
4	Bull kelp stipe	Structure	8–20 u swaying stipes with a gas float and blade fan, amber-brown; count falls as urchins graze and rises with nutrient upwelling	No
5	Prediction spline	UI-Probe	Dashed forecast line through six draggable control points spanning the forecast window	Yes: drag
6	Tolerance ribbon	Overlay	Band of adjustable half-width around the spline; the truth curve must sit inside it to score	Yes: resize
7	Mechanism card deck	UI-Probe	Twelve cards: fewer births, more deaths, emigration, diet switch, smaller bodies, delayed breeding, den or territory loss, disease spread, predator switches prey, cannibalism, dormancy, no change	Yes: place
8	Event injector	Instrument	Timeline marker the student drags to set when scarcity starts; carries type, severity and duration	Yes: drag
9	Resource stock gauge	Instrument	Vertical bar for the scarce resource in its own unit: kelp stipes/m ² , mice/ha, m ³ of pond water	No
10	Truth curve	Overlay	Solid line drawn only after Run, produced by the same agent engine used in D1.1 and D1.2	No
11	Lag marker	Overlay	Bracket between the event start and the first month the population changes measurably	No
12	Per-capita rate panel	Overlay	Two small traces, births per individual and deaths per individual, that move before the total does	No
13	Residual bars	Overlay	Prediction minus truth per month, drawn under the chart	No
14	Forecast scorer	Instrument	Scores direction, depth, lag and recovery separately, plus the mechanism cards played, and prints a written verdict	No
15	Replay scrubber	UI-Probe	Timeline handle that steps the 3D window back and forth so the student can watch what the agents did during the drop	Yes: drag
16	Locked-forecast seal	Overlay	Small padlock that greys the spline once Run is pressed, so a prediction cannot be edited after the fact	No

HOW IT WORKS — THE MODEL

The student commits first. They shape a forecast spline, choose a ribbon width, and play up to three mechanism cards; pressing Run seals all of it. The engine then runs the same individual-based population model used elsewhere in this topic, with the chosen resource stock cut by the chosen severity for the chosen duration. Populations do not respond instantly: individuals draw down body reserves, then births fail, then deaths rise, so the total lags the resource by weeks to a season, and the per-capita panel moves first. Scoring is component-wise, so a student who gets the direction right but the timing wrong is told exactly that. The engine must not tune itself toward the prediction, and the truth curve must carry its own demographic noise, because the lesson is that a good forecast is one that survives an honest run.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
System	Dropdown	Monterey kelp forest / Santa Cruz Island fox / Tomales Point elk	Monterey kelp forest	—	Loads a different ecosystem, agent set and 3D scene
Scarce resource	Dropdown	Kelp / Deer mice / Fruit crop / Drinking water / Den sites	Kelp	—	Which stock the event cuts; gauge relabels and re-units
Event severity	Slider	0–95	60	% cut to the stock	How deep the shortage goes
Event start	Timeline scrubber	Month 6–36	Month 12	month	When the shortage begins
Event duration	Slider	1–36	12	months	How long the stock stays suppressed before recovering
Recovery	Toggle	Stock recovers / Stays low	Stock recovers	—	Whether the resource returns after the event
Tolerance ribbon width	Slider	5–40	20	% of population	How generous the scoring band is
Mechanism cards played	Multi-select	The twelve deck cards, maximum three	None	—	The responses the student commits to before running
Forecast horizon	Stepper	12–60	36	months	How far past today the prediction and the run extend
Playback speed	Dial	0.5×–6×	2×	—	Simulated months per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	A shallow dip	system=Monterey kelp forest; resource=Kelp; severity=30; duration=6; recovery=Stock recovers	Predict the otter and urchin curves. Does a shallow, short shortage leave a permanent mark?
S2	The kelp does not come back	severity=85; duration=36; recovery=Stays low; horizon=60	Predict the depth of the fall and where the population settles. Why does it settle above zero rather than dying out?
S3	Two years of drought on the island	system=Santa Cruz Island fox; resource=Deer mice; severity=70; duration=24	Predict the lag. How many months pass between the mouse crash and the fox count falling?
S4	Take away the dens, not the food	system=Santa Cruz Island fox; resource=Den sites; severity=80; duration=24	With food untouched, predict which mechanism card the model plays. Does the shape differ from S3?

STUDENT ACTIVITIES

1. Drag the six control points to draw your forecast for S1 before running anything. Record the lowest value your spline reaches and the month it happens.

2. Play up to three mechanism cards and record which you chose. After the run, record which the model actually used.
3. Run S1 and record the scorer's four component results: direction, depth, lag and recovery.
4. Use the replay scrubber to step back into the month the drop began and record what the per-capita panel was doing before the total moved.
5. Repeat your best forecast in S3 with the ribbon width cut to 5 %. Record the new score and explain what a narrow ribbon actually claims.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Forecast vs truth	Line graph	animals	Sealed dashed prediction, ribbon, and the solid run outcome on one axis pair
Component score	Pass-fail badge	—	Four badges: direction, depth, lag, recovery, each green or red with a reason
Mechanism match	Data table	—	Cards played beside the mechanisms the engine actually invoked, with counts
Response lag	Live numeric	months	Gap between event start and the first month outside normal variation
Per-capita birth and death rates	Line graph	per individual per month	The two traces that move before the population total does
Resource stock	Line graph	stock unit	The scarce resource in its own units, with the event window shaded
Residuals	Bar chart	animals	Prediction minus truth per month across the horizon
Forecast log	Data table	mixed	Exportable CSV of every sealed forecast, its settings and its four scores

WHAT THE STUDENT SHOULD REALISE

Students expect a population to drop the moment its food does, and to drop to zero if the shortage is severe. Sealed forecasts scored against an honest run show a lag of weeks to months, a fall that stops at a new lower level, and responses other than death: fewer births, smaller bodies, animals leaving. The student should be able to say: "A shortage does not kill a population instantly. It changes births, deaths and movement, and the numbers follow later."

D1.5 · Competition for the same limiting resource

Experiment name: One Reef, One Ration: Urchin Against Abalone

Render mode: 3D Scene

Simulation engine: Agent-based

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-1

THEME & SCENE

A shallow Monterey reef seen from just above the rock, camera low so the animals are at eye level. Granite benches are furred with coralline algae in dusty pink, cut by crevices and surge channels. Purple urchins sit in shallow pits with their spines fanned; red abalone clamp the rock in the shadows, shells crusted, muscular feet gripping. Torn blades of giant kelp drift in on a slow surge and settle: this is the food, and there is a countable amount of it. Bubbles rise, light shafts wander. A stock gauge on the left shows drift kelp in grams per square metre; two population counters, violet and red, run beneath it.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Purple urchin agent	Actor	0.08 u test with ~60 instanced spines, violet; creeps 0.02 u/s on tube feet; traps drifting kelp with spines; carries gut fullness and body condition	Yes: drag
2	Red abalone agent	Actor	0.18 u oval shell, brick-red with respiratory pores, cream foot; moves 0.01 u/s; traps drift kelp under the shell edge; condition drives shell growth	Yes: drag
3	Drift kelp blade	Particle	0.4–1.2 u torn amber blades that arrive on the surge, tumble, then settle and are consumed gram by gram; count set by supply rate	No
4	Attached giant kelp holdfast	Structure	0.5 u rooted claw of haptera with a stipe rising out of frame; a second, harder-to-reach food source only urchins can attack	Yes: place
5	Coralline crust layer	Environment	Pink lithified algal veneer on all rock; grazed patches turn chalk-white, marking a developing barren	No
6	Crevice refuge	Structure	Dark 0.3 u rock slots; a fixed number; abalone inside are safe from otters and lose less energy to surge	Yes: place
7	Sea otter agent	Actor	1.3 u forager arriving on a schedule; selects the largest visible urchin, ignores sheltered abalone; removes prey and leaves shell fragments	Yes: place
8	Drift supply emitter	Field	Invisible upstream line releasing blades at the set grams per square metre per day; pulsed with a swell cycle	No
9	Resource stock field	Field	Per-cell grams of settled drift kelp, depleted by grazing, replenished by the emitter, decaying at 3 % per day	No
10	Body condition ring	Overlay	Ring around each animal, green to red, showing reserves; abalone shells visibly thin and urchin tests visibly shrink at low condition	No
11	Starvation threshold marker	Overlay	Two horizontal lines on the stock gauge showing the ration each species needs to break even	No
12	Barren extent overlay	Overlay	False-colour wash marking rock where coralline has been grazed white, with a live percentage	No
13	Shell fragment litter	Structure	Cracked test and shell debris accumulating where animals died, tagged by species and cause	No
14	Quadrat sampler	Instrument	Draggable 1 m ² frame that counts both species and weighs drift kelp inside it	Yes: drag
15	Population counter pair	Instrument	Two live counts plus a stacked area trace of both species over time	No
16	Surge current field	Field	Oscillating flow vector driving blade transport and adding an energy cost to any animal not in a crevice	No

HOW IT WORKS — THE MODEL

Both species eat the same settled drift kelp, and the supply arriving each day is finite. Every tick, each animal grazes from its cell, pays a metabolic cost, and updates its condition; above a growth threshold it reproduces, at zero condition it dies. Because both draw on one stock, more of either species drives the standing kelp down for both. The species that can still break even on the thinner ration keeps growing while the other keeps shrinking, so it drives the stock below its rival's threshold and the rival disappears: competitive exclusion emerges without any rule that says "urchins beat abalone". Coexistence appears only when the tie to one resource is broken, by attached kelp that only urchins can reach, by crevices that shelter abalone, or by otters cropping urchins. The engine must never remove a species by decree.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Drift kelp supply	Slider	0–60	22	g/m ² per day	Blades released by the emitter; the whole shared ration
Starting purple urchins	Stepper	0–400	60	per 100 m ²	Urchin agents instantiated at t=0
Starting red abalone	Stepper	0–120	40	per 100 m ²	Abalone agents instantiated at t=0
Urchin break-even ration	Slider	0.2–4.0	0.6	g/day per animal	Food an urchin needs to hold condition; sets its survival floor
Abalone break-even ration	Slider	0.5–8.0	2.4	g/day per animal	Food an abalone needs to hold condition
Attached kelp	Toggle	Present / Absent	Absent	—	Adds a second food source only urchins can reach
Crevice refuges	Stepper	0–60	10	count	Shelter slots that cut abalone energy cost and hide them from otters
Sea otters	Stepper	0–4	0	otters	Predators that crop the largest urchins only
Water temperature	Slider	9–20	13	°C	Metabolic costs for both species and drift decay rate
Playback speed	Dial	0.5×–8×	3×	—	Simulated days per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Fair start, one food	supply=22; urchins=60; abalone=40; attached=Absent; otters=0; refuges=0	Both start healthy on the same reef. After three simulated years, which species is left, and what number on the stock gauge explains it?
S2	Outnumbered but tougher	urchins=5; abalone=100; supply=22; attached=Absent	The urchins begin rare. Do they still take over, and what does starting number actually decide?
S3	Two different foods	attached=Present; supply=22; urchins=60; abalone=40	Give urchins something abalone cannot eat. Do both species now persist, and what happened to the drift kelp level?
S4	Otters return to the reef	otters=2; attached=Absent; supply=22; urchins=60; abalone=40	With a predator eating only urchins, which species survives that would not have? Is that competition or predation?

STUDENT ACTIVITIES

1. Run S1 for three simulated years and record both counts at years 0, 1, 2 and 3, plus the drift kelp stock at the end.
2. Record each species' break-even line from the stock gauge, then state which line the final stock sits below.
3. Set urchins to 5 and abalone to 100, re-run, and record whether the outcome changes. Write one sentence on what starting number does and does not decide.
4. Turn attached kelp on and re-run. Record both final counts and the final drift kelp stock, and compare that stock with S1.
5. Place two otters with attached kelp off. Record final counts, then argue in two sentences whether abalone survived because of the otters or because of the kelp.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Species counts	Line graph	animals per 100 m ²	Two traces, violet and red, from a monthly quadrat census
Drift kelp stock	Line graph	g/m ²	Standing food, with both break-even thresholds drawn as horizontal lines
Mean body condition	Live numeric	0–100	Population average per species, updated weekly
Barren extent	Live numeric	% of rock	Fraction of coralline crust grazed white
Deaths by cause	Bar chart	count	Starvation, predation, surge dislodgement, per species
Quadrat sample	Data table	mixed	Counts and kelp weight from wherever the student drops the frame
Outcome badge	Pass-fail badge	—	Reports Exclusion or Coexistence at the end of each run, with which species won
Run archive	Data table	mixed	Exportable CSV: settings in, both final counts and outcome out

WHAT THE STUDENT SHOULD REALISE

Students think competition is a fight, decided by size, aggression or who got there first. Nothing here fights, and starting numbers do not decide the winner. The species that can still break even on the thinner ration eats the shared stock down past what the other needs, and the other simply starves. The student should be able to say:

"Two species cannot live on one limiting resource unless something splits them apart, and the one that survives is the one that needs less."

D2 · Food webs and energy flow · MS-LS2-3

D2.1 · Producers, consumers and decomposers

Experiment name: The Floor That Eats: A Sierra Nutrient Ledger

Render mode: 2.5D Layered

Simulation engine: Agent-based + Fluid/thermal

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-3

THEME & SCENE

A cutaway of a Sierra Nevada mixed-conifer forest, sliced vertically so half the screen is air and half is soil. Above the cut, ponderosa pine trunks in cinnamon plates, sugar pine boughs, bracken fern and a scatter of ceanothus; sunlight comes down in slanted dusty shafts. Below the cut, a soil profile in three bands: brown needle litter on top, dark crumbly humus beneath, pale mineral soil at the bottom, threaded with white fungal strands and shifting mites. A chickaree carries a cone along a branch; a black-tailed deer browses. Right of the scene, a vertical nutrient ledger shows three tanks: litter, soil nitrogen, living biomass.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Ponderosa pine	Actor	14 u trunk with plated cinnamon bark and needle bundles; grows only when soil nitrogen and light are both above threshold; drops litter each autumn	Yes: place
2	Bracken fern and ceanothus	Actor	0.8–2 u understorey clumps; fast-growing producers that respond visibly within one simulated season	Yes: place
3	Sunlight photon particles	Particle	Thin gold streaks falling at 60°, absorbed on contact with any green surface; density set by the canopy gap slider	No
4	Chickaree squirrel	Actor	0.3 u rust-and-grey agent; eats cones and fungi; leaves droppings that enter the litter tank	Yes: drag
5	Black-tailed deer	Actor	1.6 u browser cropping fern and ceanothus tips; removes living tissue, returns dung	Yes: drag
6	Needle litter layer	Structure	Stackable 2D strata of brown needles above the soil line; thickness in centimetres is a live state variable and visibly grows	No
7	Fungal hyphae mat	Actor	Branching white filament network that spreads cell by cell through litter, thinning it; growth rate set by moisture and temperature	Yes: place
8	Turkey-tail bracket fungi	Structure	Banded 0.3 u fruiting shelves on fallen logs; appear only where hyphae density is high; visible marker of decomposition	No
9	Millipede and springtail agents	Actor	0.05 u detritivores shredding needles into fragments, which raises the surface area available to hyphae	Yes: place
10	Soil bacteria field	Field	Invisible per-cell population converting shredded litter into plant-available nitrogen at a temperature-dependent rate	No
11	Soil nitrogen stock	Field	Per-cell nitrogen in kg/ha, drawn as a blue saturation wash in the mineral band; drained by plant uptake, refilled by decomposers	No
12	Carbon dioxide plume	Particle	Faint grey wisps leaving every respiring organism, including hyphae and bacteria; density proportional to respiration rate	No
13	Fallen log	Structure	6 u downed trunk that decays over simulated decades, hosting hyphae, brackets and invertebrates	Yes: place
14	Role tag overlay	Overlay	Toggleable badge over every organism reading Producer, Consumer or Decomposer, with its carbon source named beneath	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Nutrient ledger tanks	Instrument	Three vertical tanks, litter, soil nitrogen and living biomass, in kg/ha, that must sum to a constant unless fire or export is on	No
16	Soil core probe	Instrument	Draggable corer returning litter depth, humus depth and nitrogen at the sampled point	Yes: drag

HOW IT WORKS — THE MODEL

Every organism is tagged by where its carbon comes from, and the tag is enforced by the engine rather than written on a label. Producers take carbon from photon particles and nitrogen from the soil stock. Consumers take carbon by removing living tissue from another agent. Decomposers take carbon only from dead material in the litter layer, and are the sole process that converts litter back into soil nitrogen. Litter grows by leaf fall, dung and death, and shrinks only through detritivore shredding and hyphal and bacterial breakdown, both scaled by moisture and temperature. Nutrients therefore circulate while energy leaves as carbon dioxide from every respiring agent, including the fungi. If decomposers are removed, the litter tank climbs, the soil nitrogen tank drains, and producers stop growing within a few simulated years. The failure to avoid is treating decomposers as tidy-up crew rather than as the return path.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Decomposer abundance	Slider	0–100	70	% of normal	Hyphal spread rate and bacterial activity across the soil
Detritivore count	Stepper	0–500	200	animals/m ²	Millipedes and springtails shredding needles into fragments
Litter input	Slider	0–6,000	2,800	kg/ha per year	Needle and branch fall from the canopy each autumn
Soil moisture	Slider	5–45	22	% by volume	Decomposition rate; below 10 % breakdown nearly stops
Soil temperature	Slider	0–30	12	°C	Bacterial and fungal rates; doubles roughly every 10 °C
Herbivore load	Stepper	0–20	6	animals/ha	Deer and squirrels removing living tissue
Canopy gap	Slider	0–80	25	% open sky	Photon density reaching the understorey
Fire	Dropdown	None / Low-intensity ground fire / Stand-replacing	None	—	Burns litter, returns nutrients fast, removes agents
Starting soil nitrogen	Slider	200–5,000	2,200	kg/ha	Size of the nutrient tank at t=0
Playback speed	Dial	1×–20×	6×	—	Simulated years per real minute

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	A working floor	decomposers=70; detritivores=200; moisture=22; fire=None	Sort every organism by where its carbon comes from. Which group is the only one that puts nitrogen back?
S2	Switch off the decomposers	decomposers=0; detritivores=0; litter input=2,800	Run 30 simulated years. Record litter depth and soil nitrogen. Why do the pines stop growing when nothing ate them?
S3	Too dry to rot	moisture=7; temperature=8; decomposers=70	Decomposers are present but barely working. Does the ledger behave more like S1 or S2, and what does that say about rate?
S4	Ground fire in the Sierra	fire=Low-intensity ground fire; decomposers=70; litter input=2,800	Fire returns nutrients in minutes. Compare its effect on the three tanks with 30 years of fungal decay.

STUDENT ACTIVITIES

1. Turn the role tag overlay on and record every organism in three columns: Producer, Consumer, Decomposer. Write each one's carbon source beside it.

2. Set decomposers and detritivores to 0. Run 30 simulated years and record litter depth and soil nitrogen at years 0, 10, 20 and 30.
3. Drag the soil corer to three different points and record litter depth, humus depth and nitrogen at each.
4. Restore decomposers to 70 % and record how many simulated years pass before pine growth restarts.
5. Run the low-intensity fire preset and record the change in all three ledger tanks in the first simulated month.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Litter depth	Live numeric	cm	Thickness of the needle layer at the corer position, updated yearly
Nutrient ledger	Line graph	kg/ha	Three traces: litter carbon, soil nitrogen, living biomass
Role tally	Bar chart	count	Number of agents currently tagged Producer, Consumer, Decomposer
Carbon dioxide released	Live numeric	kg/ha per year	Total respiration, split by producers, consumers and decomposers
Nitrogen return rate	Live numeric	kg/ha per year	Litter converted to plant-available nitrogen
Producer growth	Line graph	kg/ha per year	Standing plant production, the first thing to fail when return stops
Soil core log	Data table	mixed	Every core taken, with position, depths and nitrogen
Ledger closure	Pass-fail badge	—	Green while the three tanks sum to a constant; flags any leak when fire or export is on

WHAT THE STUDENT SHOULD REALISE

Students file decomposers under cleaning up, a tidy afterthought rather than part of the system. Removing them here leaves the forest visibly intact for a while and then starves it: litter piles metres deep, soil nitrogen drains, and pines stop growing though nothing ate them. The student should be able to say: *"Producers capture the energy, consumers pass it on, and decomposers are the only reason the raw materials ever come back."*

D2.2 · From food chain to food web

Experiment name: Draw the Web: Monterey Kelp Forest

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + Agent-based

Interaction level: Design

Session length: 18–25 min

NGSS anchor: MS-LS2-3

THEME & SCENE

A dark teal build canvas fills two thirds of the screen, faintly ruled, with a shelf of organism tokens along the bottom: giant kelp, phytoplankton, urchins, abalone, crabs, small fish, rockfish, sheephead, otters, seals, pelicans, sea stars, bacteria. Each token is a clean illustrated silhouette on a rounded card with its name and a small diet note. The student drags tokens onto the canvas and drags arrows between them; links snap and glow when accepted, flick red and spring back when refused. The right third is a live window on a real Monterey reef, kelp swaying in surge, so that every card on the canvas has a moving animal behind it. The control panel runs along the top.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Organism token	UI-Probe	90×110 px card: silhouette, common name, diet note, live population meter along its base; 18 available in the palette	Yes: drag, place
2	Giant kelp token and agent	Actor	Producer card; in the 3D window a 20 u stipe with blades and floats swaying on the surge; biomass drives everything above it	Yes: place
3	Phytoplankton token	Actor	Producer card representing the drifting green layer; in the window, a fine particle haze in the upper water	Yes: place
4	Detritus and bacteria token	Actor	Decomposer card fed by every death and every waste event on the canvas; returns nutrients to both producers	Yes: place
5	Feeding link arrow	Structure	Bezier arrow from eaten to eater, 3 px, colour-graded by how much energy flows along it; thickness set by diet fraction	Yes: connect, drag
6	Link direction validator	Instrument	Rejects any arrow drawn eater-to-eaten and shows the correction, since energy direction is the point	No
7	Plausibility checker	Instrument	Compares each proposed link against a stored diet database and refuses impossible ones, e.g. urchin eats otter, with a one-line reason	No
8	Diet fraction handle	UI-Probe	Small draggable weight on each incoming arrow setting what share of that consumer's diet it supplies; shares are normalised to 100 %	Yes: drag
9	Removal hammer	UI-Probe	Cursor tool that deletes an organism from the built web and triggers the cascade run	Yes: drag
10	Cascade propagation engine	Field	Runs the built web forward, pushing population changes along every link for 60 simulated months after a removal	No
11	Population meter	Overlay	Bar on each token, green rising, red falling, updating live during a cascade run	No
12	Live reef viewport	Environment	3D window showing the built community as real animals; species removed from the canvas visibly vanish from the reef	Yes: drag (camera)
13	Auto-arrange field	Field	Optional layout force that lifts each token to a height matching its position in the web	No
14	Web statistics panel	Instrument	Live counts of nodes, links, links per node, longest chain and number of separate paths from producer to top	No
15	Chain overlay	Overlay	Highlights one single path through the web at a time so a chain can be seen inside the web	No
16	Field guide card	UI-Probe	Flip side of every token, giving the real diet of that species in Monterey Bay with rough percentages	Yes: swap

HOW IT WORKS — THE MODEL

The canvas builds a directed graph. Each accepted arrow states that energy moves from the eaten to the eater, and each consumer's incoming arrows carry diet fractions that must total 100 %. Behind the palette sits a diet database drawn from Monterey Bay studies, so the checker can refuse links no observation supports and can score a finished web against the real one. Pressing Run passes the graph to the population engine: producers grow on light and nutrients, each consumer's intake is the sum of its links weighted by fraction, and populations update monthly. When a species is removed, the loss travels along every link it touched, and consumers with several inbound links re-weight toward their remaining prey while consumers with one link fail. That re-weighting, not any hidden robustness setting, is why a web absorbs a removal that flattens a chain.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Build mode	Dropdown	Blank canvas / Single chain (5 species) / Full web (14 species)	Blank canvas	—	Which tokens and links are pre-placed on the canvas
Organism palette	Multi-select	18 Monterey species, any subset	Kelp, urchin, otter, sheephead, bacteria	—	Which tokens are available on the shelf
Remove species	Dropdown	Any species currently on the canvas	None	—	Deletes that node and starts a cascade run
Removal type	Radio	Total removal / Reduce by half	Total removal	—	Whether the species vanishes or is only cut back
Diet fractions	Toggle	Show / Hide	Show	—	Reveals the weight handles on every incoming arrow
Plausibility checker	Toggle	On / Off	On	—	Whether impossible links are refused or allowed through
Auto-arrange by level	Toggle	On / Off	Off	—	Lifts tokens to a height matching their place in the web
Run length	Slider	12–120	60	months	How long the cascade engine runs after a change
Nutrient upwelling	Slider	20–120	70	% of normal	Producer growth rate, so the base of the web can be squeezed
Playback speed	Dial	1×–12×	4×	—	Simulated months per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Build the classic chain	mode=Single chain (5 species); checker=On	Build kelp to urchin to otter and add two more links. Which way does every arrow point, and why is that the only direction allowed?
S2	Break the chain	mode=Single chain (5 species); remove=Urchin; type=Total removal	Remove one species from a five-species chain. How many of the remaining four are affected, and how badly?
S3	Break the web at the same point	mode=Full web (14 species); remove=Urchin; type=Total removal	Remove the same species from a full web. Compare the damage with S2 and explain the difference using the arrows.
S4	Pull out the otter	mode=Full web (14 species); remove=Sea otter; run=120; upwelling=70	Remove a predator that eats several prey. Which population explodes, and what happens to the producer at the bottom?

STUDENT ACTIVITIES

1. Drag at least ten tokens onto the canvas and connect them into a working web. Record how many links the statistics panel counts and how many the checker refused.

2. Set diet fractions on every consumer with more than one prey, then record the three heaviest arrows in the whole web.
3. Use the removal hammer on the urchin in chain mode and again in web mode. Record every species whose meter changed in each case.
4. Turn on the chain overlay and trace three different paths from giant kelp to a top predator. Record the length of each path.
5. Remove the sea otter, run 120 months, and record urchin and kelp populations at months 0, 12, 60 and 120.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Web statistics	Live numeric	mixed	Nodes, links, mean links per node, longest chain, distinct producer-to-top paths
Population meters	Line graph	relative to start	One trace per species during a cascade run
Species affected count	Counter	count	How many populations moved more than 10 % after a removal
Link validity log	Data table	—	Every attempted link with accepted or refused and the reason given
Web accuracy score	Pass-fail badge	%	Overlap between the student's web and the stored Monterey diet database
Cascade comparison	Bar chart	% change	Chain result beside web result for the same removal
Energy path list	Data table	—	Every route from a producer to the selected top predator, with its length
Exported web	Data table	mixed	CSV of all nodes, links and diet fractions the student built

WHAT THE STUDENT SHOULD REALISE

Students picture feeding relationships as a single line and expect one removal to knock over everything behind it. Building both, then striking the same species out of each, shows a chain collapse and a web absorb it because consumers with several prey re-weight toward what is left. Removing a predator instead makes its prey explode and eats the base away. The student should be able to say: *"A chain is one path through a web, and having other paths is what keeps an ecosystem standing."*

D2.3 · Trophic levels

Experiment name: Half a Level Up: Where Omnivores Sit

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + Agent-based

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-3

THEME & SCENE

A tall cross-section of Elkhorn Slough and the Monterey shelf, drawn as a water column with a numbered height axis running up the left edge from 1 to 5. Eelgrass and phytoplankton sit on the line marked 1; zooplankton and grazing snails float near 2; small fish, crabs and gulls hang at awkward heights between the printed numbers. Each animal is a small live 3D model rotating slowly in place, tethered to the axis by a thin line. Beside the column stands an isotope bench: a compact grey instrument with a sample tray and a needle gauge reading parts per thousand. A diet wheel opens whenever an animal is selected.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Trophic height axis	Instrument	Vertical scale marked 1.0 to 5.0 in tenths; every organism's vertical position is its computed level, not a drawn tier	No
2	Eelgrass and phytoplankton	Actor	Producer models: 3 u ribbon blades and a fine green haze; pinned at exactly 1.0 and unmovable by the student	No
3	Detritus and bacteria node	Actor	Brown flocculent cloud with a bacterial shimmer; treated as level 1 material so detritus feeders are handled honestly	Yes: place
4	Zooplankton swarm	Actor	0.05 u copepods and larvae drifting in a loose cloud; grazes phytoplankton, so lands near 2.0	No
5	Kelp crab	Actor	0.4 u model with legs animating; diet spans algae, detritus and small animals, so its level is emphatically not a whole number	Yes: drag
6	Señorita fish	Actor	0.3 u slim orange fish picking invertebrates off blades; a near-clean level 3 for contrast	Yes: drag
7	California sheephead	Actor	0.9 u fish with a blunt head and strong jaws; crushes urchins and snails, mixing levels 2 and 3	Yes: drag
8	Sea otter	Actor	1.3 u forager holding prey on its chest; diet shifts across urchins, crabs, clams and fish by season	Yes: drag
9	Western gull	Actor	0.7 u bird; the honest awkward case, eating fish, crabs, eggs, carrion and refuse across three levels	Yes: drag
10	Diet wheel	UI-Probe	Ring of draggable wedges around the selected animal, one per prey; wedges normalise to 100 % as they are dragged	Yes: drag, resize
11	Level computation tether	Overlay	Thin line from each animal to the axis; animates smoothly to a new height whenever its diet wheel changes	No
12	Nitrogen isotope bench	Instrument	Grey benchtop unit with a sample tray, a needle gauge in parts per thousand and a printout slot; sampling any animal returns a value	Yes: drag
13	Isotope sample vial	Structure	Small labelled vial minted per sample, holding species, value and computed level; stacks in a rack	Yes: place
14	Integer tier ghost	Overlay	Optional faint horizontal bands at 1, 2, 3, 4 showing the textbook picture, so the mismatch with real animals is visible	No
15	Mismatch flag	Overlay	Amber tag on any animal more than 0.2 levels from the nearest whole number when integer mode is on	No
16	Season control ribbon	Environment	Background tint and prey availability shifting across spring upwelling, summer, autumn and winter, which moves real diets	No

HOW IT WORKS — THE MODEL

Each organism's trophic level is computed, never assigned: level equals one plus the average level of everything it eats, weighted by how much of its diet each prey supplies. Producers and detritus are fixed at 1.0, so a pure herbivore lands at 2.0 and a predator that eats only herbivores at 3.0. An animal that takes 60 % algae and 40 % crabs lands at 2.8 and sits between the printed lines. The engine recomputes the whole web whenever a wedge moves, propagating upward, and the isotope bench supplies an independent check: a baseline value plus roughly 3.4 parts per thousand of nitrogen-15 for every level, with measurement scatter added. The two methods should agree within scatter. The failure to avoid is snapping any animal to a whole number, because the misconception is precisely that every animal belongs on one clean rung.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Selected organism	Dropdown	Any of the 12 animals in the column	Kelp crab	—	Which diet wheel is open and which animal the tether highlights
Diet wedges	Drag-handle	0–100 per prey, normalised	Real Monterey values	% of diet	Rewrites that consumer's diet and moves it up or down the axis
Season	Dropdown	Spring upwelling / Summer / Autumn / Winter	Summer	—	Loads the real seasonal diet for every consumer at once
Integer tier mode	Toggle	On / Off	Off	—	Forces whole-number tiers and flags every animal that does not fit
Isotope bench	Toggle	On / Off	On	—	Enables sampling any animal for its nitrogen-15 value
Baseline isotope value	Slider	4.0–12.0	7.5	‰	Nitrogen-15 of the producers, which sets where the whole scale starts
Enrichment per level	Slider	2.0–4.5	3.4	‰ per level	How much the isotope rises with each feeding step
Samples per animal	Stepper	1–10	3	samples	Repeat measurements, which shrinks the scatter on the readout
Detritus pathway	Toggle	On / Off	On	—	Whether detritus and bacteria count as a level 1 food source
Add omnivore	Dropdown	Raccoon / Black bear / Brown pelican / None	None	—	Places an extra California omnivore into the column with its real diet

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Clean cases first	selected=Zooplankton; season=Summer; integer mode=Off	Eelgrass sits at 1.0. Where do a pure grazer and a fish that eats only grazers land, and why exactly there?
S2	The animal that will not fit	selected=Kelp crab; integer mode=On; bench=On	The crab is flagged. What is its computed level, and what would you have to believe about its diet to call it level 2?
S3	Move the diet, move the animal	selected=Western gull; wedges=drag fish to 0 %	Change one gull wedge at a time and record its level. How much diet change does one whole level take?
S4	Two methods, one answer	season=Spring upwelling; bench=On; samples=5; baseline=7.5	Sample five animals. Do the isotope levels and the diet-wheel levels agree, and what explains the gaps that remain?

STUDENT ACTIVITIES

1. Select five animals in turn and record each computed trophic level to one decimal place, without rounding.
2. Turn integer tier mode on and record every animal the mismatch flag catches, with how far off the nearest whole number it is.
3. Drag the kelp crab's algae wedge from 60 % to 20 % in four steps and record its level at each step.

4. Sample four animals with the isotope bench at three samples each. Record the measured value and the computed level side by side.
5. Load Spring upwelling and Winter in turn and record how far the sea otter's level moves between the two seasons.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Computed trophic level	Live numeric	level	One decimal place per organism, recomputed on every wedge change
Level distribution	Bar chart	count	Histogram of all organisms by level in 0.2 bands; shows how few sit on whole numbers
Diet composition	Data table	%	Every consumer's prey list with its fraction and that prey's own level
Isotope readout	Live numeric	‰	Nitrogen-15 of the sampled animal with its scatter range
Isotope versus diet level	Line graph	‰ vs level	Each sample plotted against its computed level, with the expected slope drawn
Mismatch count	Counter	count	Organisms flagged as not fitting a whole tier under integer mode
Seasonal shift	Line graph	level vs season	Selected animal's level across the four seasons
Sample rack	Data table	mixed	Exportable CSV of every vial: species, value, computed level, season

WHAT THE STUDENT SHOULD REALISE

Students believe every animal belongs to one tidy tier, so an omnivore must be filed under whichever level looks closest. Computing levels from real diets puts the crab, the gull and the otter at fractions, and an independent isotope measurement agrees with the fractions rather than the tiers. The student should be able to say: *"A trophic level is an average of what an animal eats, so animals with mixed diets genuinely sit between the levels."*

D2.4 · Why energy decreases at each level

Experiment name: Follow the Joule: Where the Energy Goes

Render mode: Hybrid 2D+3D

Simulation engine: Particle system + Agent-based

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-3

THEME & SCENE

Point Reyes coastal prairie at first light, camera low among the bunchgrass. Fog lies in the hollows, a red-tailed hawk turns slow circles overhead, and California voles run their worn runways under the thatch. Every scrap of energy in the scene is visible as a small glowing amber chip: chips sit packed in the grass blades, travel along a vole's gut as it feeds, rise off every warm body as a drifting orange heat wisp, and drop out behind the voles as dull brown pellets. To the right, a Sankey canvas draws itself in real time, one broad band entering at the grass and thin ribbons peeling off at every loss. The control panel is docked left.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Energy token	Particle	4 px amber chip, each worth exactly 1 kJ; instanced in the tens of thousands; every token has one location and one fate, and is never created or destroyed	No
2	Bunchgrass sward tile	Structure	0.5 m tiles of California oatgrass and purple needlegrass, 12–40 u blades; each tile holds a token count set by its yearly production	Yes: resize
3	Sunlight budget beam	Field	Broad gold band entering the top of the Sankey canvas; only about 1 % of it is ever captured, and the uncaptured remainder is drawn peeling straight off	No
4	California vole agent	Actor	0.12 u grey-brown rodent, running 0.5 u/s along thatch runways; visible gut segment fills with tokens as it feeds; instanced ×20–400	Yes: drag
5	Red-tailed hawk agent	Actor	1.2 u wingspan raptor, rufous tail, circling and stooping; captures a vole on a success roll and carries tokens to a perch	Yes: drag
6	Golden eagle agent	Actor	2.0 u wingspan; only present when a fourth level is added; takes hawks and large prey	Yes: place
7	Side-blotched lizard agent	Actor	0.08 u ectotherm swappable for the vole; loses far less energy to heat because it does not hold its own temperature	Yes: swap
8	Respiration heat wisp	Particle	Orange wisp rising 2 u then fading, one per respired token; density is the single most visible loss stream in the scene	No
9	Faecal pellet	Structure	0.02 u dull brown pellet dropped where an animal fed; carries the tokens that were eaten but never absorbed	No
10	Uneaten remains pile	Structure	Small heap of fur, bone and seed husk left after a kill or a meal; holds the tokens a predator did not swallow	No
11	Decomposer sink	Actor	Fungal and bacterial mat under the thatch that receives pellets, remains and dead bodies and respire their tokens away as heat	Yes: place
12	Token tracer	Instrument	Click any single token to paint it white and follow it through every transfer until it leaves as heat; leaves a visible trail	Yes: drag
13	Sankey canvas	Overlay	Self-drawing flow diagram, band width proportional to kJ, with named ribbons: respired, egested, uneaten, transferred	No
14	Thermal overlay	Overlay	False-colour camera view where every animal glows by its live respiration rate; endotherms burn bright, the lizard does not	No
15	Level calorimeter	Instrument	Boxed probe placed on any level reporting kJ per square metre per year entering, stored and leaving it	Yes: place
16	Efficiency gauge	Instrument	Dial reading the transfer percentage between the two selected levels, updated at the end of each simulated year	No

HOW IT WORKS — THE MODEL

Energy is conserved token by token, and the engine may never mint or delete one. Grass captures a small fraction of incident sunlight into tokens. A vole that eats a blade takes those tokens into its gut, then loses a fixed share straight through as faeces, and burns most of the rest on staying warm, moving and breathing, each burnt token leaving as a heat wisp. Only the tokens still in tissue at the end of the year are available to the next level, and a hawk captures only the fraction of voles it actually catches and swallows. The roughly ten per cent that reaches each next level is therefore an outcome of three visible subtractions, not a number written anywhere in the code. Warm-blooded animals lose more than cold-blooded ones, which the lizard swap makes plain. Every lost token ends as heat, so nothing recycles.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Sunlight input	Slider	1,000–8,000	5,000	MJ/m ² per year	Total light arriving; grass capture is a small fixed share of it
Grass production	Slider	200–1,200	600	g/m ² per year	Tokens loaded into the sward at the start of each year
Herbivore type	Dropdown	California vole / Side-blotched lizard / Black-tailed hare	California vole	—	Swaps the level 2 agent, changing heat loss and gut efficiency
Herbivore count	Stepper	20–400	120	animals/ha	How many level 2 agents exist
Assimilation efficiency	Slider	20–90	65	% of food absorbed	Share of eaten tokens absorbed rather than egested as pellets
Activity level	Slider	0.5–3.0	1.4	× basal rate	How hard animals work, and therefore how fast they respire tokens
Air temperature	Slider	0–35	14	°C	Heat loss for endotherms; below 10 °C respiration climbs sharply
Number of levels	Stepper	2–4	3	levels	Structural: adds or removes the hawk and the golden eagle
Decomposers	Toggle	Present / Absent	Present	—	Whether pellets, remains and bodies are respired away or pile up
Token tracing	Toggle	On / Off	Off	—	Enables clicking a single token and following its whole journey

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Grass, vole, hawk	levels=3; herbivore=California vole; grass=600; activity=1.4; temperature=14	Read the calorimeter at all three levels. What percentage of the grass energy reaches the hawk, and what are the three places the rest went?
S2	Follow one chip	tracing=On; levels=3; same as S1	Trace ten individual tokens. How many reach the hawk, and what happened to the ones that did not?
S3	Swap in a cold-blooded grazer	herbivore=Side-blotched lizard; grass=600; temperature=22	A lizard does not heat itself. Does the transfer percentage rise, and which Sankey ribbon shrank?
S4	Add a fourth level	levels=4; grass=600; herbivore=California vole	Add a golden eagle above the hawk. How much energy is left at level 4, and what does that say about long food chains?

STUDENT ACTIVITIES

1. Place the calorimeter on each of the three levels in turn and record kJ per square metre per year for each.
2. Calculate the transfer percentage from level 1 to 2 and from level 2 to 3, then compare both with the efficiency gauge.
3. Turn tracing on and follow ten tokens. Record the fate of each as respired, egested, uneaten or transferred, and tally the four columns.

4. Swap the vole for the side-blotched lizard, re-run one year, and record the new transfer percentage and the change in the respired ribbon.
5. Set levels to 4 and record the energy at each level. State how many voles-worth of grass one eagle-worth of energy required.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Energy per level	Bar chart	kJ/m^2 per year	Standing production at each level, measured by the calorimeters
Transfer efficiency	Live numeric	%	Between each adjacent pair of levels, computed at year end
Sankey flow diagram	Heat map	kJ/m^2 per year	Self-drawing bands showing captured, respired, egested, uneaten and transferred energy
Loss breakdown	Bar chart	kJ/m^2 per year	Respiration, egestion and uneaten remains side by side for each level
Traced token fates	Data table	count	Every traced token with its route and where it finally left the system
Heat released	Live numeric	kJ/m^2 per year	Total respiration across all levels including decomposers
Energy closure	Pass-fail badge	—	Green while tokens in equals tokens accounted for; flags any leak in the engine
Run archive	Data table	mixed	Exportable CSV of settings, level energies and both efficiencies per run

WHAT THE STUDENT SHOULD REALISE

Students think energy is lost at each level because animals fail to eat everything, or because it is somehow used up and stored elsewhere. Tracing individual joules shows three concrete exits, respired heat above all, and shows that heat leaves the ecosystem for good. Ten per cent is a result, not a rule. The student should be able to say: *"Most of the energy at every level is burned to stay alive and leaves as heat, so only a small part is ever left to pass on."*

D2.5 · Reading an energy pyramid

Experiment name: Build the Pyramid, Then Break It

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS2-3

THEME & SCENE

A clean charcoal studio holding one object: a stack of glowing slabs, each as wide as the quantity it represents, floating one above another and slowly rotating so the student can see the taper edge on. The bottom slab is broad and warm gold, the ones above narrow sharply, and the top slab is a sliver. Every slab carries its value on its face and its organism cards along its front edge. To the left, four dataset cards wait: the student's own kelp forest web, the Point Reyes prairie, Monterey open water, and a single valley oak in the Central Valley. To the right, three mode buttons labelled Energy, Biomass, Numbers, and a caliper tool for measuring any slab.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Pyramid tier slab	Structure	Extruded slab 0.6 u thick; width strictly proportional to the tier value; warm gold at the base cooling to pale blue at the top; animates width on any change	Yes: resize
2	Tier organism card	UI-Probe	Small card clipped to a slab's front edge naming each species on that tier with its contribution	Yes: drag
3	Imported web link	Instrument	Reads the food web the student built in the previous experiment and lays its species onto tiers automatically	Yes: swap
4	Dataset card — kelp forest	Instrument	Monterey kelp forest figures: giant kelp production, urchin and crab biomass, fish and otter standing stock	Yes: drag
5	Dataset card — coastal prairie	Instrument	Point Reyes grass, vole and hawk figures carried over from the energy-flow experiment in matching units	Yes: drag
6	Dataset card — Monterey open water	Instrument	Phytoplankton and zooplankton standing crop and annual production; the classic inverted biomass case	Yes: drag
7	Dataset card — valley oak	Instrument	One 200-year valley oak with its insect, bird and raptor community; the classic inverted numbers case	Yes: drag
8	Mode switch	UI-Probe	Three-way control rebuilding the same community as an energy, biomass or numbers pyramid with correct units	Yes: swap
9	Slab caliper	Instrument	Draggable tool that measures any slab and reports its value, its share of the tier below, and the ratio between them	Yes: drag
10	Turnover dial	Instrument	Physical dial setting how many times a year the producer stock is replaced; visibly separates standing crop from production	Yes: drag
11	Fractional tier splitter	Field	Distributes an omnivore across tiers according to its computed level from the previous experiment, so a 2.8 animal contributes to both	No
12	Inversion detector	Overlay	Red outline and a printed warning on any slab wider than the one beneath it, with the tier ratio shown	No
13	Axis scale toggle	UI-Probe	Switches slab width between linear and logarithmic so a top tier a thousand times smaller stays visible	Yes: swap
14	Unit ribbon	Overlay	Band under the pyramid showing the current units: kJ/m ² per year, g/m ² , or individuals per hectare	No
15	Claim card and scorer	UI-Probe	Blank claim with evidence slots; the scorer checks whether the chosen slab values actually support the claim written	Yes: place

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Comparison rail	Overlay	Holds up to three saved pyramids side by side at matched scale for direct comparison	Yes: drag

HOW IT WORKS — THE MODEL

One community, three questions. The engine holds each dataset as a table of species with four fields: annual energy production, standing biomass, individual count, and computed trophic level. Choosing a mode rebuilds slab widths from the matching field, using the same species and the same tier assignment, so any difference between the three pyramids comes from the quantity measured and nothing else. Omnivores are split across tiers by their fractional level rather than rounded. The turnover dial matters: phytoplankton are eaten and replaced many times a year, so a small standing crop can feed a larger consumer stock above it, which inverts a biomass pyramid while the energy pyramid stays upright. Numbers pyramids invert whenever one large producer feeds many small consumers. The engine must never quietly reorder tiers to keep a pyramid tidy.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Community	Dropdown	My kelp forest web / Coastal prairie / Monterey open water / Valley oak	My kelp forest web	—	Loads a different species table, tiers and unit set
Pyramid type	Radio	Energy / Biomass / Numbers	Energy	—	Which field sets slab width; unit ribbon relabels
Quantity basis	Toggle	Annual production / Standing crop	Annual production	—	Whether a slab shows energy passing through per year or stock present now
Producer turnover	Slider	1–120	4	replacements/year	How fast producers are eaten and regrown; drives the gap between the two bases
Omnivore handling	Dropdown	Split by fraction / Round to nearest tier	Split by fraction	—	How animals with fractional levels are placed in the stack
Include decomposers	Toggle	On / Off	Off	—	Adds a decomposer slab alongside the stack with its own flow
Axis scale	Toggle	Linear / Logarithmic	Linear	—	Slab width mapping, so a tiny top tier remains readable
Sample area	Slider	1–10,000	100	m ²	Area the counts and masses are expressed over; changes numbers, not shape
Save to comparison rail	Stepper	0–3	0	slots	Structural: stores the current pyramid beside others for comparison

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Your own web, as energy	community=My kelp forest web; type=Energy; basis=Annual production	Turn the web you built into an energy pyramid. What is the ratio between each tier and the one below it?
S2	One oak, thousands of insects	community=Valley oak; type=Numbers; area=100	Count individuals instead of energy. Why is the bottom tier now a single organism, and is anything wrong with the ecosystem?

#	Scenario	Preset	The question the student answers
S3	Upside-down ocean	community=Monterey open water; type=Biomass; basis=Standing crop; turnover=60	Weigh the plankton instead. The producers weigh less than their consumers. How can that possibly work?
S4	Same water, measured as energy	community=Monterey open water; type=Energy; basis=Annual production; turnover=60	Switch the same community to energy. Which way up is the pyramid now, and what does that tell you about which quantity can never invert?

STUDENT ACTIVITIES

1. Load your own kelp forest web as an energy pyramid and record the value on each slab, then use the caliper to record the ratio between each pair of tiers.
2. Switch the same community to Numbers and record how the shape changes without changing a single organism.
3. Load the valley oak in Numbers mode and record the count on each tier. Write one sentence explaining the inverted shape.
4. Load Monterey open water, set turnover to 60, and record producer and consumer values in both Biomass and Energy modes.
5. Build a claim card stating which pyramid type can never be inverted, fill it with three slab values as evidence, and record the scorer's verdict.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Tier values	Bar chart	kJ/m ² per year, g/m ² or individuals	The four slab values in the current mode and units
Tier ratio	Live numeric	%	Each tier as a percentage of the one below, reported by the caliper
Inversion flag	Pass-fail badge	—	Red on any tier wider than the one beneath it, with the ratio printed
Standing crop versus production	Line graph	g/m ² and kJ/m ² per year	The two bases plotted together as turnover is changed
Pyramid comparison	Bar chart	mixed	Up to three saved pyramids at matched scale on the comparison rail
Species contribution	Data table	mixed	Every species with its tier, fractional level and contribution to its slab
Claim score	Pass-fail badge	—	Whether the selected evidence supports the written claim, with a reason
Export	Data table	mixed	CSV of all tiers, all three quantities and the active settings

WHAT THE STUDENT SHOULD REALISE

Students think a pyramid is the shape ecosystems have, so an upside-down one means something is broken. Rebuilding one community three ways shows that numbers and biomass pyramids invert quite legitimately, when one oak feeds thousands of insects or when fast-growing plankton are eaten as quickly as they grow. Only energy per year cannot invert. The student should be able to say: *"The shape depends on what you measured, and only the energy pyramid must always be widest at the bottom."*

D3 · Cycling matter through an ecosystem · MS-LS2-3

D3.1 · Matter cycling between organisms and the environment

Experiment name: The Sealed Jar: Nothing Leaves, Nothing Arrives

Render mode: Hybrid 2D+3D

Simulation engine: Agent-based + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-3

THEME & SCENE

A one-litre glass jar stands on a pale birch bench, lit from the left by a small desk lamp, sealed with a clamped steel lid and an orange silicone gasket. Inside: two centimetres of washed grit, three strands of pondweed leaning toward the light, a green algal bloom fogging the glass on the lamp side, one ramshorn snail grazing, and a slow snow of detritus settling. Bubbles cling to the pondweed leaves and release one at a time. Behind the jar, floating in dark slate, is the Atom Ledger: five stacked bars showing where every tagged atom currently sits. The control panel is docked right; a loupe on the cursor zooms any patch to microbe scale with a 20 µm scale bar.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Glass jar body	Environment	Cylinder 120 mm tall, 95 mm across, 3 mm wall, refractive glass shader with a faint green cast; interior volume is the whole simulated world	Yes: resize
2	Lid and gasket seal	Structure	Steel disc with a knurled clamp and an orange gasket ring; when unsealed the lid tilts 20° and a dotted boundary line opens across the jar mouth	Yes: swap
3	Grit substrate	Environment	20 mm bed of 1–3 mm quartz grains, warm grey with mica flecks; holds the buried-detritus reservoir	No
4	Water column	Environment	Translucent teal volume with slow convection swirl, 0.3 mm/s drift; carries dissolved gases and ions	No
5	Pondweed strand	Actor	Stem capsule 70 mm long bearing 24 lance-shaped leaves 6 mm long, jade green with a darker midrib; leaves tilt toward the lamp over 30 s; grows one leaf node per simulated day when photosynthesising	Yes: place, resize
6	Algal film	Actor	Instanced 0.4 mm discs, 400–3000 of them, sprayed on the lamp-side glass; density tracks light and nutrients	Yes: swap
7	Ramshorn snail	Actor	Flat coiled shell 9 mm across, amber banded, foot in dull cream; crawls the glass at 2 mm/s, rasps algae, drops a pellet every 40 s	Yes: place
8	Ostracod grazers	Actor	0.9 mm bean-shaped shells, translucent straw; 0–60 of them, jittering random walk; eat suspended detritus	Yes: place
9	Decomposer bacteria colonies	Actor	Rod cells 2 µm long, rendered as slate-blue haze patches over dead matter until the loupe resolves them into rods; population doubles every 90 simulated minutes on available detritus	Yes: swap
10	Detritus particles	Particle	Irregular 0.2–1.5 mm brown flecks, 200–1200 instanced; sink at 0.5 mm/s, settle into the grit, shrink as bacteria consume them	No
11	Dissolved gas field	Field	Two invisible scalar fields (O ₂ , CO ₂) on a 12×12×12 lattice inside the water; drives bubble formation and snail stress; shown as a blue-to-amber wash when overlay is on	No
12	Tagged atom set	Particle	60 individually numbered atoms, drawn as small glowing beads with a comet trail 3 s long: carbon gold, nitrogen violet, oxygen cyan; each carries a reservoir label that updates when it changes hands	Yes: place
13	Atom Ledger panel	Overlay	Five stacked bars (Air space · Water · Producers · Consumers · Decomposers + soil) plus a TOTAL strip that must never change height	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Desk lamp	Instrument	40 mm shaded bulb on a hinged arm, warm-white cone; intensity and daily on-hours are slider-driven	Yes: drag
15	Gas and temperature probe	Instrument	Slim 60 mm rod dipped through a port, tip glowing; reports dissolved O ₂ , CO ₂ and temperature at its tip depth	Yes: drag
16	Boundary flux meter	Instrument	Invisible plane across the jar mouth counting atoms crossing in and out per hour; reads exactly zero while the lid is sealed	No

HOW IT WORKS — THE MODEL

The jar holds a fixed inventory of tagged atoms. Every process is written as a transfer, never a creation: photosynthesis moves carbon and oxygen atoms from the dissolved-gas reservoir into the pondweed and algae reservoirs; grazing moves them from producers to the snail; respiration moves them back to the gas field; death and excretion move them to detritus; bacterial decay moves them from detritus back to gas and to dissolved ions. Each transfer is a rate rule of the form $\text{flux} = k \cdot (\text{source pool}) \cdot (\text{limiting factor})$, where the limiting factor is light for photosynthesis, food density for grazing, and temperature for decay. The ledger sums all five reservoirs every tick and asserts the total is unchanged. The engine must never spawn or delete a tagged atom while the lid is sealed; a "matter used up" bug is the exact misconception the experiment exists to kill. Biomass is reported in atom counts, not grams, so growth always visibly costs some other reservoir.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Lid seal	Toggle	Sealed / Open	Sealed	—	Opens or closes the boundary; when open, atoms may leave and enter
Snails	Stepper	0–6	1	count	How many grazers exist; sets grazing and respiration load
Pondweed strands	Stepper	0–5	3	count	Producer biomass and therefore photosynthetic capacity
Decomposers	Toggle	Present / Removed	Present	—	Whether detritus is broken down or simply accumulates
Daily light hours	Slider	0–24	12	h/day	Length of the lit phase each simulated day
Lamp intensity	Slider	0–1200	400	μmol/m ² /s	Photosynthesis rate ceiling
Water temperature	Slider	8–32	21	°C	Respiration and decay rates; snail stress threshold
Tracer element	Dropdown	Carbon / Nitrogen / Oxygen	Carbon	—	Which element the 60 tagged atoms represent
Time compression	Dial	1×–2000×	500×	—	Simulated hours per real second
Gas overlay	Toggle	On / Off	Off	—	Shows the dissolved O ₂ / CO ₂ colour wash

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Balanced jar	lid=Sealed; snails=1; pondweed=3; decomposers=Present; light hours=12	Run 30 simulated days. Which reservoirs gain atoms and which lose them, and what happens to the TOTAL bar?
S2	Overstocked	snails=5; pondweed=1; light hours=12	The snails die by day 9. Did their atoms disappear from the jar, and where can you find them afterwards?
S3	Decomposers removed	decomposers=Removed; snails=2; run 60 days	Detritus piles up on the grit. Which reservoir bar stops refilling, and why does the pondweed stall?
S4	Lid off	lid=Open; snails=1; pondweed=3	With the boundary open, the total is no longer fixed. What does the boundary flux meter show, and what changed about your accounting?

STUDENT ACTIVITIES

1. Set the tracer to Carbon, pin tag #7 onto a pondweed leaf, and record every reservoir it visits over 20 simulated days, in order, with the day number.
2. Run S1 for 30 days and record the five reservoir bar heights at day 0, 10, 20 and 30. Add the five numbers each time and write down the totals.
3. Predict, before running S2, whether the total atom count will fall when the snails die. Run it, record the answer, and write one sentence explaining what happened instead.
4. Set decomposers to Removed and run 60 days. Measure the detritus reservoir and the producer reservoir every 10 days and describe the shape of each curve.
5. Open the lid and run 10 days. Record the boundary flux meter, then re-seal and explain in one sentence why the sealed jar was easier to account for.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Atom Ledger	Bar chart	atoms	Five reservoir bars, updated every simulated hour
Total atom count	Pass-fail badge	atoms	Green while the total is unchanged; amber the instant the lid is opened
Tagged atom itinerary	Data table	reservoir vs day	One row per reservoir change for the pinned atom, exportable
Dissolved O ₂ and CO ₂	Line graph	mg/L	Two traces vs simulated days, from the probe tip
Living biomass by group	Line graph	atoms	Producers, consumers, decomposers on shared axes
Detritus on the grit	Live numeric	atoms	Updated hourly; the pile that grows when decomposers are off
Boundary flux	Counter	atoms/h	In, out and net across the jar mouth; reads 0 when sealed
Snail survival	Timer	simulated days	How long each grazer lasted before starvation or low oxygen

WHAT THE STUDENT SHOULD REALISE

Students believe matter is used up: that a dead snail's substance is gone and that plants make new material out of nothing. A sealed jar with a fixed, numbered atom inventory makes that untenable, because the total never moves while every individual atom changes address many times. Growth in one reservoir is always a withdrawal from another. The student should be able to say: *"The same atoms keep getting reused by different living things; nothing is made and nothing is destroyed, it just changes hands."*

D3.2 · Decomposers closing the loop

Experiment name: Rot Chamber: Who Unlocks the Nutrients

Render mode: 3D Scene

Simulation engine: Agent-based + Fluid/thermal

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-3

THEME & SCENE

A cutaway litter box the size of a shoebox sits under soft north light, its front wall a pane of glass so the student sees the profile: a top layer of crisp brown oak leaves, then a matted black horizon of half-rotted fragments, then crumbly dark soil. A fallen coast live oak branch lies across the surface, its bark split. White fungal hyphae creep out from the split in thin threads that visibly lengthen. Wood lice trundle in the litter, an earthworm slides through a burrow behind the glass, and a faint warm shimmer rises from the middle layer. On the right a nutrient-release chart draws itself day by day. A loupe zooms to a 10 µm scale bar showing bacterial rods swarming a leaf edge.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Chamber and glass front	Environment	300×180×160 mm box, matte grey walls, one optically clear front pane with a 10 mm grid etched faintly for depth reference	No
2	Fresh leaf litter layer	Structure	60–200 instanced oak leaf cards 35 mm long, curled, russet to tan, individually tracked with a remaining-mass value; visibly perforate and shrink as they are consumed	Yes: place
3	Fragmented litter horizon	Structure	25 mm band of 2–8 mm dark brown flecks, produced when a detritivore chews a leaf; grows as leaves shrink	No
4	Humus and soil layer	Environment	70 mm of crumb-textured near-black soil with visible pore spaces; holds the mineral nutrient pool	No
5	Oak branch	Structure	180 mm cylinder tapering 22 mm to 14 mm, greyish bark shell over pale sapwood, one lengthwise split; decays an order of magnitude slower than leaves	Yes: place, swap
6	Fungal hyphae network	Actor	Branching filaments 4 µm wide, chalk white with a faint pearl sheen, growing 3 mm per simulated day from inoculation points, branching every 8 mm; only fungi can attack the lignin in bark and wood	Yes: place
7	Fungal fruiting body	Actor	Small tan bracket or 12 mm toadstool that erupts once a hyphal network passes a size threshold; releases a spore puff	No
8	Decomposer bacteria colonies	Actor	Rod cells 2 µm long in slate-blue clusters on wet leaf surfaces; population doubles every 40 simulated minutes in warm, moist conditions and collapses when dry	Yes: swap
9	Wood louse (detritivore)	Actor	11 mm segmented armoured body, slate grey, 14 legs, ambling at 4 mm/s; shreds whole leaves into fragments, multiplying the surface area available to microbes	Yes: place
10	Earthworm	Actor	90 mm segmented tube, dull pink with a clitellum band, peristaltic crawl through a visible burrow; drags surface leaves down and mixes horizons	Yes: place
11	Moisture field	Field	Scalar field over the profile, 0–100 % of water-holding capacity; rendered as a darkening of soil colour and a glisten on leaf surfaces	No
12	Heat field and thermal plume	Field	Temperature field driven by microbial respiration; a translucent orange shimmer rises from any cell above ambient by 2 °C	No
13	Mineral nutrient pool	Field	Three tracked ion stocks in the soil layer (nitrate, phosphate, potassium) with a colour-ramp overlay; only rises when decomposers work	No
14	Seedling bioassay tray	Instrument	Six 40 mm radish seedlings planted in soil drawn from the chamber; their height is the biological read of nutrient release	Yes: place

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Mass balance and CO ₂ probe	Instrument	Load cell under the chamber plus a gas port in the lid; reports remaining litter mass and CO ₂ evolution rate	No
16	Loupe	UI-Probe	6× circular lens following the cursor, with a 10 µm scale bar and a microbe-count badge	Yes: drag

HOW IT WORKS — THE MODEL

Every piece of dead matter carries a mass, a toughness value and a nutrient content. Detritivores reduce particle size without releasing many nutrients; microbes release nutrients but their rate depends on exposed surface area, so shredding by wood lice and worms multiplies microbial output. Bacteria attack soft, wet, sugary tissue fast; fungi grow slowly but are the only agents with the enzymes to break lignin, so bark and wood decay only where hyphae have reached. Decay rate per patch is $\text{rate} = k \cdot \text{surface} \cdot f(\text{moisture}) \cdot Q_{10}^{((T-20)/10)}$, with a moisture optimum near 60 % and near-zero rate below 15 % or above 95 %. Released nutrients enter the soil mineral pool with a lag, producing the characteristic curve: slow at first, steep in the middle, then flattening as the easy material runs out. Respiration adds CO₂ to the headspace and heat to the profile, which is how the student sees that decomposers are living things eating, not chemicals dissolving.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Decomposer crew	Multi-select	Bacteria / Fungi / Wood lice / Earthworms	Bacteria + Fungi	—	Which agent types exist in the chamber
Litter load	Slider	20–300	120	g	How much dead leaf material starts on the surface
Litter type	Dropdown	Oak leaves / Pine needles / Grass clippings / Oak branch	Oak leaves	—	Toughness and nutrient content of the dead matter
Soil moisture	Slider	5–100	60	% of capacity	Microbial activity; below 15 % decay nearly stops
Temperature	Slider	2–40	20	°C	Decay rate through the Q ₁₀ rule
Fungal inoculation points	Stepper	0–8	2	count	Where hyphal networks begin; only these can reach the branch
Detritivore count	Stepper	0–20	6	count	Shredding rate and therefore exposed surface area
Run length	Slider	10–180	90	simulated days	Length of the run drawn on the nutrient curve
Seedling bioassay	Toggle	On / Off	On	—	Plants the radish tray and reads soil nutrients biologically

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Full crew	crew=all four; litter=120 g; moisture=60 %; temp=20 °C; run=90 d	On which simulated day does the nutrient-release curve rise most steeply, and what is the litter mass by day 90?
S2	Sterile floor	crew=none; litter=120 g; run=180 d	With no decomposers at all, what happens to litter mass, soil nutrients and seedling height over 180 days?
S3	Bacteria only, wood added	crew=Bacteria; litter type=Oak branch; run=180 d	Bacteria alone barely touch the branch. Which crew member is missing, and what can it break that bacteria cannot?
S4	Mojave dry heat	moisture=10 %; temp=35 °C; crew=all four; run=180 d	The chamber is hot but bone dry. Why does the leaf litter stay almost intact for half a year?

STUDENT ACTIVITIES

1. Set the crew to Bacteria + Fungi only and run 90 days. Record litter mass and soil nitrate at days 0, 15, 30, 60 and 90, and sketch the release curve.
2. Add wood lice and earthworms and re-run the identical settings. Compare the day-30 nitrate value with your first run and state, in one sentence, why shredding sped up the microbes.
3. Swap the litter type to Oak branch, remove fungi, and run 180 days. Measure how much branch mass is lost, then re-run with fungi restored and record the difference.
4. Drag the moisture slider to 10 %, run 60 days, and record the CO₂ evolution rate. Then return to 60 % and record it again at the same run length.
5. Read the six radish seedlings' heights at day 90 in your best and worst runs, and write the number that shows nutrients actually returned to the soil.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Remaining litter mass	Line graph	g	Falling curve vs simulated days, from the load cell
Nutrient release curve	Line graph	mg/kg soil	Nitrate, phosphate and potassium on shared axes; the lag, rise and plateau are the point
CO ₂ evolution rate	Live numeric	mg/kg/day	Updated daily; the breathing of the decomposers
Microbial population	Line graph	cells ×10 ⁶	Bacteria and fungal biomass traced separately
Chamber temperature rise	Live numeric	°C above ambient	Heat produced by respiration in the middle horizon
Seedling height	Bar chart	mm	Six radish bioassay plants at run end
Decay half-life	Live numeric	simulated days	Days for litter mass to fall by half in the current settings
Run comparison	Data table	mixed	One row per run with crew, conditions and day-90 values, exportable

WHAT THE STUDENT SHOULD REALISE

Students believe dead things simply "rot away" on their own, and that decomposers are unpleasant extras rather than a step the ecosystem depends on. Removing the crew leaves the litter almost intact after 180 days and the nutrient pool flat, while the seedlings stay stunted, so decay is visibly the work of living organisms feeding. The student should be able to say: *"Decomposers are the ones that hand the atoms back to the soil and the air; without them the material stays locked in dead bodies and the plants get nothing."*

D3.3 · The carbon cycle, introduced

Experiment name: Carbon on Tour: Reservoirs and Fluxes

Render mode: 2.5D Layered

Simulation engine: Data-driven model + Particle system

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS2-3

THEME & SCENE

A wide, stylised California cross-section runs from Monterey surf on the left to Sierra granite on the right, drawn in flat layered planes: pale sky, a redwood stand, ploughed Central Valley fields, a city with a freeway, the coastal ocean, and a dark rock basement. Over each part of the landscape floats a translucent glass tank labelled with its carbon stock, filled to a level in gigatonnes: Atmosphere, Land plants, Soil and litter, Ocean, Fossil rock. Between the tanks, curved arrow-pipes carry visible carbon beads; the pipes thicken and thin as fluxes change. One bead glows gold, trails a ribbon, and is the atom the student follows. The control panel is docked right; a scrubber timeline runs along the bottom.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Landscape backplate	Environment	Five parallax layers, coast to Sierra, muted teal-ochre palette, slow cloud drift; purely scenic context for the tanks	No
2	Atmosphere tank	Structure	Wide flat glass tank across the sky band, filled with pale blue gas at a level in GtC; a ppm label sits on the rim	Yes: resize
3	Land plant tank	Structure	Tall green-tinted tank behind the redwood stand; level rises with photosynthesis and drops with harvest, fire and respiration	Yes: resize
4	Soil and litter tank	Structure	Broad dark-brown tank set into the ground plane, the largest fast-cycling land store	Yes: resize
5	Ocean tank	Structure	Deep blue tank under the sea layer, split into a thin surface compartment and a thick deep compartment with a slow exchange valve between them	Yes: resize
6	Fossil rock tank	Structure	Sealed charcoal tank in the basement layer with a valve icon; opening the valve is what burning coal and oil means	Yes: swap
7	Flux pipes	Structure	12 curved translucent tubes between tanks, width mapped linearly to flux in GtC/yr, each labelled with its process name	Yes: connect
8	Carbon beads	Particle	3 px gold-brown spheres flowing along pipes, one bead per 0.1 GtC/yr, spacing shows rate at a glance	No
9	Tagged carbon atom	Particle	One bead scaled 3×, glowing, with a 4 s ribbon trail and a floating label naming its current reservoir	Yes: place
10	Photosynthesis engine	Field	Invisible rate node on the plant tank; draws from the atmosphere as a function of plant stock, light and temperature	No
11	Respiration and decay nodes	Field	Two invisible return nodes from plants and from soil back to the atmosphere; decay is temperature-sensitive	No
12	Combustion node	Actor	Freeway and power-station sprite with an animated exhaust plume; its rate is the fossil-fuel slider	Yes: swap
13	Wildfire event	Actor	Animated flame front across the redwood stand that transfers a slug of carbon from plants to atmosphere in one step	Yes: place
14	Ocean exchange valve	Instrument	Two-way gate between atmosphere and surface ocean; rate depends on the difference in partial pressure and on water temperature	No
15	Flux meter clamp	Instrument	Draggable clamp that snaps onto any pipe and reads its instantaneous flux and cumulative total	Yes: drag
16	Timeline scrubber	UI-Probe	Bottom-docked track, year 1750 to 2100, with a playhead and pinnable event markers	Yes: drag

HOW IT WORKS — THE MODEL

Five reservoirs hold stocks in gigatonnes of carbon; twelve fluxes move carbon between them each simulated year. Each stock updates by $S(t+1) = S(t) + \sum \text{inflows} - \sum \text{outflows}$, so the sum across all five tanks is constant unless the student explicitly opens the fossil valve, which moves carbon out of a store that had been sealed for millions of years. Photosynthesis scales with plant stock and light; plant and soil respiration scale with their own stocks and with temperature; ocean uptake scales with the atmosphere-to-surface difference and falls as the water warms. Default stocks and fluxes are rounded from published global estimates so the numbers are realistic without being fussy. The critical honesty is that fossil carbon is not a new kind of matter: it is the same element, previously parked. The sim must never draw carbon appearing from nowhere, and the ledger strip must always show the grand total flat while the valve is shut.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Fossil fuel burning	Slider	0–15	9	GtC/yr	Opens the fossil valve; adds carbon to the atmosphere
Forest cover	Slider	0–150	100	% of 1750 area	Plant stock and photosynthesis capacity
Global temperature offset	Slider	–2 to +5	0	°C	Decay rate, respiration rate and ocean solubility
Ocean mixing	Slider	0.2–4.0	1.0	× baseline	Surface-to-deep exchange speed
Wildfire	Stepper	0–5	0	events/decade	Pulses of plant carbon into the atmosphere
Reservoir editor	Drag-handle	Resize any tank 0.5×–2×	1×	× baseline	Structural: changes a starting stock and redraws the pipes
Tagged atom start	Dropdown	Atmosphere / Plants / Soil / Ocean / Fossil rock	Atmosphere	—	Where the glowing atom begins its tour
Run to year	Timeline scrubber	1750–2100	2025	year	How far forward the model integrates
Flux labels	Toggle	On / Off	On	—	Shows numeric GtC/yr on every pipe

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Pre-industrial balance	fossil=0; forest=100 %; temp=0; run to 1850	With the fossil valve shut, do the tank levels drift? What does that say about inflow and outflow for the atmosphere?
S2	Valve open	fossil=9 GtC/yr; forest=100 %; run to 2100	Which tank empties and which fills, and where does roughly a quarter of the added carbon end up instead of the air?
S3	Clear the forest	fossil=9; forest=40 %; run to 2100	Cutting forests changes two fluxes at once. Name both, and say what happens to the atmosphere tank compared with S2.
S4	Warm soil	fossil=6; temp=+3 °C; run to 2100	Warming speeds decay. Does the soil tank act as a store or a source in this run, and how can you tell from the pipe widths?

STUDENT ACTIVITIES

1. Set the tagged atom to start in the Atmosphere, run to 1850, and record every reservoir it enters with the year of each move.
2. Clamp the flux meter onto the photosynthesis pipe and onto the plant-respiration pipe in S1. Record both numbers and explain why the plant tank level barely moves.
3. Open the fossil valve to 9 GtC/yr, run to 2100, and record all five tank levels at 2025, 2050 and 2100. Add them and compare with the 1750 total.

4. Drag the Land plant tank handle to 0.5× and re-run S2. Record the 2100 atmosphere level and compare it with the unedited run.
5. Predict, before running S4, whether warming makes the soil tank rise or fall. Run it, record the soil flux at 2100, and correct your prediction in one sentence.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Reservoir levels	Bar chart	GtC	Five bars updated each simulated year
Atmospheric CO ₂	Line graph	ppm	Converted from the atmosphere stock, 1750 to the run end
Flux table	Data table	GtC/yr	All twelve fluxes with names, exportable
Clamped pipe reading	Live numeric	GtC/yr and cumulative GtC	Instantaneous and running total on the selected pipe
Tagged atom itinerary	Data table	reservoir vs year	The gold atom's route, one row per move
Grand total carbon	Pass-fail badge	GtC	Green while the five-tank total is constant; flags the fossil valve as the only true source
Land and ocean uptake share	Bar chart	% of emissions	How much of the added carbon each sink absorbs

WHAT THE STUDENT SHOULD REALISE

Students believe carbon dioxide is "made" by cars and "destroyed" by trees. The tank-and-pipe ledger makes that untenable, because the grand total is flat until fossil rock is opened, and trees are shown parking carbon temporarily rather than removing it. Emissions are a transfer from a sealed store into a fast-moving loop. The student should be able to say: *"Burning fuel does not create carbon; it moves carbon that was locked in rock into the air, where the cycle then has to deal with it."*

D3.4 · The nitrogen cycle, introduced

Experiment name: Under the Alfalfa: Four Microbe Crews

Render mode: 3D Scene

Simulation engine: Agent-based + Molecular

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-3

THEME & SCENE

A soil cross-section stands like an aquarium slice, 400 mm wide and 500 mm deep, seen head-on. Above the surface, three alfalfa shoots and one young corn plant catch low warm light. Below, the root systems fan out through a dark crumb structure of soil aggregates, air pockets and glinting water films. Pink nodules the size of peppercorns bulge along the alfalfa roots, cut away on the nearest ones to show a coral-red interior packed with bacteria. In the air band above, blue N_2 molecules drift as tightly bonded pairs with a visible triple bond. Four microbe crews work in visibly different places, each with its own silhouette and colour. The control panel is docked right; a molecular loupe reads a 100 nm scale bar.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Soil profile block	Environment	400×500×120 mm slice, front face clear; three horizons with distinct crumb sizes, 3 mm topsoil aggregates over 8 mm subsoil peds	No
2	Soil aggregates and pore space	Structure	Irregular clumped meshes with 20 % visible air-filled porosity; pore water drawn as glossy meniscus films that thicken with the moisture slider	No
3	Alfalfa plant	Actor	180 mm stem with trifoliate leaves, taproot descending 320 mm with fine laterals; grows in visible increments when nitrogen supply allows	Yes: place
4	Corn plant (non-fixer)	Actor	220 mm stem, strap leaves, fibrous root mat; has no nodules and depends entirely on soil nitrate	Yes: place, swap
5	Root nodule	Structure	3 mm pink ovoid on the alfalfa root, cutaway variant showing coral-red leghaemoglobin interior with packed bacteroids; forms over 6 simulated days after infection	Yes: place
6	Crew 1 — Nitrogen fixers (<i>Rhizobium</i>)	Actor	1.5 μm curved rods, deep magenta, living inside nodules and as free cells near roots; animation: grabs a drifting N_2 pair, the triple bond flashes and splits, and two NH_4^+ badges are released	Yes: place
7	Crew 2 — Ammonia oxidisers	Actor	1 μm spherical cells, amber, clustered on aggregate surfaces in well-aerated pores; convert NH_4^+ to NO_2^- with a small oxygen bubble consumed per conversion	Yes: swap
8	Crew 3 — Nitrite oxidisers	Actor	1.2 μm short rods, orange, always found beside Crew 2; convert NO_2^- to NO_3^- in a visibly separate second step	Yes: swap
9	Crew 4 — Denitrifiers	Actor	2 μm fat rods, slate blue, active only inside waterlogged, oxygen-poor pores; convert NO_3^- back to N_2 gas which rises as bubbles and leaves the soil	Yes: swap
10	Decomposer crew (ammonifiers)	Actor	Grey-green rods and fine hyphae on dead root fragments; release NH_4^+ from organic nitrogen	Yes: swap
11	N_2 molecules	Particle	Two blue spheres $r=0.30$ u joined by a three-cylinder bond bundle, drifting above the surface and in air-filled pores; the bond bundle is deliberately conspicuous	No
12	Nitrogen ion badges	Particle	NH_4^+ as a blue-white tetrahedral cluster with a plus halo; NO_2^- as one blue and two red spheres; NO_3^- as one blue and three red spheres; organic N as a violet chain segment	No
13	Oxygen availability field	Field	Scalar field through the profile, driven by moisture and compaction; rendered as a pale wash, with anoxic pockets outlined in blue	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Fertiliser applicator	Instrument	Hopper above the surface that broadcasts NH_4NO_3 granules; granules dissolve into badges over 2 simulated days	Yes: drag
15	Leaching drain and gauge	Instrument	Collection tray at 500 mm depth measuring nitrate washed below the root zone	No
16	Step-through stage controller	UI-Probe	Four-stop dial that isolates fixation, nitrification, assimilation or denitrification and dims the other three	Yes: drag

HOW IT WORKS — THE MODEL

Nitrogen atoms are tagged and conserved across six chemical forms: N_2 gas, NH_4^+ , NO_2^- , NO_3^- , organic N in living tissue, and organic N in dead matter. Each conversion is performed by a named microbe crew and only where that crew's conditions are met. Fixation runs inside nodules and needs plant sugar, so it stops if the plant is shaded. Nitrification is two separate steps by two separate crews and requires oxygen, so it collapses in waterlogged soil. Assimilation moves NO_3^- or NH_4^+ into root tissue at a rate set by root surface area. Ammonification returns organic N to NH_4^+ as dead matter decays. Denitrification runs only in anoxic pockets and is the one path that returns nitrogen to the air. Rates follow $\text{flux} = k \cdot \text{crew population} \cdot \text{substrate} \cdot f(\text{O}_2, \text{moisture}, \text{temperature})$. The sim must never let a plant absorb N_2 directly from the air, because "plants take nitrogen from the air" is precisely the misconception in play.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Crop planted	Dropdown	Alfalfa (nodulating) / Corn / Bare soil / Alfalfa + corn	Alfalfa	—	Structural: whether nodules and fixation exist at all
Microbe crews present	Multi-select	Fixers / Ammonia oxidisers / Nitrite oxidisers / Denitrifiers / Ammonifiers	All five	—	Structural: switch any crew out and break one step of the chain
Soil moisture	Slider	10–100	45	% of capacity	Oxygen field; above ~80 % anoxic pockets spread and denitrifiers wake
Soil temperature	Slider	4–35	18	°C	All microbial rates
Fertiliser application	Slider	0–250	0	kg N/ha	Adds NH_4NO_3 badges directly to the soil
Soil compaction	Slider	0–100	30	%	Pore space and therefore oxygen supply
Nodule count per plant	Stepper	0–40	18	count	Fixation capacity of the alfalfa
Stage isolation	Dial	All / Fixation / Nitrification / Assimilation / Denitrification	All	—	Dims every process but one for close inspection
Run length	Slider	5–120	60	simulated days	Length of the run

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Alfalfa field	crop=Alfalfa; crews=all; moisture=45 %; fertiliser=0	With no fertiliser at all, where does the nitrogen in the alfalfa's leaves come from? Trace one atom from air to leaf and name every crew that handled it.
S2	Corn without nodules	crop=Corn; fertiliser=0; run=60 d	The corn yellows and stalls. Nitrogen gas is all around its leaves. Why can it not use it?
S3	Flooded Central Valley field	moisture=95 %; crop=Corn; fertiliser=150 kg N/ha	Track the nitrate level after fertilising a waterlogged field. Which crew is thriving, and where is the nitrogen going?
S4	Broken chain	crews=Fixers + Ammonifiers only (oxidisers removed); crop=Corn; run=60 d	Ammonium accumulates but nitrate stays near zero. Which two steps are missing, and which plant suffers most?

STUDENT ACTIVITIES

1. Set the stage dial to Fixation and watch one nodule for 60 s. Record how many N_2 molecules are split per simulated day and what molecule leaves the nodule.
2. Step the dial through Nitrification and record the two separate products in order, naming the crew responsible for each.
3. Plant corn with fertiliser at 0, run 60 days, and record leaf nitrogen and plant height. Then plant alfalfa with the same settings and record both again.
4. Set moisture to 95 %, run 30 days, and measure the denitrification flux and the leaching gauge. Record where the nitrogen went, in two numbers.
5. Remove the ammonia oxidisers, run 60 days, and record the NH_4^+ and NO_3^- levels. Restore them and re-record to show which step you had broken.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Nitrogen form inventory	Bar chart	mg N/kg soil	Six bars: N_2 , NH_4^+ , NO_2^- , NO_3^- , living organic N, dead organic N
Process flux panel	Live numeric	mg N/kg/day	Fixation, nitrification step 1, step 2, assimilation, ammonification, denitrification
Soil nitrate over time	Line graph	mg N/kg	Trace vs simulated days, with fertiliser events marked
Tagged nitrogen atom route	Data table	form vs day	One row per conversion, naming the crew that did it
Plant nitrogen and height	Bar chart	mg N per plant, mm	Per plant, updated daily
Nitrogen lost to air	Counter	mg N/kg cumulative	Running total of denitrified nitrogen leaving the soil
Nitrate leached below root zone	Live numeric	mg N/kg	From the drain gauge at 500 mm
Crew populations	Line graph	cells $\times 10^6$ /g	One trace per microbe crew

WHAT THE STUDENT SHOULD REALISE

Students believe plants take nitrogen straight from the air, since the air is four-fifths nitrogen and plants are surrounded by it. Watching corn starve in a nitrogen-rich atmosphere while alfalfa thrives on the output of bacteria inside its own nodules makes that untenable. The N_2 triple bond has to be broken by a living crew before any plant can use the atom. The student should be able to say: *"Nitrogen gas is useless to plants until certain bacteria convert it, and different bacteria run each step of the loop that eventually sends it back to the air."*

D3.5 · Modeling matter and energy together

Experiment name: The Loop and the One-Way Street

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + Particle system

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS2-3

THEME & SCENE

The screen splits. On the left, a Sierra foothill clearing sits under a low glass dome: blue oak, bunchgrass, a mule deer browsing, a bobcat resting on granite, fungi fruiting on a fallen limb, all in warm late-afternoon light. The dome is sealed at the sides but its roof is an open grille, and faint orange shimmer escapes through it. On the right, the same scene is redrawn live as a flow board: five reservoir nodes joined by pipes. Two tracer types move through both halves at once. Blue matter tokens travel a closed circuit and always come back. Gold energy tokens enter from a sun icon, travel one way up the chain, and shed dull-red heat quanta that rise and vanish through the grille. The control panel is docked right; a heat-loss counter sits under the grille.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Glass dome and heat grille	Environment	Hemisphere 600 mm across in clear glass; sides opaque to matter, roof a fine louvre through which only heat quanta may pass; louvre can be shut by the student	Yes: swap
2	Sun source	Instrument	Warm disc at the upper left with a beam cone; emits gold energy tokens at a rate set by the irradiance slider; can be switched off entirely	Yes: drag
3	Blue oak and bunchgrass (producers)	Actor	320 mm oak with lobed leaves plus a 60 mm grass sward, sage-green; leaves flash faintly each time they capture an energy token	Yes: place, resize
4	Mule deer (primary consumer)	Actor	140 mm quadruped, dun coat, black-tipped tail; browses grass on a 20 s cycle, one bite transferring both token types	Yes: place
5	Bobcat (secondary consumer)	Actor	90 mm spotted cat, tawny with barred forelegs; hunts on a 3-simulated-day cycle	Yes: place
6	Fungi and soil microbes (decomposers)	Actor	Cream bracket fungi 30 mm plus a slate haze on the fallen limb; take tokens from any carcass or litter	Yes: place
7	Matter tokens	Particle	4 px blue-white cubes, 240 of them, numbered; move between reservoirs and can never leave the dome while the sides are sealed	Yes: place
8	Energy tokens	Particle	4 px gold teardrops with a soft bloom; created only at the sun, destroyed only by conversion into heat quanta	No
9	Heat quanta	Particle	3 px dull-red motes, wobbling upward at 20 mm/s, deleted when they pass the grille; every trophic transfer spawns them	No
10	Flow board nodes	Overlay	Five rounded panels on the right half (Sun, Producers, Primary consumers, Secondary consumers, Decomposers + soil), each showing its current matter and energy totals	No
11	Matter pipes	Overlay	Closed circuit of blue pipes with arrowheads, returning from decomposers to producers, so the loop visibly closes	No
12	Energy pipes	Overlay	Gold pipes running strictly one way, each with a wide red side-branch labelled "lost as heat", none returning to the sun	No
13	Trophic transfer engine	Field	Invisible node on each feeding link applying the transfer efficiency and spawning the corresponding heat quanta	No
14	Heat grille counter	Instrument	Digital readout mounted on the louvre totalling heat quanta that have left the dome	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Dual ledger strip	Overlay	Two horizontal bars along the bottom: total matter inside the dome, and total usable energy inside the dome, side by side	No
16	Token inspector	UI-Probe	Click any token to pin it and show a scrolling history of every reservoir it has occupied	Yes: drag

HOW IT WORKS — THE MODEL

Two tracer systems run on the same food web with deliberately different rules. Matter tokens obey strict conservation: every feeding, death and decay event moves a token from one reservoir to another, and the decomposer node returns tokens to the soil pool that producers draw from, so the circuit closes and the matter ledger bar never changes length while the dome sides are sealed. Energy tokens obey a one-way rule: at every transfer, a fixed efficiency (default 10 %) passes forward and the remainder is immediately converted into heat quanta that rise and exit. Producers convert incoming sunlight at about 1 %. Nothing in the model can turn heat quanta back into gold tokens, and there is no pipe from any reservoir back to the sun. If the sun is switched off, the matter loop keeps turning briefly while the energy ledger drains to zero and the system dies. That asymmetry, shown by two tracers in one scene, is the whole experiment.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Sunlight input	Slider	0–1600	800	W/m ²	Rate of gold token creation at the producers
Dome sides	Toggle	Sealed / Open	Sealed	—	Whether matter tokens may leave the system
Heat grille	Toggle	Open / Shut	Open	—	Structural: shutting it traps heat and raises dome temperature
Trophic transfer efficiency	Slider	2–25	10	%	How much energy passes to the next level per feeding event
Food web members	Multi-select	Producers / Deer / Bobcat / Decomposers	All four	—	Structural: removes a node and its pipes from both halves
Decomposer return	Toggle	On / Off	On	—	Whether the matter circuit closes or dead-ends in litter
Tracer view	Radio	Matter only / Energy only / Both	Both	—	Which token type is drawn
Run length	Slider	10–400	120	simulated days	Length of the run
Time compression	Dial	10×–5000×	800×	—	Simulated days per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Sun on, dome sealed	sunlight=800; sides=Sealed; grille=Open; web=all four	Run 120 days. Which ledger bar stays the same length and which one only holds up while the sun shines?
S2	Sun off	sunlight=0; sides=Sealed; run=60 d	The matter total has not changed but everything dies. Explain what ran out, using the heat counter as evidence.
S3	Loop broken	decomposer return=Off; sunlight=800; run=200 d	Matter piles up in the litter node. What happens to the producers, and is any matter actually lost from the dome?
S4	Trap the heat	grille=Shut; sunlight=800; run=60 d	With heat unable to leave, dome temperature climbs and organisms fail. Does trapped heat become usable energy again?

STUDENT ACTIVITIES

1. Set the tracer view to Matter only, pin token #12, and record every reservoir it visits over 60 days. Then set the view to Energy only and try to pin one gold token for the same length of time. Record what happens to it.

- Run S1 and record both ledger bars at days 0, 40, 80 and 120. State which one is flat and which one depends on the sun.
- Switch the sun off and record the heat grille counter and the energy ledger every 10 days until the ledger hits zero. Record the day it does.
- Turn decomposer return off, run 200 days, and record the matter held in the litter node and in the producers. Confirm from the matter ledger that none of it left the dome.
- Compare the energy arriving at the bobcat with the energy captured by the producers in S1, and write the ratio as a single number.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Matter ledger	Live numeric	tokens	Total matter inside the dome; flat while the sides are sealed
Energy ledger	Live numeric	tokens	Total usable energy inside the dome; falls whenever the sun dips
Heat lost through the grille	Counter	quanta cumulative	Running total of energy leaving as heat, updated every tick
Matter by reservoir	Bar chart	tokens	Five bars, showing redistribution without change in the total
Energy by trophic level	Bar chart	tokens	The pyramid shape appears without being drawn as a pyramid
Population survival	Line graph	count	Producers, deer, bobcat, decomposers vs simulated days
Pinned token history	Data table	reservoir vs day	Route of a chosen matter token; energy tokens end at "heat, gone"
Dome temperature	Live numeric	°C	Rises when the grille is shut, showing where the energy actually went

WHAT THE STUDENT SHOULD REALISE

Students believe energy is recycled like matter, and that a food chain "passes energy round". Running both tracers in one scene makes that untenable: the blue circuit closes and its total never changes, while gold tokens march one way and end as heat that physically leaves through the roof and never comes back. The student should be able to say: *"Matter goes round and round in an ecosystem, but energy only passes through once and leaves as heat, so the sun has to keep resupplying it."*

D4 · Patterns of interaction among organisms · MS-LS2-2

D4.1 · Competition and predation

Experiment name: Chase, Lag and Crowd-Out

Render mode: Hybrid 2D+3D

Simulation engine: Agent-based + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-2

THEME & SCENE

A dry Sierra foothill meadow seen from a low, wide angle: golden bunchgrass, scattered blue oaks, a granite outcrop, dust hanging in warm light. Black-tailed jackrabbits graze in loose groups and freeze when a coyote crests the ridge; the coyote quarters the grass with its nose down. Along a fenceline, two small seed-eaters work the same patch of buckwheat seed: a kangaroo rat that hops and a pocket mouse that scurries. Docked below the meadow is a dark graph panel where two population traces climb and fall out of step, with a movable caliper that measures the gap between their peaks. The control panel is docked right; a census hoop can be dropped anywhere on the meadow to count animals inside it.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Meadow terrain	Environment	200×140 m plot, low-poly rolling ground, straw-gold grass shader with wind ripple at 0.3 Hz	Yes: resize
2	Bunchgrass and buckwheat patches	Structure	300–900 instanced tussocks 400 mm high plus 120 seed-bearing buckwheat clumps; regrow at a slider-set rate and thin visibly when overgrazed	Yes: place
3	Blue oak and granite outcrop	Structure	Three 6 m oaks and a 4 m granite slab providing shade and cover patches that alter capture probability	Yes: drag
4	Black-tailed jackrabbit	Actor	480 mm body, tan with black ear tips, bounding gait at 3 m/s, freeze-then-flee behaviour, breeds when local grass is above threshold	Yes: place
5	Coyote	Actor	900 mm body, grizzled grey-buff, trotting search at 1.6 m/s with a 12 m detection radius and a 40 m sprint; starves if its energy store empties	Yes: place
6	Kangaroo rat	Actor	110 mm body with a 150 mm tufted tail, sandy; bipedal hops, caches seed, tolerates dry conditions well	Yes: place
7	Pocket mouse	Actor	80 mm body, greyish; quadrupedal scurry, forages faster per second but needs more water	Yes: place
8	Seed resource field	Field	Scalar seed-density field on a 2 m grid; consumed by both rodents, regrows logistically; the shared resource that drives competition	No
9	Grass biomass field	Field	Second scalar field on the same grid feeding the jackrabbits; regrowth rate is slider-driven	No
10	Predator detection cone	Overlay	Translucent wedge in front of the coyote, 12 m long and 120° wide, brightening when prey enters	No
11	Territory and cover overlay	Overlay	Toggleable hatching showing high-cover cells where capture probability drops by half	No
12	Coupled population graph	Overlay	Two-axis dark panel plotting predator and prey counts against simulated weeks, traces in tan and slate	No
13	Phase-lag caliper	Instrument	Draggable two-legged marker on the graph; snaps to peaks and reads the delay between them in weeks	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Census hoop	Instrument	20 m translucent ring dropped on the meadow; counts each species inside it and reports density per hectare	Yes: place, resize
15	Niche overlap dial (resource partitioning)	Field	Sets what fraction of seed sizes both rodents can eat; at 100 % they eat identically, at 0 % they use separate seed sizes	No
16	Event timeline	UI-Probe	Bottom strip with pinnable markers for drought, predator removal and introductions	Yes: drag

HOW IT WORKS — THE MODEL

Two coupled systems share one meadow. The predator-prey pair runs agent-by-agent: rabbits eat grass and breed above an energy threshold; coyotes search, detect within a cone, capture with a probability reduced by cover, and starve without kills. Because a coyote must first eat before it can breed, and a rabbit population must first grow before it can feed many coyotes, the two curves cannot peak together. The engine deliberately produces that delay from the agents' own energy budgets rather than imposing it, and the caliper measures it. The rodent pair runs on a shared seed field: each species has a consumption rate and a set of seed sizes it can handle. Where the overlap is total, the species with the marginally better rate drives the other to zero, slowly and without any fighting animation, because exclusion is a matter of arithmetic, not aggression. Where overlap is partial, both persist at lower densities.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Starting jackrabbits	Slider	0–200	60	count	Initial prey population
Starting coyotes	Slider	0–20	4	count	Initial predator population
Grass regrowth rate	Slider	0.1–5.0	1.5	% biomass/day	Carrying capacity for prey
Cover density	Slider	0–80	25	% of cells	Fraction of the meadow where capture probability halves
Competitor pair	Multi-select	Kangaroo rat / Pocket mouse	Both	—	Structural: which seed-eaters are present
Niche overlap	Slider	0–100	100	% of seed sizes shared	How completely the two rodents use the same food
Predator removal	Toggle	Coyotes present / Removed at week 20	Present	—	Structural: deletes all coyotes mid-run
Drought	Slider	0–80	0	% cut in regrowth	Reduces both grass and seed production
Run length	Slider	20–300	150	simulated weeks	Length of the run drawn on the graph

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Coyotes and hares	rabbits=60; coyotes=4; grass=1.5; run=150 wk	Measure the delay between each rabbit peak and the coyote peak that follows it. Is the delay roughly the same every cycle?
S2	Remove the predator	predator removal=Removed at week 20; run=150 wk	Rabbits surge, then crash without a coyote in sight. What ran out, and what does that show about who else limits prey numbers?
S3	Identical rivals	pair=Both; overlap=100 %; coyotes=0; run=300 wk	Two seed-eaters, exactly one food supply. What happens to the second species, and how long does it take?
S4	Split the seeds	pair=Both; overlap=30 %; run=300 wk	With the seed sizes partly divided, can both species persist? Record both final densities.

STUDENT ACTIVITIES

1. Run S1 for 150 weeks. Drag the caliper onto the first rabbit peak and the following coyote peak, and record the lag in weeks. Repeat for two more cycles.

- Predict which curve leads and which follows before running. Record your prediction, then correct it in one sentence using the caliper reading.
- Remove the coyotes at week 20 and record rabbit numbers and grass biomass every 10 weeks to week 100. Note the week of the crash.
- Set niche overlap to 100 % with no predators, run 300 weeks, and record the week each rodent species falls below 5 individuals.
- Drop the census hoop on a high-cover patch and on open ground. Record rabbit density per hectare in each and explain the difference in one sentence.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Coupled population graph	Line graph	count vs weeks	Predator and prey traces on shared axes, the out-of-step pattern being the point
Phase lag	Live numeric	weeks	Caliper reading between paired peaks
Competitor populations	Line graph	count vs weeks	Kangaroo rat and pocket mouse traces
Grass and seed biomass	Line graph	g/m ²	Both resource fields vs time
Kills per week	Counter	kills/week	Coyote hunting success, updated weekly
Census hoop density	Live numeric	animals/ha	Local density inside the placed hoop
Time to exclusion	Timer	simulated weeks	Weeks until a competitor drops below 5 individuals
Run comparison	Data table	mixed	One row per run: settings, peak values, lag, outcome; exportable

WHAT THE STUDENT SHOULD REALISE

Students believe predators simply wipe out prey, and that competition means animals fighting. Neither survives this meadow. Predator and prey numbers rise and fall together with a measurable delay, each one limiting the other, and when the coyotes are removed the rabbits crash anyway once the grass runs out. The rodents never touch each other, yet one still disappears. The student should be able to say: *"Predators and prey control each other on a delay, and competition can push a species out with no fighting at all, just by taking the same food."*

D4.2 · Mutualism, commensalism and parasitism

Experiment name: The Fitness Ledger: Who Gains, Who Pays

Render mode: 3D Scene

Simulation engine: Agent-based + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-2

THEME & SCENE

A long dark-walled gallery holds five lit habitat pods, each a glass drum about a metre across, arranged like specimen tanks. Pod 1 is a warm reef nook where a clownfish shelters among the stinging tentacles of a sea anemone. Pod 2 is a soil block with a blue oak seedling whose roots are sleeved in pale fungal threads. Pod 3 is a Mojave night: a yucca in flower with a small white moth working the blossoms. Pod 4 is a Central Valley pasture with a cow and a cattle egret stepping beside its hooves. Pod 5 is a foothill scene with a mule deer carrying ticks and an oak burdened with mistletoe. Above each pod hang two fitness dials, one per partner. The control panel is docked right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Habitat pod frame	Environment	Glass drum 1000 mm across on a brushed steel plinth, individually lit, with a species label plate and a swap socket	Yes: swap
2	Sea anemone	Actor	180 mm column, ochre, crowned with 120 pale tentacles 60 mm long, each carrying animated stinging cells that flash on contact with any fish but the clownfish	Yes: place
3	Clownfish	Actor	70 mm ovoid body, orange with three white bands, mucus coat rendered as a faint sheen; darts out to chase butterflyfish and returns; carries scraps back to the anemone	Yes: place
4	Blue oak seedling and root system	Actor	200 mm shoot, taproot with 40 fine laterals; root tips glow faintly where fungal contact is active	Yes: place, resize
5	Mycorrhizal fungal sheath and hyphae	Actor	Pale cream sleeve 0.2 mm thick over root tips, extending 60 mm of 5 µm hyphae into the soil; the hyphae reach soil volume the roots cannot	Yes: swap
6	Yucca and flower cluster	Actor	900 mm rosette of stiff blades with a 600 mm flower spike bearing 30 cream bells; ovules visible in a cutaway ovary	Yes: place
7	Yucca moth	Actor	18 mm white moth with specialised mouth tentacles; collects pollen into a ball, flies to another flower, packs pollen onto the stigma, then lays eggs among the ovules	Yes: place
8	Cow and cattle egret	Actor	1.4 m cow with a plodding walk that flushes insects from the grass; 500 mm white egret stepping alongside, snatching flushed insects; the cow's behaviour is unaffected by the egret's presence	Yes: place
9	Mule deer with ticks	Actor	1.2 m deer; 0–60 tick sprites 3 mm long attached at ears and flanks, each drawing a fixed blood volume per day and shown swelling	Yes: place
10	Oak with mistletoe	Actor	5 m oak carrying 0–8 mistletoe clumps 400 mm across, olive-green, each with haustoria drawn tapping into a branch and drawing water and nutrients	Yes: place
11	Insect prey swarm	Particle	200 instanced 6 mm insects in the pasture pod, flushed upward in a burst when the cow's hoof lands	No
12	Fitness dials	Instrument	Paired analogue dials above each pod reading –100 to +100 % change against a solo baseline, with a coloured sign badge (+, 0 or –)	No
13	Partner removal switch	Instrument	Toggle beside each pod that deletes one partner and re-runs the pod against the same baseline	Yes: swap
14	Baseline solo control pod	Structure	Sixth dimmed pod containing each organism alone; supplies the reference values the dials compare against	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Sign table	Overlay	Live five-row grid with a +/0/- cell per partner, filled in automatically as each pod finishes a run	No
16	Interaction inspector loupe	UI-Probe	Click any pair to open a slow-motion close-up of the exchange with an annotated cost and benefit list	Yes: drag

HOW IT WORKS — THE MODEL

Every organism carries a fitness score built from three measured quantities: survival probability, growth rate and offspring produced. Each pod is run twice, once with both partners and once against a stored solo baseline, and the dials display the percentage change. The interaction type is never declared by a label; it is derived from the pair of signs the model produces. Clownfish gain shelter while the anemone gains cleaning and defence, so both dials read positive. The fungal sheath supplies phosphorus from soil the roots cannot reach and takes sugar in return, again both positive. The yucca moth is the only pollinator that fits the flower and its larvae eat a portion of the seeds, so the score is positive on both sides but visibly costly. The egret gains insects while the cow's numbers are statistically unchanged, giving plus and zero. Ticks and mistletoe take blood, water and nutrients, so one dial rises while the other falls in proportion to the parasite load.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Active pod	Dropdown	Anemone + clownfish / Oak + mycorrhizae / Yucca + moth / Cow + egret / Deer + ticks / Oak + mistletoe	Anemone + clownfish	—	Which pair is centre stage and instrumented
Partner present	Toggle	Both partners / Partner removed	Both partners	—	Structural: deletes one organism and re-runs against baseline
Parasite load	Slider	0–80	20	ticks per deer, or clumps per oak	Size of the cost imposed on the host
Soil phosphorus	Slider	1–60	8	mg/kg	How much the oak actually needs its fungal partner
Predator pressure in reef pod	Slider	0–10	4	attacks/day	How valuable the anemone's shelter is to the clownfish
Moth egg load	Stepper	0–20	6	eggs per flower	Seeds eaten by larvae against seeds set by pollination
Run length	Slider	30–365	180	simulated days	Length of each pod run
Show cost and benefit tags	Toggle	On / Off	On	—	Floating labels on each exchange in the pod
Compare pods	Multi-select	Any two to five pods	Pods 1 and 5	—	Puts chosen pods side by side in the sign table

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Both gain	pod=Anemone + clownfish; predator pressure=4; run=180 d	Record both dials. What does each partner actually give the other, and what are the two signs?
S2	One gains, one pays	pod=Deer + ticks; parasite load=60; run=180 d	Put a number on the cost: how much does the deer's fitness fall, and does the parasite benefit from killing its host?

#	Scenario	Preset	The question the student answers
S3	One gains, one unaffected	pod=Cow + egret; run=180 d	The egret's dial is clearly positive. Read the cow's dial. What does a reading of zero mean about this relationship?
S4	Partner removed	pod=Oak + mycorrhizae; soil phosphorus=4; partner=Partner removed	With the fungus gone from poor soil, record the oak's growth. Which partner turns out to have needed the other more?

STUDENT ACTIVITIES

1. Run each of the five pods for 180 days and record both fitness dials for each. Fill in the sign table with +, 0 or – for every partner.
2. Remove the anemone from pod 1 and re-run. Record the clownfish's survival, then restore it and record again.
3. Set the tick load to 0, 20, 40 and 60 in turn, running 180 days each time, and plot deer fitness against parasite load.
4. Set soil phosphorus to 4 mg/kg and then to 50 mg/kg with the fungus present, and record the oak's growth in both. Write one sentence about when the partnership matters most.
5. Compare pod 3 and pod 5 side by side and explain, using the moth's egg load, why a relationship can be a partnership even when one side takes something.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Fitness change per partner	Live numeric	% vs solo baseline	The two dials above the active pod, updated daily
Sign table	Data table	+, 0 or –	Five rows, two cells each; the classification the student builds from evidence
Host cost curve	Line graph	% fitness vs parasite load	Built point by point as the student steps the load slider
Survival probability	Bar chart	% surviving 180 days	Each organism with and without its partner
Offspring produced	Counter	count per run	Direct fitness measure for both partners
Nutrient or food transferred	Live numeric	mg/day or items/day	Phosphorus to the oak, sugar to the fungus, insects to the egret, blood to the tick
Seeds set against seeds eaten	Bar chart	seeds per flower	The yucca pod's balance sheet
Run comparison	Data table	mixed	One row per pod run with settings and both dial values; exportable

WHAT THE STUDENT SHOULD REALISE

Students believe relationships between species are either friendly or hostile, and that a partner that takes anything must be an enemy. Reading two fitness dials per pod makes that untenable: the categories are simply the pairs of signs, and the yucca moth takes seeds while remaining essential. Cost and benefit are measured, not assumed. The student should be able to say: *"You classify a relationship by measuring what happens to each partner: both up is mutualism, up and unchanged is commensalism, up and down is parasitism."*

D4.3 · The same pattern in very different ecosystems

Experiment name: Three Places, One Pattern

Render mode: 2.5D Layered

Simulation engine: Agent-based + State machine

Interaction level: Design

Session length: 18–25 min

NGSS anchor: MS-LS2-2

THEME & SCENE

Three tall viewports stand side by side like windows in a gallery wall, each running its own living scene. Left: a Monterey kelp forest in green underwater light, giant kelp fronds swaying upward from holdfasts, urchins on the rock, a sea otter rolling at the surface. Centre: a Sierra foothill oak woodland in dry golden light, blue oaks and acorn woodpeckers, ground squirrels, a gopher snake in the leaf litter. Right: a Mojave bajada at dusk, creosote and yucca on pale gravel, kangaroo rats, a sidewinder, a kit fox. Below the viewports runs a rail of blank role cards. Drag a card into a viewport slot and the matching organism is spotlighted in all three scenes at once. The control panel is docked right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Viewport frames	Environment	Three 420×700 px windows with brushed dark surrounds, each independently pan and zoom capable, each running its own agent population	No
2	Kelp forest backdrop	Environment	Six parallax layers of green water with god-rays, a rocky reef floor, and drifting particulate; surface shimmer on the top layer	No
3	Oak woodland backdrop	Environment	Five layers, straw-gold understorey, scattered blue oaks, granite knuckles, heat haze near the ground	No
4	Mojave bajada backdrop	Environment	Five layers of pale alluvial gravel, evenly spaced creosote, distant ranges in dusk violet	No
5	Producer set	Actor	Giant kelp blades 8 m with pneumatocysts; blue oak canopy with acorn crop; creosote with resinous leaves and yucca rosette; each grows and is grazed	Yes: swap
6	Primary consumer set	Actor	Purple urchin 70 mm with moving spines; California ground squirrel 250 mm; Merriam's kangaroo rat 110 mm with tufted tail; each feeds on its scene's producer	Yes: place
7	Predator set	Actor	Sea otter 1.2 m rolling and hammering urchins on its chest; gopher snake 1.4 m in a searching crawl; sidewinder 600 mm with its looping sideways track and kit fox 500 mm with oversized ears	Yes: place
8	Mutualist pair set	Actor	Kelp holdfast with encrusting bryozoans; oak roots with mycorrhizal sheath; yucca with yucca moth; each pod animates its exchange	Yes: place
9	Role cards	UI-Probe	Six draggable cards labelled Producer, Primary consumer, Predator, Decomposer, Mutualist, Competitor, blank of any species name	Yes: drag, place
10	Role slots	Structure	Five empty sockets beneath each viewport that accept a card and bind it to a species in that scene	Yes: connect
11	Pattern-match scorer	Instrument	Compares the interaction network the student has assembled across the three scenes and scores structural equivalence out of 100	No
12	Interaction arrow overlay	Overlay	Coloured arrows drawn between filled slots: red for eats, green for mutual benefit, grey for competes; identical arrow shapes appear in all three viewports when the pattern matches	No
13	Synchronised perturbation bar	Instrument	Single control that removes the predator, or halves the producers, in all three viewports simultaneously	Yes: swap
14	Population strip charts	Overlay	Three stacked mini-graphs, one per ecosystem, on a shared time axis so the shapes can be compared directly	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Species dossier card	UI-Probe	Hover any organism for a card with its name, diet, size, and its role in that ecosystem	Yes: drag
16	Scale reference bar	Overlay	Per-viewport bar showing metres, since the three scenes differ enormously in physical scale	No

HOW IT WORKS — THE MODEL

One interaction engine runs three parameter sets. Each ecosystem holds the same abstract nodes (producer, primary consumer, predator, decomposer, mutualist pair) filled by different species with different rates: kelp grows fast and is grazed hard, oak grows slowly with a pulsed acorn crop, creosote grows very slowly with long dormancy. Because the structure is shared, a perturbation applied to all three at once produces the same qualitative response with different timescales: remove the otter and urchins boom and strip the kelp within a season; remove the snake and squirrels boom over a few years; remove the kit fox and kangaroo rats boom over a decade. The scorer checks whether the student's assembled arrow network is structurally identical across scenes, not whether the species names match. The honest simplification is that only one strand of each real food web is modelled, and the sim says so on the dossier cards.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Ecosystems shown	Multi-select	Kelp forest / Oak woodland / Mojave bajada	All three	—	Structural: which viewports are live
Role assignment	Drag-handle	Six cards into five slots per viewport	Empty	—	Structural: builds the interaction network the scorer grades
Synchronised perturbation	Dropdown	None / Remove predator / Halve producers / Add competitor / Remove mutualist	None	—	Applies the same shock to every live viewport at once
Run length	Slider	1–40	10	simulated years	Length of the run on the strip charts
Time alignment	Toggle	Real time / Normalised time	Real time	—	Normalised stretches each scene's axis so shapes can be overlaid
Producer growth multiplier	Slider	0.2–3.0	1.0	× baseline	Scales productivity in every scene at once
Predator starting number	Stepper	0–20	6	count per scene	Initial predator population in each viewport
Arrow overlay	Toggle	On / Off	On	—	Draws the interaction arrows between filled slots
Dossier cards	Toggle	On / Off	On	—	Hover information for every species

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Fill the slots	ecosystems=all three; perturbation=None; run=10 yr	Assign all six role cards in each viewport. What score does the pattern matcher give, and which slot was hardest to fill in the Mojave?
S2	Pull the predator	perturbation=Remove predator; run=20 yr	The same shock in three places. Compare the three strip charts: what is the same about the shapes, and what is different about the timing?
S3	Normalise the clock	perturbation=Remove predator; time alignment=Normalised	With each scene's time axis stretched to the same length, how similar do the three responses look now?
S4	Lean years	producer growth=0.4; run=40 yr	Cut productivity everywhere. Which ecosystem's populations fall furthest, and what does its producer's growth rate have to do with it?

STUDENT ACTIVITIES

1. Drag the Producer, Primary consumer and Predator cards into all three viewports and record which species you chose for each slot.
2. Run S1 for 10 years and record the pattern-match score. Fix any mismatched arrow and record the improved score.
3. Apply Remove predator to all three at once, run 20 years, and record the peak primary-consumer number and the year it occurs in each ecosystem.
4. Switch time alignment to Normalised and sketch the three curves on one axis. Write one sentence on how alike they are.
5. Set producer growth to 0.4, run 40 years, and record final populations in all three. Rank the ecosystems by how hard they were hit.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Pattern-match score	Pass-fail badge	0–100	How structurally identical the three assembled networks are
Population strip charts	Line graph	count vs years	Three stacked panels on a shared or normalised axis
Peak consumer number and year	Data table	count, year	Per ecosystem after a perturbation
Response time to perturbation	Live numeric	simulated years	Time from the shock to the peak in each scene
Producer biomass	Line graph	relative units	Kelp, oak and creosote biomass on common relative axes
Role slot audit	Data table	filled / empty	Which roles the student assigned in each viewport
Arrow network map	Vector overlay	—	The interaction diagram the student built, exportable as an image

WHAT THE STUDENT SHOULD REALISE

Students believe ecosystems are lists of local species and that a kelp forest has nothing to do with a desert. Running one engine behind three unrelated scenes and applying the identical shock makes that untenable: the same roles are present in each, the same arrows connect them, and the response curves have the same shape once the clocks are matched. Only the names and the timescales differ. The student should be able to say:

"Different places have completely different species, but the same patterns of interaction repeat, so what I learn in one ecosystem helps me predict another."

D4.4 · Interactions that shift over time or condition

Experiment name: When Partners Turn: Finding the Tipping Point

Render mode: Hybrid 2D+3D

Simulation engine: State machine + Agent-based

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS2-2

THEME & SCENE

Two lit bays face each other across a dark bench. The left bay is a soil column: a blue oak seedling with its roots sleeved in pale mycorrhizal threads, a fertiliser hopper poised above the surface, and a phosphorus dial set into the glass. The right bay is a warm reef ledge where a bluestreak cleaner wrasse works a queue of client fish, darting into gills and mouths; a small counter above it tallies parasites removed and mucus bites stolen. Between the bays hangs the Sign Board: a single large plus-or-minus badge for each partner that visibly flips as conditions cross a threshold. Below both bays lies a two-axis sign map that fills in with colour as the student sweeps conditions. The control panel is docked right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Soil column bay	Environment	300×600 mm glass column, crumb-structured soil, front face clear, root zone lit from a low angle	No
2	Blue oak seedling	Actor	240 mm shoot with four leaf pairs, taproot to 400 mm with 60 fine laterals; leaf area and stem height are its measured fitness	Yes: place, resize
3	Mycorrhizal sheath and hyphal net	Actor	Cream 0.2 mm sleeve on root tips extending 80 mm of 5 µm hyphae; hyphae glow faint blue when delivering phosphorus and faint amber when drawing sugar	Yes: swap
4	Sugar and phosphorus exchange tokens	Particle	3 px amber tokens travelling root to fungus, 3 px blue tokens travelling fungus to root; token counts are the raw exchange data	No
5	Fertiliser hopper	Instrument	90 mm hopper on a rail above the column; broadcasts phosphate granules that dissolve over two simulated days	Yes: drag
6	Soil phosphorus field	Field	Scalar field through the column, 1–120 mg/kg, rendered as a violet wash of increasing saturation	No
7	Reef ledge bay	Environment	500 mm coralline ledge in warm blue-green water with drifting motes and a soft caustic pattern	No
8	Cleaner wrasse	Actor	90 mm slender fish, pale blue with a black lateral stripe, hovering dance to advertise, darting into gill covers; carries a hunger state that rises between meals	Yes: place
9	Client fish queue	Actor	6–24 clients 200–400 mm long in three species, hovering in a loose line, posing with fins spread; each carries a parasite count and a mucus reserve	Yes: place
10	Gill parasites	Particle	2 mm pale isopod sprites clinging to gills and flanks, 0–40 per client, removed one at a time by the cleaner	No
11	Mucus bite marker	Overlay	Small red flash on a client's flank each time the cleaner cheats and takes mucus instead of a parasite; clients jolt and may leave the queue	No
12	Sign Board	Overlay	Two large badges, one per partner, reading + or – with the interaction name beneath; flips live and logs the tick at which it flipped	No
13	Sign map	Overlay	Two-axis heat map, condition on x, second condition on y, each cell shaded green for mutualism, grey for neutral and red for parasitism as it is sampled	No
14	Condition sweep dial	Instrument	Dial that ramps the chosen condition smoothly across its whole range while the run continues, drawing a live trace	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Season timeline	Instrument	Twelve-month strip that can drive conditions automatically, so the same pair changes sign across a year	Yes: drag
16	Fitness meters	Instrument	Paired dials per bay reading percentage change against a partner-free baseline, identical in style to the ledger the student met earlier	No

HOW IT WORKS — THE MODEL

Each partnership is a state machine with two states, Mutualism and Exploitation, and the transition is driven by a measured net benefit rather than by a label. In the soil bay, the oak pays a fixed sugar cost per unit of hyphal network and receives phosphorus that the hyphae reach beyond the root zone. When soil phosphorus is low the delivered phosphorus is worth far more than the sugar, so both fitness meters are positive. As fertiliser raises soil phosphorus, the roots can get phosphorus unaided while the sugar bill stays the same, so the oak's meter crosses zero and turns negative: the same fungus, unchanged, is now a parasite. In the reef bay, the cleaner chooses between eating a parasite and biting mucus, weighted by hunger and by queue length; cheating raises the cleaner's fitness and lowers the client's. Cheating is punished when clients can leave, which is why long queues make cleaners behave. Nothing about either partner's identity changes; only the conditions do.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Active bay	Dropdown	Oak + mycorrhizae / Reef cleaner + clients / Both	Oak + mycorrhizae	—	Which partnership is instrumented
Soil phosphorus	Slider	1–120	6	mg/kg	The condition that flips the fungal partnership
Fungal network size	Slider	10–200	80	mm of hyphae per root tip	Sugar cost paid by the oak
Light available to the oak	Slider	50–1500	600	μmol/m ² /s	How easily the oak can afford the sugar bill
Client queue length	Stepper	2–24	10	fish	Whether cheating cleaners are punished by clients leaving
Cleaner hunger	Slider	0–100	30	%	Probability the cleaner takes mucus instead of a parasite
Client escape allowed	Toggle	Clients may leave / Clients captive	May leave	—	Structural: removes the punishment mechanism entirely
Condition sweep	Dial	Off / Sweep phosphorus / Sweep hunger / Sweep queue	Off	—	Ramps one condition across its range during a single run
Season driver	Timeline scrubber	Jan–Dec	Off	month	Drives conditions automatically through a year
Run length	Slider	20–365	120	simulated days	Length of the run

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Poor soil partnership	bay=Oak + mycorrhizae; phosphorus=6; light=600; run=120 d	Record both fitness meters. Which partner gains, and what exactly does each one hand over?
S2	Fertilised to the flip	bay=Oak + mycorrhizae; sweep=Sweep phosphorus; run=120 d	Sweep phosphorus from 1 to 120 mg/kg. At what value does the oak's meter cross zero, and what is the fungus called after that point?

#	Scenario	Preset	The question the student answers
S3	The cheating cleaner	bay=Reef; hunger=80; queue=3; escape=May leave; run=60 d	Count mucus bites per hour. Does the client's fitness stay positive, and what happens to the queue?
S4	Captive clients	bay=Reef; hunger=80; queue=3; escape=Clients captive	With clients unable to leave, how does the cheating rate change, and which way does each sign badge point?

STUDENT ACTIVITIES

1. Run S1 for 120 days and record both fitness meters and the number of phosphorus tokens delivered per day.
2. Set the sweep dial to Sweep phosphorus and run once. Record the exact phosphorus value at which the Sign Board flips, and repeat with the fungal network at 200 mm to see whether the threshold moves.
3. Sample nine points on the sign map by setting phosphorus to 5, 30 and 100 against light at 100, 600 and 1400, recording the oak's sign in each cell.
4. In the reef bay, set hunger to 20, 50 and 80 with a queue of 10, and record mucus bites per hour and client fitness at each setting.
5. Switch clients to captive, re-run the hunger=80 case, and write one sentence comparing the cheating rate with and without the ability to leave.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Partner fitness meters	Live numeric	% vs partner-free baseline	Both partners in the active bay, updated daily
Sign Board state	Pass-fail badge	+/- pair	The current interaction classification, with a log of every flip
Flip threshold	Live numeric	mg/kg or % hunger	The condition value at which the sign last changed
Sign map	Heat map	—	Interaction outcome across two conditions, filled cell by cell
Exchange tokens	Line graph	tokens/day	Sugar out and phosphorus in, on shared axes
Mucus bites and parasites removed	Bar chart	count/hour	The cleaner's honest and dishonest behaviour side by side
Queue length over time	Line graph	fish vs hours	How clients respond to being cheated
Sweep trace	Line graph	fitness vs condition	Continuous record from a condition sweep, showing the zero crossing

WHAT THE STUDENT SHOULD REALISE

Students believe a species is permanently a helper or permanently a pest, as though the label lived inside the organism. Sweeping one condition and watching the identical fungus cross from mutualist to parasite, with no change to the fungus at all, makes that untenable. The relationship is a running balance of what each side gains and pays under current conditions. The student should be able to say: *"Whether an interaction helps or harms depends on the conditions, so the same two species can be partners one week and a parasite and host the next."*

D4.5 · Interaction patterns and population change

Experiment name: Draw the Curve: From Interaction to Population

Render mode: Data Dashboard

Simulation engine: Data-driven model + State machine

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS2-2

THEME & SCENE

A dark analyst's console fills the screen. Top left, a small live stage shows the two chosen species as animated tokens in a stylised habitat strip so the abstraction stays anchored to organisms. Top right sits the Interaction Setter: a three-by-three sign grid where the student clicks a cell to declare what each species does to the other, plus a strength dial. The lower two-thirds is the Prediction Canvas: a blank grid with population on the vertical axis and weeks on the horizontal, where the student draws two freehand curves with the cursor before pressing Run. On Run, the model's own curves are drawn over the sketch in a contrasting colour and a match score appears. The control panel is docked right; a scenario library sits along the bottom.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Console shell	Environment	Dark slate panel layout with three docked regions and a thin amber accent rule; all typography high-contrast for graph reading	No
2	Live stage strip	Overlay	420×160 px habitat band showing species A and B as instanced tokens whose numbers track the model; token count redraws every simulated week	No
3	Species A token	Actor	14 px organism sprite, teal, swappable between eight presets (kelp, blue oak, jackrabbit, kangaroo rat, urchin, sea otter, mycorrhizal fungus, tick)	Yes: swap
4	Species B token	Actor	14 px organism sprite, amber, same swappable preset list, with a warning if the pair is biologically absurd	Yes: swap
5	Interaction Setter grid	UI-Probe	3×3 grid of cells labelled by the effect of A on B and B on A (+, 0, -); selecting a cell names the interaction automatically	Yes: connect
6	Interaction strength dial	Instrument	0–100 dial setting how hard each species presses on the other; feeds the coupling coefficients	Yes: drag
7	Prediction Canvas	UI-Probe	900×420 px grid, 0–500 individuals against 0–200 weeks, with a freehand pen that captures two sketched traces at 5-week resolution	Yes: drag
8	Sketched prediction traces	Overlay	Two dashed lines in the student's colours, locked once Run is pressed so they cannot be quietly redrawn	Yes: drag
9	Model output traces	Overlay	Two solid lines drawn over the sketch after Run, teal and amber, with peaks and troughs auto-marked	No
10	Match scorer	Instrument	Compares sketch and model on three features (shape, lead-lag order, final level) and reports a score out of 100 with a per-feature breakdown	No
11	Coupled population model	Field	Invisible pair of difference equations updating both populations each simulated week from the sign grid and strength dial	No
12	Carrying capacity line	Overlay	Dashed horizontal rule on the canvas at the resource limit for species A	Yes: drag
13	Perturbation injector	Instrument	Drop a marker anywhere on the time axis to fire a one-off event: disease, drought, removal or introduction	Yes: place
14	Scenario library rail	UI-Probe	Bottom rail of preset cards, each loading a sign-grid setting and a species pair	Yes: place

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Evidence claim card	UI-Probe	Panel where the student writes a claim and drags a graph region onto it as evidence; the panel checks that the region actually shows what the claim states	Yes: drag, connect
16	Run log	Overlay	Scrolling table of every run with the sign grid used, the strength, the match score and the outcome	No

HOW IT WORKS — THE MODEL

Two populations update each simulated week by $N(t+1) = N(t) + r \cdot N(t) \cdot (1 - N(t)/K) + c \cdot N(t) \cdot M(t) / 1000$, where c carries the sign the student chose in the grid and its magnitude comes from the strength dial. Setting minus and zero gives simple exploitation; minus and minus gives competition, which drives one population to zero when strengths are unequal; plus and plus gives mutualism, where both settle above their solo carrying capacities; plus and minus with a lag term gives the classic out-of-step predator-prey oscillation. The engine adds a one-week reproductive delay to the consumer so that lag emerges rather than being drawn on. The match scorer never rewards a lucky line: it grades shape category, which curve peaks first, and the final level, separately. The point of the prediction step is that a wrong sketch is useful data, so the model output is never previewed before Run and the sketch is locked on submission.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Interaction type	Radio	Predation / Competition / Mutualism / Commensalism / Parasitism / No interaction	Predation	—	Structural: sets the sign pair and relabels the axes legend
Species pair	Dropdown	Eight preset organisms for A and for B	Jackrabbit / Coyote	—	Structural: swaps the stage sprites and the sensible starting numbers
Interaction strength	Dial	0–100	45	—	Coupling magnitude between the two populations
Starting population A	Slider	5–500	120	individuals	Initial value of the teal trace
Starting population B	Slider	1–200	20	individuals	Initial value of the amber trace
Carrying capacity for A	Drag-handle	50–600	300	individuals	Structural: moves the dashed resource ceiling on the canvas
Perturbation event	Dropdown	None / Disease in A / Drought / Remove B / Introduce competitor	None	—	Fires a one-off shock at the marker's week
Perturbation week	Slider	5–195	60	week	Where on the time axis the shock lands
Run length	Slider	40–200	150	simulated weeks	Horizontal extent of the canvas

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Predict predation	interaction=Predation; pair=Jackrabbit / Coyote; strength=45; run=150 wk	Sketch both curves first. Which peaks first, and by how many weeks, and how close was your sketch?
S2	Predict competition	interaction=Competition; pair=Kangaroo rat / Kangaroo rat variant; strength=70	Sketch what happens to both. Does either curve reach zero, and does the winner end above or below its solo capacity?
S3	Predict mutualism	interaction=Mutualism; pair=Blue oak / Mycorrhizal fungus; strength=60	Sketch both. Where do the two curves settle compared with the carrying capacity line?

#	Scenario	Preset	The question the student answers
S4	Shock the system	interaction=Predation; perturbation=Disease in A; week=60; run=200 wk	Sketch the recovery. How long does the oscillation take to return to its earlier size, and does the predator recover first or second?

STUDENT ACTIVITIES

1. Choose Predation, sketch both curves on the canvas without running anything, and record your match score and the three feature scores.
2. Re-sketch the same scenario after seeing the model once, run again, and record the improvement. Write one sentence on what you got wrong the first time.
3. Set Competition with strength 70, predict which species survives, run, and record the week the loser reaches zero.
4. Switch to Mutualism, drag the carrying capacity line from 300 to 150, run, and record both final populations against the line.
5. Place a disease perturbation at week 60 in the predation run, sketch the recovery, then record the actual recovery time and build one claim card citing the graph region that supports it.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Model population curves	Line graph	individuals vs weeks	Both species drawn over the locked sketch
Prediction match score	Pass-fail badge	0–100	Overall score with shape, lead-lag and final-level components shown separately
Lead-lag between peaks	Live numeric	weeks	Auto-measured from marked peaks in the model output
Final populations	Live numeric	individuals	Both species at the end of the run, with the capacity line for reference
Interaction sign pair	Data table	+, 0 or –	What the student declared and what the resulting curves actually show
Recovery time after shock	Timer	simulated weeks	Weeks to return within 10 % of pre-shock amplitude
Claim card verdict	Pass-fail badge	—	Whether the cited graph region supports the written claim
Run log	Data table	mixed	Every run with settings, scores and outcomes, exportable as CSV

WHAT THE STUDENT SHOULD REALISE

Students believe population graphs are things to be read off rather than things that follow from how species treat each other, and they expect every curve to rise or fall smoothly. Committing a sketch before the model runs exposes that: predation gives out-of-step waves, competition gives one curve to zero, mutualism lifts both above their solo ceilings, and the shape is decided by the sign pair. The student should be able to say: *"If I know what each species does to the other, I can predict the shape of both population curves before I run anything."*

D5 · Ecosystem disruption and change · MS-LS2-4

D5.1 · Physical disruptions

Experiment name: Fire, Flood and Slide: The Landscape Stress Test

Render mode: Hybrid 2D+3D

Simulation engine: Agent-based + Fluid/thermal

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-4

THEME & SCENE

A tilted three-quarter view of one square kilometre of Californian foothill country, rendered as a low-poly terrain block with a visible soil cutaway down the front face. An oak woodland fills the upper slope, chaparral the mid-slope, a willow-lined creek the valley floor, and a grassy terrace beside it. Colour is late-summer: straw gold, olive scrub, dusty green oaks, a thin silver creek. Animal agents move as small coloured markers grazing, hunting and drinking. A wind sock on a ridge pole shows direction and strength. The control panel docks right; a slim before/after biodiversity strip runs along the bottom edge, greyed until a disruption has been run.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Terrain block	Environment	1 km × 1 km heightfield, 4 m grid, vertical exaggeration ×2, cutaway front face showing 40 cm topsoil over pale weathered bedrock	Yes: resize
2	Blue oak stand	Actor	180 instanced trees, 6–9 m crowns, olive canopy; each carries a health value 0–1 that drives leaf colour from green to grey	Yes: place
3	Chaparral patch	Actor	900 low domes 0.8–1.8 m, dusty sage-green, high oil content flagged in the fuel model	Yes: place
4	Perennial grass sward	Structure	Tinted ground layer, straw gold, regrows on a 60-day clock after removal	Yes: swap
5	Creek and riparian willows	Structure	Polyline channel 3 m wide with 40 willow sprites; channel widens and browns during flood	No
6	Soil column	Structure	Layered slab: 12 cm dark A-horizon, 28 cm brown B-horizon, bedrock; root mesh drawn as fine white threads through the top two layers	No
7	Deer agents	Actor	24 tan markers, browse oak and shrub, flee fire fronts at 4 m/s	No
8	Ground squirrel agents	Actor	140 small brown markers, burrow icons; survive surface fire in burrows	No
9	Scrub jay agents	Actor	30 blue markers, cache acorns, disperse after canopy loss	No
10	Stream invertebrate agents	Particle	500 fine specks in the creek, killed by sediment above a turbidity threshold	No
11	Fire front	Field	Cellular spread field over the fuel grid; flame sprites 0.4–4 m tall scaled by fuel and wind	No
12	Flood water body	Field	Height-based water surface with sediment tint; deposits or scours soil cells on recession	No
13	Landslide mass	Particle	3,000 debris particles released when slope saturation exceeds a threshold; buries cells beneath	No
14	Soil moisture field	Field	Invisible per-cell scalar 0–1, driven by rainfall and drought controls; feeds fire, flood and slide rules	No
15	Wind sock and vane	Instrument	Ridge pole with striped cone; angle and inflation show direction and speed	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Quadrat survey frame	UI-Probe	50 m × 50 m draggable square that counts species and individuals inside it	Yes: drag

HOW IT WORKS — THE MODEL

Each terrain cell holds fuel load (t/ha), moisture (0–1), slope and vegetation type. Fire spreads cell to cell each tick with probability rising with fuel load and wind speed, falling with moisture; spread is biased downwind and uphill, so a 30 km/h upslope wind roughly triples front speed. Rainfall fills the moisture field; above field capacity the excess becomes runoff, and runoff above the channel capacity floods valley cells. Slope failure triggers where saturation exceeds 0.85 on slopes steeper than 25°. Drought simply holds rainfall at zero and drains moisture on an evaporation curve. Every agent carries a survival rule tied to the disruption type: burrowers survive surface fire, canopy nesters do not; stream invertebrates die to turbidity, not to heat. Species richness and each population are sampled before the event and at chosen intervals after, so the readouts are always a comparison, never a single number.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Disruption type	Dropdown	Wildfire / Flood / Landslide / Drought / None	Wildfire	—	Which physical event the Run button releases
Fuel load	Slider	2–40	18	t/ha	Density of shrub and litter; flame height and spread rate
Wind speed	Slider	0–60	15	km/h	Fire front speed and direction bias
Wind direction	Dial	0–359	225	degrees	Which way the front runs across the slope
Rainfall intensity	Slider	0–120	0	mm/h	Fills soil moisture; drives flood depth and slide risk
Slope steepness	Slider	5–40	22	degrees	Terrain tilt; landslide threshold and runoff speed
Drought length	Stepper	0–36	0	months	Baseline moisture before the event
Vegetation layout	Drag-handle	Place or delete oak, chaparral, grass, willow	Mixed foothill	—	Structural: rebuilds the fuel and habitat map
Recovery window	Timeline scrubber	0–10	1	years	How long after the event the survey is taken
Playback speed	Dial	0.5×–8×	2×	—	Time scaling of the event itself

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Cool grass fire	disruption=Wildfire; fuel load=6; wind speed=8; drought length=0	Which populations drop and which barely change when a low fire runs through grass?
S2	Diablo wind event	disruption=Wildfire; fuel load=34; wind speed=55; wind direction=45; drought length=18	Same landscape, same ignition. Why does this fire remove the oak canopy when S1 did not?
S3	Winter storm on a bare slope	disruption=Flood; rainfall intensity=95; slope steepness=34; vegetation layout=grass removed	The creek invertebrates die but the deer survive. What killed the invertebrates?
S4	Three dry years	disruption=Drought; drought length=36; rainfall intensity=0; recovery window=3	Nothing burns and nothing washes away. Explain how the ecosystem still loses species.

STUDENT ACTIVITIES

1. Run S1, then record the before and after richness count and the deer, squirrel and jay populations from the survey strip.
2. Set fuel load to 34 and wind to 55 km/h, re-run, and record the same four numbers beside the first set.

3. Drag the quadrat frame onto the burnt oak stand and onto unburnt chaparral; record species counts in each and state which patch recovered faster.
4. Predict, before running S3, whether flood or fire removes more species; then run both and compare your prediction to the richness bars.
5. Set the recovery window to 1 year, then 5, then 10 for the same fire, and record how richness changes across the three surveys.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Species richness before/after	Bar chart	species count	Paired bars per survey, updated when the recovery window changes
Population by species	Data table	individuals	One row per species, columns for before, immediately after, and end of window
Area disturbed	Live numeric	hectares	Burnt, flooded or buried area, updated each tick during the event
Fire front speed	Live numeric	m/min	Instantaneous spread rate; responds visibly to the wind slider
Soil loss	Live numeric	cm of A-horizon	Depth of topsoil scoured or buried, sampled from the cutaway face
Canopy cover	Line graph	% vs months	Cover trace from event to end of recovery window
Turbidity in creek	Live numeric	NTU	Sediment load; the killer variable in flood runs
Run comparison log	Data table	mixed	Exportable CSV, one row per run, all settings and all outcomes

WHAT THE STUDENT SHOULD REALISE

Students believe a disruption is bad in proportion to how dramatic it looks. Here a quiet three-year drought removes more species than a fast grass fire, and the same ignition produces a survivable burn or a stand-replacing one depending only on fuel and wind. Damage tracks conditions and the traits of each species, not spectacle. The student should be able to say: *"The same event can be small or catastrophic — what decides is the state the ecosystem was already in."*

D5.2 · Biological disruptions

Experiment name: Uninvited: Invaders, Blights and a Missing Keystone

Render mode: Hybrid 2D+3D

Simulation engine: Agent-based

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-4

THEME & SCENE

The screen splits into three linked stages the student can switch between: a Central Valley grassland terrace in bleached summer gold, a slice of Monterey kelp forest in green-blue water with light shafts, and a Sierra reservoir shore with pale concrete intake pipes. Each stage holds its own living community drawn as recognisable silhouettes rather than icons: bunchgrass tussocks, purple starthistle rosettes, otters rolling at the surface, urchin spines bristling on rock, mussel crusts thickening on pipe steel. A translucent invader release capsule floats above whichever stage is active. The panel docks right; a running species ledger sits along the bottom, one row per species, filling in as the years advance.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Grassland stage	Environment	200 m terrace, straw ground plane, 400 native bunchgrass tussocks 30 cm tall, seed-head sway 0.3 Hz	Yes: resize
2	Yellow starthistle agents	Actor	Grey-green rosette 20 cm wide with a spined yellow flower head; taproot drawn 1.2 m into the soil cutaway; spreads by seed rings	Yes: place
3	Native forb agents	Actor	250 low broadleaf sprites in six colours, outcompeted where starthistle density exceeds 4 per m ²	No
4	Kelp forest stage	Environment	Water column 18 m deep, 60 giant kelp stipes with gas bladders, canopy shimmer at the surface	Yes: resize
5	Sea otter agents	Actor	12 brown floating markers, forage dives every 90 s, each removes 1–3 urchins per dive	Yes: place
6	Purple urchin agents	Actor	800 spiked violet discs 6 cm across, graze kelp holdfasts, breed when kelp is plentiful	No
7	Urchin barren overlay	Overlay	Pink coralline wash that replaces kelp texture where holdfast loss exceeds 80%	No
8	Reservoir stage	Environment	Shoreline slab, submerged intake pipe 1.2 m diameter, boat ramp, launch trailer prop	Yes: place
9	Quagga mussel colony	Actor	Striped 2 cm shells instanced as crust layers on pipe and rock; layer thickness grows with time and calcium	Yes: place
10	Plankton field	Field	Invisible scalar in the reservoir, drawn as green haze; drained by mussel filtration	No
11	Tanoak and coast live oak stand	Actor	90 trees, canopy 5–8 m, trunk bleeding cankers appear as dark seeps as infection advances	Yes: place
12	Sudden oak death pathogen field	Field	Spore concentration grid; spreads on rain splash and along walking and vehicle routes	No
13	Vector paths	Structure	Dashed lines for trail, road and boat ramp; carry propagule pressure into each stage	Yes: connect
14	Species ledger	Instrument	Bottom strip, one row per species with a live count and a trend arrow	No
15	Time dial	Instrument	Circular dial marked 0–25 years, drives all population updates	Yes: drag
16	Sampling transect	UI-Probe	25 m draggable line that counts every agent it crosses	Yes: drag

HOW IT WORKS — THE MODEL

Each stage runs coupled logistic populations on an annual tick. Natives grow toward a carrying capacity; an invader added at some starting density grows at its own rate and reduces the natives' effective capacity in proportion to its cover, which is why starthistle can occupy a terrace without ever "eating" anything. Otters and urchins run a predator–prey pair: remove the otters and urchin numbers rise until kelp holdfast loss crosses the barren threshold, at which point kelp cannot recover even if urchins later decline — a deliberate hysteresis, because the misconception is that removing a predator only affects its prey. Sudden oak death spreads as a spore field with a rain-splash term and a human-transport term, killing tanoak faster than coast live oak. Quagga mussels filter plankton at a fixed rate per individual, so their effect on everything else is indirect: fewer plankton, fewer larval fish.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Stage	Dropdown	Grassland / Kelp forest / Reservoir / Oak woodland	Grassland	—	Structural: swaps the whole community and its species set
Disruption	Dropdown	Invasive plant / Invasive mussel / Disease / Keystone removal / None	Invasive plant	—	Which biological event runs
Invader starting density	Slider	0–50	5	plants/m ² or shells/m ²	Size of the initial introduction
Invader spread rate	Slider	0.1–3.0	1.2	×/year	How fast the invader's range expands
Otter population	Stepper	0–40	12	individuals	Predation pressure on urchins; 0 models the fur-trade removal
Pathogen introduction	Toggle	On / Off	Off	—	Seeds the spore field at the trailhead
Human traffic	Slider	0–500	80	visits/week	Propagule and spore transport along vector paths
Native seed bank	Slider	0–100	60	% of original	How much native recovery capacity remains in the soil
Years elapsed	Timeline scrubber	0–25	0	years	Advances all populations; scrub back and forth freely
Control action	Multi-select	Mow / Graze / Boat wash station / Trail closure / None	None	—	Structural: adds a management intervention mid-run

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Starthistle takes a terrace	stage=Grassland; disruption=Invasive plant; invader starting density=5; years elapsed=12	Starthistle eats nothing. How does it still reduce native forb numbers by more than half?
S2	The otters are gone	stage=Kelp forest; disruption=Keystone removal; otter population=0; years elapsed=15	Only one species was removed. Count how many species end up affected, and explain the chain.
S3	Putting the otters back	stage=Kelp forest; otter population=20; years elapsed=25	Return the otters after a barren has formed. Why does kelp not simply come back?
S4	Mussels on the intake	stage=Reservoir; disruption=Invasive mussel; human traffic=400; control action=Boat wash station	Compare plankton and larval fish with the wash station on and off. What did the mussels actually take?

STUDENT ACTIVITIES

1. Set the grassland stage with starthistle at 5 plants/m², scrub the timeline to year 12, and record native forb richness before and after.

2. Drag the sampling transect across a starthistle patch and an uninvaded patch; record counts for both and compare.
3. Set otters to 0 in the kelp stage, run to year 15, then raise otters to 20 and scrub to year 25; record kelp cover at all three points.
4. Switch to the oak woodland, turn pathogen introduction on, and record tanoak and coast live oak deaths separately at years 5 and 15.
5. Run the reservoir stage twice, once with the boat wash station and once without, and record final plankton and mussel density for each.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Species ledger	Data table	individuals	Live count and trend arrow for every species in the active stage
Richness before/after	Bar chart	species count	Paired bars at year 0 and at the scrubbed year
Population trajectories	Line graph	individuals vs years	One trace per species, native traces solid and invader dashed
Invaded area	Live numeric	% of stage	Fraction of cells above the invader density threshold
Kelp canopy cover	Live numeric	%	Falls as urchins graze; the barren threshold is marked on the gauge
Plankton concentration	Line graph	mg/m ³ vs years	Drops as the mussel crust thickens
Infected tree count	Counter	trees	Split by species so the difference in susceptibility is visible
Run comparison log	Data table	mixed	Exportable CSV of settings and end-state values for every run

WHAT THE STUDENT SHOULD REALISE

Students believe an introduced species harms only what it eats, and that a disruption can be reversed by undoing it. Here starthistle crowds without eating, one missing otter reshapes an entire kelp community, and returning the otters does not restore the kelp because a barren holds itself in place. The student should be able to say: *"A living disruption spreads through relationships, and putting the original piece back is not the same as putting the ecosystem back."*

D5.3 · Constructing an argument from evidence

Experiment name: The Evidence Docket: Prove What Wrecked the River

Render mode: Data Dashboard

Simulation engine: Data-driven model

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS2-4

THEME & SCENE

A case-file desk viewed straight down, lit like a reading lamp on dark wood. On the left, a stack of evidence cards fans out: monitoring graphs, a photo pair, a survey table, a rainfall record, a land-use map of a Sierra foothill river reach. Centre desk holds an empty claim board with three ruled slots labelled Claim, Evidence and Reasoning. Right side shows the river reach itself as a small live map with a scrub bar, so any card can be checked against the year it came from. A brass scoring plate at the top right stays dim until a claim is assembled, then lights amber, then green.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Case desk	Environment	Dark oak surface 1600×900 px, warm pooled lamp light, soft paper shadows under every card	No
2	Evidence card — fish counts	Instrument	300×200 px card carrying a 20-year line graph of trout and sucker counts; flips to show source note	Yes: drag
3	Evidence card — water temperature	Instrument	Same card frame, monthly mean temperature 1998–2024 with a summer maximum trace	Yes: drag
4	Evidence card — turbidity record	Instrument	Bar series of monthly turbidity with two spikes matching logging years	Yes: drag
5	Evidence card — rainfall record	Instrument	Annual precipitation bars; deliberately shows no trend, so it is the tempting irrelevant card	Yes: drag
6	Evidence card — canopy photo pair	Instrument	Two aerial photographs, 2004 and 2019, riparian shade visibly thinned	Yes: drag
7	Evidence card — invertebrate survey	Instrument	Table of mayfly, stonefly and midge counts by year, with a tolerance rating column	Yes: drag
8	Evidence card — upstream land use	Instrument	Map card, hatched areas for logged parcels, dated	Yes: drag
9	Distractor card set	Instrument	Three cards holding true but irrelevant facts: a road resurfacing date, a nearby town's population, a fishing licence count	Yes: drag
10	Claim board	Structure	Three ruled slots with magnetic snap; Claim takes one card, Evidence up to four, Reasoning takes typed text	Yes: place
11	Claim card deck	Actor	Six pre-written claim cards, one correct, two partly supported, three unsupported by the available data	Yes: swap
12	Reasoning text field	UI-Probe	Free text box, 40–120 words, with sentence starters that can be toggled off	Yes: swap
13	Scoring plate	Instrument	Brass plate returning three sub-scores: relevance, sufficiency, and match between evidence and claim	No
14	Rebuttal generator	Actor	Produces one counter-argument card that the student must answer with a further piece of evidence	No
15	River reach mini-map	Overlay	Live 6 km reach with shade, turbidity and temperature layers, tied to the year scrubber	Yes: drag
16	Year scrubber	UI-Probe	Timeline 1998–2024 that redraws every card's highlighted year and the mini-map	Yes: drag

HOW IT WORKS — THE MODEL

The dataset behind the cards is a single generated 26-year record for one river reach, built so that riparian canopy loss drives summer water temperature up by 3.4 °C, which drives trout counts down while tolerant midges rise; turbidity spikes are real but transient and explain none of the long-term decline; rainfall is flat. The scorer does not check whether the student's claim is the one the author preferred. It checks three things per selected card: is this variable causally linked to the claimed cause in the underlying model, does the card's time span overlap the change being explained, and does the card discriminate between the claim and its rivals. A card that is true but explains nothing scores zero on discrimination, which is how the sim teaches that evidence must do work. Reasoning text is checked for whether it names the mechanism linking each selected card to the claim.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Case file	Dropdown	Foothill river / Coastal lagoon / Sierra meadow	Foothill river	—	Structural: swaps the whole dataset and card deck
Claim selection	Radio	Six claim cards	None	—	Which claim the evidence must support
Evidence slots	Drag-handle	1–4 cards	Empty	count	Structural: which cards sit on the board
Year window	Timeline scrubber	1998–2024	Full range	year	Restricts every card to the selected span
Show source notes	Toggle	On / Off	On	—	Reveals who collected each dataset and how
Distractor cards	Toggle	On / Off	On	—	Adds or removes the three true-but-irrelevant cards
Rebuttal difficulty	Stepper	1–3	1	level	How pointed the generated counter-argument is
Scoring detail	Toggle	Score only / Full breakdown	Score only	—	Shows or hides per-card relevance and discrimination marks
Reasoning aids	Toggle	Sentence starters on / off	On	—	Scaffolding for the written reasoning
Peer comparison	Toggle	On / Off	Off	—	Overlays the class's most common evidence choices

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The obvious suspect	case file=Foothill river; distractor cards=Off; claim selection=Turbidity killed the trout	Turbidity spiked and trout fell. Which single card shows that the spikes cannot explain the decline?
S2	Full docket	case file=Foothill river; distractor cards=On; evidence slots=4	Build the best-supported claim you can. Which two cards carry almost all the score, and why?
S3	Blame the weather	claim selection=A drought caused the decline; evidence slots=rainfall + fish counts	The rainfall card is genuine data. Why does the scorer give it nothing?
S4	Answer the rebuttal	case file=Coastal lagoon; rebuttal difficulty=3	The rebuttal says the fish moved rather than died. What evidence settles it?

STUDENT ACTIVITIES

1. Select a claim card, drag three evidence cards into the slots, and record the three sub-scores from the brass plate.
2. Swap one high-scoring card for a distractor, re-score, and record how much the total falls and which sub-score dropped.
3. Set the year window to 2004–2019 and record whether the fish count card still supports your claim over that shorter span.
4. Write a reasoning paragraph that names the mechanism linking canopy to temperature to trout, then record the reasoning sub-score before and after adding the word "because".

5. Run the rebuttal generator at level 3 and record which single extra card raises your score enough to clear the green threshold.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Argument score	Live numeric	0–100	Total across relevance, sufficiency and discrimination, updated on every card change
Per-card breakdown	Data table	mixed	Each selected card with its three marks and a one-line explanation
Claim strength badge	Pass-fail badge	—	Grey, amber or green; green requires at least two discriminating cards
Evidence coverage map	Heat map	—	Which years and which variables the selected cards actually cover
Rebuttal status	Pass-fail badge	—	Whether the counter-argument has been answered
Reasoning check	Live numeric	0–20	Marks for naming a mechanism rather than restating the data
Argument export	Data table	mixed	Claim, cards, reasoning and scores as an exportable record

WHAT THE STUDENT SHOULD REALISE

Students believe evidence is anything true that they can find, and that more evidence is always stronger. Here true-but-irrelevant cards score zero and a four-card pile can beat a two-card argument by nothing at all, because only cards that separate one explanation from another do work. The student should be able to say: *"Evidence has to rule something out, not just sound related — otherwise I have collected facts, not built an argument."*

D5.4 · Succession after a disruption

Experiment name: The Comeback: Ninety Years on a Burnt Slope

Render mode: 2.5D Layered

Simulation engine: State machine + Procedural geology

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-4

THEME & SCENE

A wide side-on band of Sierra foothill slope, drawn as layered parallax: charred soil and blackened snags in the foreground, a receding ridge in soft haze behind, sky washing from smoke-grey at year zero to clear blue by mid-run. Under the ground line a soil cutaway shows the organic layer thickening and root threads deepening as time passes. A thick timeline scrubber runs the full width beneath the scene, marked in years 0 to 90, with faint stage bands labelled Pioneer, Early seral, Mid seral and Climax. Dragging the scrubber redraws the slope continuously; the panel docks right, and a species turnover ribbon tracks the top edge.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Burnt slope base layer	Environment	1400 px parallax band, charcoal ground with ash streaks at year 0, greening progressively	Yes: resize
2	Standing snags	Structure	26 blackened trunks 4–11 m, bark sloughing over years 3–12, falling one by one after year 15	Yes: place
3	Soil cutaway	Structure	60 cm profile: ash cap, mineral soil, weathering bedrock; organic horizon grows 0 to 14 cm across the run	No
4	Fire-following annuals	Actor	Whispering bells and golden eardrops sprites, 8–25 cm, dense in years 1–3 then near-absent	No
5	Lupine and other nitrogen fixers	Actor	Blue-purple spikes 40 cm with visible root nodules in the cutaway; peak years 2–8	Yes: place
6	Perennial bunchgrasses	Actor	Tussocks 30–60 cm, establish from year 3, persist in gaps thereafter	No
7	Chamise and manzanita shrubs	Actor	Domes growing 0.4 m to 2.4 m across years 4–30; manzanita drawn with red bark	Yes: place
8	Ponderosa pine cohort	Actor	Seedlings from year 8, 25 m crowns by year 70, self-thinning as canopy closes	No
9	Black oak cohort	Actor	Resprouts from surviving root crowns in year 1, slower, shade-tolerant in later stages	Yes: place
10	Seed source edge	Structure	Unburnt forest strip at one end supplying propagules; distance is adjustable	Yes: drag
11	Animal guild markers	Actor	Bird, small mammal and insect icons that appear and vanish by stage: bluebird early, woodpecker at snag stage, flying squirrel at climax	No
12	Litter and coarse woody debris	Particle	Needle mat and fallen logs accumulating from year 12; feeds the organic horizon	No
13	Canopy light field	Field	Invisible transmitted-light scalar per column; falls as canopy closes and gates which species can establish	No
14	Species turnover ribbon	Overlay	Top-edge stacked band showing each species' relative cover across the whole 90 years	No
15	Timeline scrubber	UI-Probe	Full-width control, 0–90 years, scrubbable in both directions with stage bands marked	Yes: drag
16	Stage inspector	UI-Probe	Draggable magnifier that reports stage name, dominant species and light level at any point	Yes: drag

HOW IT WORKS — THE MODEL

Succession runs as a state machine over four stages, but the transitions are driven by two continuous variables rather than a clock: available light at ground level, and soil organic content. Each species carries an establishment window defined by the light and soil ranges it tolerates, plus a growth curve and a lifespan. Fire-followers germinate only on ash chemistry and vanish once litter covers the mineral soil. Nitrogen fixers raise the soil nitrogen term, which is what allows shrubs to grow faster after year 6, so the sim shows early species building the conditions for their own replacement. Shrubs shade out the annuals; pines overtop the shrubs; oaks persist beneath the pines because their tolerance window includes low light. Distance to the nearest seed source delays every arrival, and severity sets how much root crown and seed bank survived. Scrubbing backwards recomputes state rather than replaying an animation, so the timeline is genuinely explorable.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Disturbance severity	Slider	0–100	65	% biomass removed	How much seed bank and root crown survives year 0
Starting substrate	Dropdown	Ash over soil / Bare mineral soil / Bare rock / Flood silt	Ash over soil	—	Structural: sets whether this is primary or secondary succession
Distance to seed source	Slider	10–2000	150	m	Arrival delay for every wind and animal dispersed species
Rainfall regime	Slider	250–1400	800	mm/year	Growth rates and how long each stage lasts
Timeline position	Timeline scrubber	0–90	0	years	Scrubs the whole scene forward and backward
Repeat disturbance	Stepper	0–4	0	events	Structural: inserts further fires at chosen years
Repeat interval	Slider	3–40	15	years	Spacing of those repeat events
Grazing pressure	Slider	0–3	0	animal units/ha	Suppresses shrub and seedling establishment
Species turnover ribbon	Toggle	On / Off	On	—	Shows or hides the stacked cover band
Soil cutaway	Toggle	On / Off	On	—	Reveals the organic horizon and root depth

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Secondary succession after fire	starting substrate=Ash over soil; disturbance severity=65; distance to seed source=150	At which year does shrub cover overtake annual cover, and what changed to cause the switch?
S2	Starting from bare rock	starting substrate=Bare rock; distance to seed source=150	Compare year 30 with S1. Why is this slope so far behind?
S3	Fire every eight years	repeat disturbance=4; repeat interval=8	Does the slope ever reach the climax stage? Name the stage it keeps returning to.
S4	Isolated patch	distance to seed source=1800; rainfall regime=450	Which species never arrive at all, and what does that do to the final community?

STUDENT ACTIVITIES

1. Scrub to years 1, 5, 15, 40 and 80 and record the dominant species and total species count at each stop.
2. Turn the soil cutaway on and record organic horizon depth at the same five years beside your species counts.
3. Set the substrate to bare rock, re-run, and record how many years later the shrub stage begins compared with S1.
4. Predict which animal markers appear at the snag stage, then scrub to year 12 and record which actually appear.

5. Set repeat disturbance to 4 at 8-year intervals and record the highest stage the slope reaches across the full 90 years.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Current stage	Live numeric	stage name	Pioneer, early seral, mid seral or climax at the scrubbed year
Species turnover ribbon	Bar chart	% cover	Stacked relative cover for every species across 0–90 years
Species richness	Line graph	species vs years	Total richness trace; typically peaks mid-seral rather than at climax
Canopy light at ground	Live numeric	% of full sun	The gate variable, updated as the scrubber moves
Soil organic depth	Live numeric	cm	Read from the cutaway; grows through the run
Above-ground biomass	Line graph	t/ha vs years	Rises smoothly and dips at each repeat disturbance
Animal guild present	Data table	species	Which animal markers are active at the scrubbed year
Stage timing table	Data table	years	Start and end year of each stage, exportable for comparison across runs

WHAT THE STUDENT SHOULD REALISE

Students believe succession is a fixed queue of species marching to a final forest, and that the last stage holds the most life. Here richness peaks in the middle stages, the same slope takes twenty years or ninety depending on substrate and seed distance, and repeat fire parks the community permanently in the shrub stage. Early species change the soil that dooms them. The student should be able to say: *"Succession is not a schedule — each stage builds the conditions that decide what comes next."*

D5.5 · Short-term versus long-term change

Experiment name: One Year Later, Fifty Years Later

Render mode: 2.5D Layered

Simulation engine: Data-driven model

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: MS-LS2-4

THEME & SCENE

The screen is a split diptych of the same Sierra meadow-and-forest basin, rendered twice side by side from an identical camera. The left panel is locked to a short horizon, the right to a long one, and a shared disruption is applied to both. At load, the left shows a blackened basin one year after fire — bare ground, standing snags, a silt-choked stream — while the right shows the same basin at fifty years, with a mixed-age forest, a meadow edge and a clear stream. A brass slider between the panels sets each panel's year independently. Below, twin readout strips mirror each other so any number can be compared at a glance. The control panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Basin scene, left panel	Environment	700 px parallax basin: meadow floor, forested slope, stream, granite outcrop; state redrawn from its own year value	Yes: resize
2	Basin scene, right panel	Environment	Identical geometry and camera to the left, driven by a separate year value	Yes: resize
3	Snag field	Structure	40 blackened trunks that stand years 0–20, fall progressively, and are gone by year 35	No
4	Regenerating conifer cohort	Actor	Seedlings from year 4, 3 m at year 20, 24 m at year 60; density thins with age	Yes: place
5	Meadow sward	Structure	Sedge and forb ground layer; expands into the burn scar in years 2–15, then retreats under shade	Yes: swap
6	Stream channel	Structure	2 m polyline with adjustable sediment tint; runs brown for the first 4 years, clears thereafter	No
7	Downed wood and litter	Particle	Log and needle instances accumulating from year 12; feeds the soil carbon readout	No
8	Wildlife markers	Actor	Six guild icons — woodpecker, bluebird, deer, marten, trout, beetle — appearing and vanishing by decade	No
9	Soil carbon column	Instrument	Vertical bar embedded in each panel's foreground, filled to the current soil carbon value	No
10	Divergence graph	Overlay	Shared strip beneath both panels plotting a chosen metric across 0–100 years with both panel years marked	No
11	Twin year dials	UI-Probe	Two brass dials, 0–100 years each, one per panel, linkable	Yes: drag
12	Metric selector chips	UI-Probe	Row of draggable chips — richness, biomass, carbon, water clarity, timber value, deer numbers	Yes: drag
13	Verdict cards	Actor	Two cards, one per panel, each carrying a student-set judgement of Catastrophic / Damaging / Neutral / Beneficial	Yes: swap
14	Baseline ghost overlay	Overlay	Faint pre-disruption outline drawn over each panel for direct comparison	No
15	Difference meter	Instrument	Centre gauge showing the signed gap between the two panels on the selected metric	No

HOW IT WORKS — THE MODEL

One 100-year trajectory is computed per metric for the chosen disruption, and both panels sample that same trajectory at their own year. Nothing differs between panels except the sampling time, which is the entire point. Trajectories are shaped so that different metrics recover at genuinely different rates: water clarity returns within 5 years, species richness overshoots the pre-fire value around year 20, above-ground biomass takes 60 years, and soil carbon dips for a decade before exceeding its starting value. Some disruptions have no long-term recovery at all — a reservoir inundation and a paved conversion hold flat forever — so the student cannot learn the false rule that everything heals. The difference meter reports the signed gap, and the verdict cards are stored per panel so a single disruption can carry two honest and opposite judgements at once.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Disruption	Dropdown	Wildfire / Flood / Landslide / Clear-cut logging / Reservoir flooding / Paved development	Wildfire	—	Structural: swaps every trajectory and the scene furniture
Left panel year	Dial	0–100	1	years	Which year the left basin is drawn from
Right panel year	Dial	0–100	50	years	Which year the right basin is drawn from
Metric on display	Multi-select	Richness / Biomass / Soil carbon / Water clarity / Timber value / Deer numbers	Richness	—	Which trajectory drives the graph and the difference meter
Disruption severity	Slider	10–100	65	% biomass removed	Depth of the initial drop and the length of recovery
Repeat frequency	Slider	0–50	0	years between events	Structural: adds recurring disruptions to the trajectory events
Baseline ghost	Toggle	On / Off	On	—	Draws the pre-disruption outline over both panels
Link panels	Toggle	On / Off	Off	—	Locks the two dials a fixed number of years apart
Verdict card, left	Radio	Catastrophic / Damaging / Neutral / Beneficial	None	—	Records the student's short-horizon judgement
Verdict card, right	Radio	Catastrophic / Damaging / Neutral / Beneficial	None	—	Records the student's long-horizon judgement

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The same fire, twice	disruption=Wildfire; left panel year=1; right panel year=50	Both panels show one event. Which verdict is correct, and can both be?
S2	Race of the metrics	disruption=Wildfire; metric on display=Water clarity, then Biomass	Which metric recovers first and which last? Record the year each returns to baseline.
S3	No coming back	disruption=Reservoir flooding; left panel year=1; right panel year=100	Run the long panel to year 100. Why does this trajectory never return to baseline?
S4	Fire every ten years	disruption=Wildfire; repeat frequency=10; right panel year=50	Compare with S1 at the same year. What has repetition done to the long-term picture?

STUDENT ACTIVITIES

1. Set the left dial to 1 and the right to 50 for a wildfire, and record species richness in each panel.
2. Step the right dial through 5, 10, 20, 40 and 80 years, recording richness at each stop, and mark the year it first exceeds baseline.
3. Swap the display metric to biomass and repeat the same five stops; record which metric crosses baseline first.

- Set both verdict cards for the wildfire, then switch the disruption to reservoir flooding and set them again; record which verdicts changed.
- Turn repeat frequency to 10 years and record the year-50 richness beside your original year-50 value.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Panel values	Live numeric	metric units	The selected metric in each panel, side by side
Difference meter	Live numeric	signed metric units	Right panel minus left panel, updated as either dial moves
Recovery trajectory	Line graph	metric vs years	Full 0–100 year curve with both panel years marked as vertical pins
Time to baseline	Live numeric	years	Year at which the selected metric first returns to its pre-disruption value, or "never"
Metric race table	Data table	years	Time-to-baseline for all six metrics in one column each
Verdict pair	Pass-fail badge	—	Shows whether the two recorded verdicts conflict, which is expected rather than an error
Wildlife guild timeline	Data table	decade	Which guild markers are present in each panel
Run log	Data table	mixed	Exportable CSV of disruption, severity, both years and all metric values

WHAT THE STUDENT SHOULD REALISE

Students believe a disruption is simply good or bad, judged from the immediate aftermath. Here the identical fire reads as catastrophic at one year and as regenerative at fifty, different measurements recover at wildly different rates, and some disruptions never recover at all. The judgement depends on the timescale and the metric chosen. The student should be able to say: *"Before I call a change harmful, I have to say how long I looked and what I measured."*

D6 · Biodiversity and ecosystem services · MS-LS2-5

D6.1 · Naming ecosystem services

Experiment name: The Invisible Payroll: What the Land Does for Free

Render mode: Data Dashboard

Simulation engine: Data-driven model

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: MS-LS2-5

THEME & SCENE

A ledger-styled dashboard laid over a soft aerial illustration of a Central Coast watershed: orchard blocks, oak savanna, a marsh at the river mouth, a working harbour. Five service dials sit across the top like meters on a plant room wall — Pollination, Water filtration, Carbon storage, Flood buffering, Fisheries — each with a needle, a monetary figure beneath it and a small non-monetary caption above. The aerial view stays interactive: clicking any land parcel highlights every service that parcel contributes to, drawing thin gold threads from the parcel up to the dials it feeds. The palette is warm paper and ink with the dials in brass. The control panel docks right; a running annual account sits bottom-left.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Watershed aerial base	Environment	1200×800 px illustrated plan, 20 km across, parcels outlined in fine ink	Yes: resize
2	Orchard parcels	Actor	14 hatched blocks of almond and apple rows; yield depends on the pollination dial	Yes: swap
3	Oak savanna parcels	Actor	22 stippled parcels with scattered canopy dots; feed carbon and pollinator habitat	Yes: swap
4	Riparian corridor	Structure	Green ribbon 30–90 m wide along the river; width is directly draggable	Yes: resize
5	Tidal marsh	Structure	Cross-hatched delta polygon; area drives flood buffering and fish nursery capacity	Yes: resize
6	Wetland treatment cells	Structure	Three shallow ponds with reed fringes; nitrogen removal proportional to area and residence time	Yes: place
7	Native bee nesting habitat	Actor	Hedgerow and bare-ground patches shown as dashed edges; sets wild pollinator population	Yes: place
8	Managed hive stacks	Actor	Boxed hive icons in threes; the paid substitute for wild pollination	Yes: place
9	Harbour and fishing fleet	Actor	Quay with 12 boat icons; landings scale with marsh nursery area	No
10	Service dials	Instrument	Five brass gauges with needle, monetary figure and non-monetary caption; animate on any change	No
11	Gold contribution threads	Overlay	Bezier threads from a selected parcel to every dial it feeds, thickness $\propto$ contribution	No
12	Substitution cost calculator	Instrument	Panel that prices the engineered replacement for any service switched off	No
13	Non-monetary card set	Instrument	Six cards holding values that resist pricing: cultural significance, drinking water safety, species existence, recreation, sense of place, Tribal harvest rights	Yes: drag
14	Annual account ledger	Instrument	Bottom-left running table of service value per year, in dollars and in physical units	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Land-use brush	UI-Probe	Paint tool that converts any parcel between orchard, savanna, marsh, wetland and housing	Yes: place
16	Parcel inspector	UI-Probe	Draggable loupe reporting a parcel's area, land use and every service it supplies	Yes: drag

HOW IT WORKS — THE MODEL

Each service is computed from land cover using published-style coefficients simplified to one line each: pollination value scales with the area of orchard within 800 m of nesting habitat; water filtration removes nitrogen at a fixed kg/ha/year for wetland and riparian cover; carbon storage sums a per-hectare stock by cover type; flood buffering converts marsh and floodplain area into peak-flow reduction and then into avoided damage; fisheries landings scale with nursery marsh area on a saturating curve. Every service reports twice, once in dollars per year and once in a physical unit, and the substitution calculator prices what it would cost to buy the same outcome from engineering — hives trucked in, a filtration plant, a levee. Some values deliberately have no dollar figure at all: the non-monetary cards can be attached to a service but never convert, so the ledger always ends with a section that does not add up.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Land-use brush	Drag-handle	Orchard / Savanna / Marsh / Wetland / Housing	Off	—	Structural: repaints parcels and recomputes every dial
Riparian corridor width	Slider	0–120	45	m	Nitrogen removal and shade; drives water filtration
Marsh area	Slider	0–900	380	hectares	Flood buffering and fish nursery capacity
Wild pollinator habitat	Slider	0–30	12	% of farm edge	Wild bee population and orchard set rate
Managed hives	Stepper	0–600	0	hives	Bought pollination; appears as a cost, not a service
Oak canopy cover	Slider	0–60	28	%	Carbon stock and shade
Valuation mode	Radio	Dollars / Physical units / Both	Both	—	Which figure each dial shows
Storm severity	Dropdown	2-year / 10-year / 100-year storm	10-year	—	Which flood event the buffering service is tested against
Non-monetary cards	Multi-select	Six value cards	None	—	Attaches unpriceable values to the ledger
Show contribution threads	Toggle	On / Off	On	—	Draws the gold parcel-to-dial links

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The watershed as found	All defaults; valuation mode=Both	Which two services carry the largest annual value, and which parcels supply them?
S2	Pave the marsh	land-use brush=Housing over marsh; marsh area=0; storm severity=100-year	The housing has a sale value. What does the ledger lose, and what does the flood dial do?
S3	Buy the bees back	wild pollinator habitat=0; managed hives=450	Hedgerows removed, hives hired. Is the orchard as productive, and what does it now cost?
S4	The unpriceable column	non-monetary cards=Drinking water safety + Tribal harvest rights + Sense of place	Attach three cards. Why does the total at the bottom of the ledger refuse to be one number?

STUDENT ACTIVITIES

1. Click each of the five dials in turn and record which parcels light up gold for each service.
2. Set marsh area to 0 and record all five dial values before and after; note which services change and which do not.

3. Set wild pollinator habitat to 0, then raise managed hives until orchard yield returns to its original figure; record the number of hives and the cost.
4. Run the flood dial at the 2-year, 10-year and 100-year storm settings and record avoided damage for each.
5. Attach three non-monetary cards, then write one sentence explaining why the ledger total is now incomplete.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Service dials	Live numeric	\$/year and physical units	All five services, both valuations, updated on any land change
Pollination delivered	Live numeric	% of orchard flowers set	Physical measure independent of price
Nitrogen removed	Live numeric	kg/year	Water filtration in mass, not money
Carbon stored	Live numeric	tonnes C	Standing stock across the watershed
Avoided flood damage	Live numeric	\$/year	Depends on the chosen storm severity
Substitution cost	Bar chart	\$/year	Engineered replacement cost beside each natural service
Annual account ledger	Data table	mixed	Full account, with a separate unpriced section for the non-monetary cards
Land-use summary	Data table	hectares	Area by cover type after every brush edit

WHAT THE STUDENT SHOULD REALISE

Students believe nature is scenery and that the working parts of a landscape are the farms and buildings. Here removing a marsh drops flood protection and fish landings at once, replacing wild pollinators with hired hives costs real money for the same crop, and some of the most important values will not convert into dollars at all. The student should be able to say: *"Healthy ecosystems already do jobs we would otherwise have to pay for, and a few of those jobs have no price at all."*

D6.2 · Threats to biodiversity

Experiment name: Pressure Stack: Layering the Threats on a Range Map

Render mode: 2.5D Layered

Simulation engine: Field/vector + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: MS-LS2-5

THEME & SCENE

A cool cartographic view of coastal and inland California from Monterey Bay to the western Sierra, drawn as a muted relief map with rivers in pale blue and city footprints in warm grey. Hovering above the map are six semi-transparent threat layers, drawn as stacked sheets a few pixels apart so the student can see them as physically separate: habitat loss, fragmentation, pollution, overharvest, invasive species and climate shift. Each sheet can be lifted on or off. Below the map sits a species tray holding eight Californian species cards, each draggable onto the map to see its range clipped by whichever pressures are active. The palette is restrained so the threat washes read clearly. The control panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Relief base map	Environment	1400×900 px shaded relief, Albers projection, 400 km across, coastline and rivers inked	Yes: resize
2	Habitat loss layer	Field	Semi-transparent sheet, ochre wash where cover is converted; intensity 0–1 per 1 km cell	Yes: swap
3	Fragmentation layer	Field	Sheet drawn as a lattice of roads, canals and fence lines; carries an edge-density value per cell	Yes: swap
4	Pollution layer	Field	Blue-grey plume sheet from outfalls, farm runoff and urban stormwater, decaying with distance	Yes: swap
5	Overharvest layer	Field	Red hatch over fishing grounds and logging blocks, scaled by extraction rate	Yes: swap
6	Invasive species layer	Field	Violet stipple with expanding fronts from ports, marinas and highways	Yes: swap
7	Climate shift layer	Field	Warm gradient sheet plus arrows showing the direction suitable climate is moving	Yes: swap
8	Species cards	Actor	Eight cards: sea otter, Chinook salmon, valley elderberry longhorn beetle, giant garter snake, monarch butterfly, Sierra yellow-legged frog, burrowing owl, blue oak	Yes: drag
9	Range polygon	Overlay	Each species' potential range drawn as a soft-edged polygon, clipped live by the active layers	No
10	Pressure stack meter	Instrument	Right-edge column showing each active layer's contribution to total pressure at the pinned cell	No
11	Interaction highlighter	Overlay	Cross-hatch that appears only where two or more layers overlap above threshold, marking compound pressure	No
12	Corridor pinch points	Structure	Auto-detected narrow passages between habitat blocks, drawn as amber chevrons	No
13	Map pin probe	UI-Probe	Draggable pin returning per-layer values, total pressure and predicted local extinction risk	Yes: drag
14	Time slider ghosts	Overlay	Faint 1970 and 2070 outlines of each layer for before-and-after comparison	No
15	Threat ranking board	Instrument	Sortable list of the six pressures ranked by area affected and by species affected	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Legend and scale bar	UI-Probe	Movable legend block with colour keys, a 50 km scale bar and a north arrow	Yes: drag

HOW IT WORKS — THE MODEL

Every 1 km cell carries six pressure values from 0 to 1. A species card holds a potential range polygon and a sensitivity weight for each pressure, so a salmon is heavily weighted to pollution and fragmentation while a blue oak is weighted to habitat loss and climate. Realised range is the potential range with cells removed wherever the weighted sum of pressures exceeds that species' tolerance. Crucially the sum is not linear: where two or more layers exceed 0.4 in the same cell, the model applies a compound multiplier, so two moderate pressures together remove more range than either severe pressure alone. Fragmentation is computed from edge density rather than area lost, which lets the sim show a landscape that keeps most of its habitat but loses its connectivity. Climate shift moves the suitable band northward and upslope each decade, so range loss can occur in cells where nothing was built and nothing was polluted.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Active threat layers	Multi-select	Habitat loss / Fragmentation / Pollution / Overharvest / Invasives / Climate shift	Habitat loss	—	Structural: which sheets are stacked on the map
Species on map	Drag-handle	Any of eight species cards	Sea otter	—	Structural: swaps range polygon and sensitivity weights
Habitat conversion	Slider	0–90	35	% of cells converted	Extent of the ochre wash
Road and canal density	Slider	0–4.0	1.4	km/km ²	Edge density driving the fragmentation layer
Pollutant load	Slider	0–200	60	kg N/km ² /year	Plume strength and reach
Harvest rate	Slider	0–80	25	% of stock per year	Overharvest layer intensity
Invasive front speed	Slider	0–40	8	km/year	How fast the violet stipple expands from ports and roads
Warming by 2070	Slider	0.0–4.0	2.0	°C	Distance the suitable climate band shifts
Decade	Timeline scrubber	1970–2070	2020	year	Redraws every layer at that decade
Compound highlighting	Toggle	On / Off	On	—	Shows where overlapping pressures multiply

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	One threat at a time	active threat layers=Habitat loss only; species on map=Burrowing owl; habitat conversion=35	How much range does habitat loss alone remove? Record the percentage.
S2	Two moderate pressures	active threat layers=Habitat loss + Fragmentation; road and canal density=2.2	Add fragmentation at moderate strength. Why is the loss larger than the two separate losses added together?
S3	Different species, same map	active threat layers=All six; species on map=Chinook salmon, then Blue oak	Same pressures, two species. Which pressure dominates for each, and why do they differ?
S4	Nothing built, range still lost	active threat layers=Climate shift only; warming by 2070=3.0; decade=2070	No habitat was converted here. Explain how the species still loses its range.

STUDENT ACTIVITIES

1. Place the sea otter card on the map with only the overharvest layer active and record realised range as a percentage of potential range.
2. Add layers one at a time in any order, recording realised range after each addition, and mark where the drop is larger than expected.
3. Drag the map pin to a cross-hatched compound cell and to a single-pressure cell; record the per-layer values at both.
4. Swap the species card to the monarch butterfly and repeat the full layer stack; record which two pressures matter most for it.
5. Scrub the decade slider from 1970 to 2070 with all layers on and record realised range at 1970, 2020 and 2070.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Realised range	Live numeric	% of potential range	Updates whenever a layer or slider changes
Range area	Live numeric	km ²	Absolute area remaining for the selected species
Pressure stack at pin	Bar chart	0–1 per pressure	Six stacked bars for the pinned cell
Compound-pressure area	Live numeric	km ²	Area where two or more pressures exceed threshold
Threat ranking board	Data table	mixed	Six pressures ranked by area affected and by species affected
Extinction risk band	Pass-fail badge	—	Low, elevated or high, from remaining range and fragmentation
Range over time	Line graph	km ² vs decade	1970–2070 trace for the selected species
Species comparison table	Data table	mixed	Realised range for all eight species under the current layer stack

WHAT THE STUDENT SHOULD REALISE

Students believe biodiversity is lost mainly through one obvious cause, usually hunting or bulldozers, and that threats simply add up. Here two moderate pressures together strip more range than one severe pressure, a species can lose habitat where nothing was built because the climate band moved, and the leading threat is different for a salmon than for an oak. The student should be able to say: *"Threats stack and multiply, and which one matters most depends on the species you are asking about."*

D6.3 · Named solution categories

Experiment name: The Solutions Bench: Six Tools, One Watershed

Render mode: Hybrid 2D+3D

Simulation engine: Agent-based + Data-driven model

Interaction level: Design

Session length: 18–25 min

NGSS anchor: MS-LS2-5

THEME & SCENE

A workbench view: along the bottom of the screen sits a physical-looking rack of six labelled tool trays — Protected areas, Wildlife corridors, Habitat restoration, Invasive control, Regulation, Incentives — each holding draggable tokens with their own icon and material, from a green boundary ribbon to a paper permit card. Above the bench, filling most of the screen, is a working Central Valley and foothill landscape in three-quarter view: farmland grid, a river with a levee, remnant riparian woodland, a highway cutting the valley, a hill range on the horizon. Animals move as markers along whatever connectivity exists. Dropping a token onto the landscape places a real object there. The control panel docks right; an outcomes strip runs across the top.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Valley landscape	Environment	25 km × 15 km three-quarter terrain, farm grid, river, levee, highway, foothills; 2 m vertical exaggeration	Yes: resize
2	Protected area token	Actor	Green boundary ribbon that snaps to parcel edges and encloses any area drawn; ranger station prop at its gate	Yes: place, resize
3	Wildlife corridor token	Actor	60–400 m planted strip or a highway underpass box culvert with fencing wings	Yes: place, connect
4	Habitat restoration token	Actor	Restoration patch that converts a farm parcel to riparian or grassland over a visible 8-year growth animation	Yes: place
5	Invasive control token	Actor	Crew truck and treatment zone disc; repeats annually while funded, otherwise the invader front resumes	Yes: place
6	Regulation token	Actor	Permit card that applies a rule to a whole class of parcels, e.g. a 30 m no-clear buffer on every stream	Yes: place
7	Incentive token	Actor	Contract card offering a per-hectare payment; landowner agents accept or refuse by their own economics	Yes: place
8	Landowner agents	Actor	40 farm markers with a profit value and a willingness threshold; change land use when an incentive beats their margin	No
9	Wildlife agents	Actor	200 markers across four guilds; movement is blocked by highway and levee unless a corridor exists	No
10	Invasive front	Field	Expanding perimeter from the port and highway; retreats inside funded control zones	No
11	Connectivity graph	Overlay	Node-and-link diagram over the landscape; links appear where corridors join habitat blocks	No
12	Budget meter	Instrument	Top-strip gauge showing annual cost against the available budget; turns red when overdrawn	No
13	Compliance meter	Instrument	Gauge showing what fraction of landowners are actually following a placed regulation	No
14	Tool tray rack	UI-Probe	Bench along the bottom holding all six token types with counts remaining	Yes: drag
15	Outcome inspector	UI-Probe	Draggable probe reporting local richness, connectivity and cost for any spot	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Year clock	Instrument	Dial 0–30 years driving growth, invasion, compliance drift and agent decisions	Yes: drag

HOW IT WORKS — THE MODEL

Each tool type acts on a different variable, and the sim refuses to let one tool do another's job. Protected areas set conversion probability to zero inside their boundary but do nothing about an invasive front or a barrier. Corridors add edges to the connectivity graph, and wildlife population is computed from effective connected habitat area rather than raw habitat area, so a corridor can raise numbers without adding a hectare. Restoration adds habitat on a delay: newly planted parcels count at 20% value for the first five years. Invasive control is the only recurring cost and its gains reverse within three years of defunding. Regulation applies landscape-wide at low cost but generates a compliance value below 100% that drifts with enforcement spending. Incentives work through landowner agents, who accept only when the payment exceeds their margin, so uptake is patchy and voluntary. Every placement bills against the budget meter.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Tool selection	Drag-handle	Six token types from the bench	Protected area	—	Structural: which object the next placement creates
Annual budget	Slider	0.5–20	6	\$ million/year	How much can be placed and maintained
Protected area size	Slider	50–5000	800	hectares	Extent enclosed by the next boundary ribbon
Corridor width	Slider	30–500	120	m	Passage quality; narrow corridors carry fewer guilds
Restoration intensity	Slider	0–100	50	% of a parcel replanted	How quickly a restored parcel reaches full habitat value
Invasive control funding	Slider	0–100	0	% of area treated per year	Whether the invasive front retreats or resumes
Regulation strength	Dropdown	None / Stream buffer / Grading permit / Take prohibition	None	—	Which landscape-wide rule applies
Incentive payment	Slider	0–900	250	\$/hectare/year	Landowner uptake rate
Enforcement spending	Slider	0–30	10	% of budget	Compliance level for any active regulation
Year clock	Timeline scrubber	0–30	0	years	Advances growth, invasion, uptake and compliance drift

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	A park and nothing else	tool selection=Protected area; protected area size=3000; invasive control funding=0; year clock=15	The reserve is large and legally safe. Why does richness inside it still fall by year 15?
S2	Connect what exists	tool selection=Wildlife corridor; corridor width=200; annual budget=6	Add no new habitat, only corridors. How much does wildlife population rise, and why?
S3	Rules versus payments	regulation strength=Stream buffer; enforcement spending=5, then incentive payment=600	Compare riparian hectares protected by rule and by payment at the same cost. Which delivers more, and which is more certain?
S4	Stop paying the weed crew	invasive control funding=80 for 10 years, then 0; year clock=20	Fund control for a decade then stop. What happens between years 10 and 20?

STUDENT ACTIVITIES

1. Drag one protected area token of 3000 hectares onto the foothills, run to year 15, and record richness inside and outside the boundary.
2. Place two corridor tokens joining the riparian woodland to the foothills, run again, and record wildlife population before and after.
3. Set a stream buffer regulation with enforcement at 5%, then at 25%, and record compliance and riparian hectares for each.
4. Raise the incentive payment in steps of 150 \$/ha/year and record landowner uptake at each step until it stops rising.
5. Spend the entire budget on one tool type, record the outcomes strip, then rebuild using three tool types for the same money and compare.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Species richness	Live numeric	species	Landscape total, updated as the year clock advances
Connected habitat area	Live numeric	hectares	Effective area after connectivity is accounted for, not raw area
Wildlife population by guild	Bar chart	individuals	Four guilds, showing which respond to corridors and which to area
Invaded area	Line graph	hectares vs years	Rises, falls under funded control, and rebounds when funding stops
Annual cost	Live numeric	\$ million/year	Placement plus maintenance against the budget meter
Compliance rate	Live numeric	% of landowners	Only meaningful when a regulation is active
Incentive uptake	Live numeric	% of eligible parcels	Voluntary acceptance at the current payment
Solution mix log	Data table	mixed	Exportable record of every token placed, its cost and its measured effect

WHAT THE STUDENT SHOULD REALISE

Students believe conservation means one thing, usually drawing a line around a park, and that a good solution works everywhere. Here a large reserve fails against an invasion, corridors raise populations without adding habitat, regulation is cheap but only as strong as its compliance, and incentives are willing but patchy. Each tool fixes one specific problem. The student should be able to say: *"Every solution type has a job it is good at, so the real question is which problem I am trying to fix."*

D6.4 · Evaluating competing solutions

Experiment name: Weights Change Winners: The Trade-off Bench

Render mode: Data Dashboard

Simulation engine: Data-driven model

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS2-5

THEME & SCENE

A cool analyst's bench in slate and pale ink. The centre is a matrix: five candidate solutions for one degraded Sacramento Valley floodplain run down the rows, and six criteria run across the columns — biodiversity gain, cost, time to benefit, community support, water security, and maintenance burden. Each cell holds a small filled bar rather than a bare number, so the pattern is readable at a glance. Above each column sits a physical-looking weight dial with a brass knob. Down the right edge, a ranking column shows the five solutions ordered by weighted score, and the rows visibly slide and reorder whenever a weight is turned. A small floodplain thumbnail sits top-left, updating to show whichever solution is selected.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Decision matrix grid	Structure	5 rows × 6 columns, 1100×520 px, ruled slate panel with ink gridlines and column headers	Yes: resize
2	Solution row cards	Actor	Five cards: setback levee, engineered levee raise, riparian restoration, upstream reservoir release rules, buy-out and retreat; each carries an icon and a thumbnail	Yes: swap
3	Criterion columns	Structure	Six headed columns, each with its own unit and its own better-is-higher or better-is-lower direction	Yes: swap
4	Score cells	Instrument	30 cells, each a filled bar 0–10 with the raw value printed beneath in real units	No
5	Weight dials	UI-Probe	Six brass knobs above the columns, 0–5 each, with a live percentage-of-total readout	Yes: drag
6	Ranking column	Instrument	Right-edge ordered list with animated row reordering and the weighted score to 1 dp	No
7	Sensitivity strip	Overlay	Thin band under each dial showing the weight range across which the top-ranked solution stays the same	No
8	Floodplain thumbnail	Overlay	320×220 px plan of the reach that redraws for the selected solution: levee line, restored acres, houses removed	No
9	Stakeholder viewpoint chips	Actor	Five draggable chips — farmer, city flood manager, Tribal council, biologist, taxpayer — each carrying a preset weight profile	Yes: drag
10	Raw data drawer	Instrument	Slide-out panel holding the underlying figure and source note behind every cell	Yes: drag
11	Normalisation switch	UI-Probe	Control that swaps between raw units, 0–10 scaling and rank scoring, so the effect of scaling is visible	Yes: swap
12	Tie and near-tie flag	Overlay	Amber marker on any pair of solutions within 0.3 of each other, warning against over-reading small gaps	No
13	Constraint bar	Instrument	Budget and schedule limits drawn as a red line across the matrix; violating solutions grey out	Yes: drag
14	Decision log	Instrument	Bottom strip recording every weight profile tried and the winner it produced	No
15	Justification field	UI-Probe	Text box requiring the student to name the weights they chose and why, scored for consistency with their ranking	Yes: swap

HOW IT WORKS — THE MODEL

Each solution has a fixed set of real-unit performance figures: hectares of habitat gained, capital cost, years to first benefit, a survey-style support percentage, acre-feet of water security, and annual maintenance cost. Those figures never change when a weight moves, which is the central honesty of the sim. Normalisation converts each column to a 0–10 scale, flipping direction for cost-like criteria so that higher always means better, and the weighted score is the sum of $\text{weight} \times \text{normalised score}$ divided by the total weight. Turning any dial recomputes and reorders instantly. The sensitivity strip solves, for each dial, the range over which the current winner remains the winner, so a fragile result is visibly fragile. Stakeholder chips are stored weight vectors, not opinions layered on top of the data, so choosing a viewpoint is exactly the same operation as turning the dials by hand.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Biodiversity weight	Dial	0–5	3	—	Influence of habitat gain on the weighted score
Cost weight	Dial	0–5	3	—	Influence of capital cost, direction-flipped
Time-to-benefit weight	Dial	0–5	2	—	Influence of how soon results appear
Community support weight	Dial	0–5	2	—	Influence of the support survey figure
Water security weight	Dial	0–5	2	—	Influence of acre-feet secured
Maintenance weight	Dial	0–5	1	—	Influence of annual upkeep, direction-flipped
Stakeholder profile	Dropdown	None / Farmer / Flood manager / Tribal council / Biologist / Taxpayer	None	—	Structural: loads a stored weight vector into all six dials
Solutions in play	Multi-select	Five solution cards	All five	—	Structural: adds or removes rows from the matrix
Normalisation	Radio	Raw units / 0–10 scale / Rank score	0–10 scale	—	How cells are converted before weighting
Budget constraint	Slider	5–250	120	\$ million	Greys out any solution above the line

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Balanced start	All weights=3; normalisation=0–10 scale; budget constraint=250	With every criterion weighted equally, which solution ranks first and by what margin?
S2	Cheap and fast	cost weight=5; time-to-benefit weight=5; biodiversity weight=0	Which solution wins now? Did any cell value change when the ranking changed?
S3	Two stakeholders	stakeholder profile=Biologist, then Farmer	The same five solutions and the same data. Why do the two viewpoints rank them differently?
S4	Fragile winner	All weights=3, then move one dial by a single step	Use the sensitivity strip to find the criterion whose smallest change flips first place.

STUDENT ACTIVITIES

1. Set all six weights to 3 and record the full ranking with each weighted score.
2. Turn the cost dial from 3 to 5 and the biodiversity dial from 3 to 0; record the new ranking and note which raw cell values changed.
3. Load the biologist profile, then the farmer profile, and record the top two solutions under each.
4. Open the raw data drawer for the cost column and record the actual figure behind each solution's bar.
5. Write in the justification field which weights you chose and why, then record the consistency score the field returns.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Weighted score	Live numeric	0–10	One score per solution, recomputed on every dial move
Ranking order	Data table	rank	Five solutions ordered, with animated reordering
Sensitivity range	Live numeric	weight units	The span over which the current winner survives, per criterion
Criterion contribution	Bar chart	score points	Stacked bars showing which criteria are carrying each solution's score
Raw performance table	Data table	mixed units	The unweighted figures behind every cell, exportable
Near-tie flag	Pass-fail badge	—	Amber when the top two are within 0.3
Constraint status	Pass-fail badge	—	Which solutions remain feasible under the budget line
Decision log	Data table	mixed	Every weight profile tried and the winner it produced

WHAT THE STUDENT SHOULD REALISE

Students believe one solution is objectively best and that the numbers will simply reveal it. Here no cell value ever changes while the winner changes three times, because the ranking depends on what the decider chose to value. That does not make the exercise arbitrary: the data constrains the argument, and the sensitivity strip shows when a result is fragile. The student should be able to say: *"The evidence tells me what each option does — the weights tell you what I decided mattered, and I have to say them out loud."*

D6.5 · A biodiversity solution for a real ecosystem

Experiment name: Delta Commission: A Budget, a Room, and Thirty Years

Render mode: Hybrid 2D+3D

Simulation engine: Agent-based + Data-driven model

Interaction level: Design

Session length: 18–25 min

NGSS anchor: MS-LS2-5

THEME & SCENE

A working commission table viewed at a slight angle. Most of the screen is a living map of the Sacramento–San Joaquin Delta: a braid of sloughs and channels, diked islands sitting below sea level behind their levees, tule marsh fringes, pumping plants at the south end, Suisun Bay to the west. Boats, birds and fish markers move across it. Along the left edge sit five stakeholder seats, each an illustrated place card with a mood indicator: Delta farmer, Southern California water agency, commercial fishing association, a Tribal representative, and a levee district engineer. Along the bottom sits a project tray with costed interventions. A thirty-year clock and a budget bar run across the top. The control panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Delta map base	Environment	90 km × 70 km plan, channels in blue, 18 diked islands with subsidence depth shown in a side profile strip, levee lines picked out in grey	Yes: resize
2	Diked island parcels	Actor	18 polygons, each with land elevation from +1 m to –5 m, current use, and a flood-risk value	Yes: swap
3	Tidal marsh restoration project	Actor	Placeable patch that breaches a levee and converts an island to marsh over a 12-year vegetation animation	Yes: place
4	Levee reinforcement project	Actor	Placeable segment thickening a levee cross-section; reduces failure probability for that island only	Yes: place
5	Fish screen and passage project	Actor	Placeable structure at pump intakes and channel junctions; cuts entrainment losses	Yes: place
6	Invasive water hyacinth control	Actor	Treatment barge and zone disc; the mat regrows within three years if unfunded	Yes: place
7	Flow and salinity field	Field	Coupled scalar field for freshwater outflow and salinity intrusion from the west; drives habitat suitability	No
8	Chinook salmon agents	Actor	400 migrating markers routed through channels; survival depends on screens, temperature and marsh rearing area	No
9	Delta smelt population	Actor	Population marker cloud in the low-salinity zone; highly sensitive to outflow and hyacinth cover	No
10	Waterbird flock agents	Actor	12 flock markers using marsh and flooded-field habitat seasonally	No
11	Stakeholder seat cards	Actor	Five illustrated cards with a 0–100 satisfaction meter and a stated red line that turns the card red when crossed	Yes: drag
12	Water export gauge	Instrument	Pumping plant dial in acre-feet per year, coupled inversely to freshwater outflow	Yes: drag
13	Budget bar	Instrument	Top-strip bar showing capital spend and committed annual maintenance against the cap	No
14	Thirty-year clock	Instrument	Timeline scrubber 0–30 years driving vegetation growth, subsidence, populations and levee failure rolls	Yes: drag
15	Project tray	UI-Probe	Bottom rack of costed project tokens with unit price and maintenance cost printed on each	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Commission report card	Instrument	Final scoring panel: biodiversity outcome, budget compliance, stakeholder acceptance and a written rationale field	Yes: swap

HOW IT WORKS — THE MODEL

The Delta runs as a coupled system rather than a set of independent projects. Freshwater outflow and export pumping trade off directly: raising exports pulls the low-salinity zone eastward, which shrinks smelt habitat and raises the water agency's satisfaction at the same time. Breaching an island for tidal marsh adds rearing habitat that raises salmon survival and buffers flood peaks, but permanently removes that island's farmland and drops the farmer's meter. Subsidence continues on every farmed island at roughly 1–2 cm per year, so the cost of defending a levee rises through the run and the model never lets a do-nothing plan stay level. Each levee faces an annual failure probability, reduced by reinforcement; a failure floods that island regardless of any other plan. Every project carries capital plus annual maintenance, and maintenance is charged for all thirty years, so an over-built year-one plan runs out of money by year twelve. Stakeholder meters move from measured outcomes, not from opinion, and each seat has one red line that fails the plan if crossed.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Ecosystem case	Dropdown	Sacramento–San Joaquin Delta / Monterey Bay kelp / Sierra meadow	Delta	—	Structural: swaps the whole map, species set and stakeholder seats
Total budget	Slider	50–1200	400	\$ million over 30 years	Capital plus all maintenance available
Project placement	Drag-handle	Marsh restoration / Levee reinforcement / Fish screens / Invasive control	Marsh restoration	—	Structural: which project the next placement creates
Water exports	Slider	1.0–6.5	4.2	million acre-feet/year	Freshwater outflow, salinity intrusion and smelt habitat
Marsh restoration area	Slider	0–20000	0	hectares	How much diked island is converted to tidal marsh
Levee investment	Slider	0–100	30	% of levee length reinforced	Annual failure probability across the Delta
Invasive control funding	Slider	0–100	0	% of channel area treated per year	Water hyacinth cover and channel habitat quality
Sea level rise by 2055	Slider	0.1–1.0	0.4	m	Levee load, salinity intrusion and marsh drowning risk
Stakeholder view	Multi-select	Five seat cards	All five	—	Which satisfaction meters and red lines are displayed
Year clock	Timeline scrubber	0–30	0	years	Advances subsidence, growth, populations and failure rolls

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Hold the line	levee investment=90; marsh restoration area=0; water exports=4.2; year clock=30	Defend every island and restore nothing. What happens to salmon and smelt by year 30, and what does maintenance cost?

#	Scenario	Preset	The question the student answers
S2	Restore at scale	marsh restoration area=14000; levee investment=20; total budget=400	Breach a third of the islands. Which two stakeholder meters collapse, and which species recover?
S3	Maximum exports	water exports=6.3; marsh restoration area=6000	Push exports to the top of the range. Which red line is crossed first?
S4	Rising sea, same budget	sea level rise by 2055=0.9; total budget=400; year clock=30	Run your best plan again under high sea level rise. What fails, and what would you change?

STUDENT ACTIVITIES

1. Place projects until the budget bar is full, run the clock to year 30, and record the four report card scores.
2. Set marsh restoration to 14,000 hectares and record salmon survival, smelt population and the farmer's satisfaction before and after.
3. Step water exports from 3.0 to 6.3 in four moves, recording the water agency meter and the smelt population at each step.
4. Rebuild your plan for half the budget and record which projects you dropped and what each drop cost in biodiversity.
5. Write a 60-word rationale in the report card naming your two biggest trade-offs, then record the acceptance score it returns.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Species richness and key populations	Data table	individuals and species	Salmon survival, smelt index, waterbird numbers, total richness at the scrubbed year
Tidal marsh area	Live numeric	hectares	Functioning marsh, discounted for the first 12 years of establishment
Budget spent and committed	Live numeric	\$ million	Capital plus all remaining maintenance against the cap
Stakeholder satisfaction	Bar chart	0–100	Five meters, with red-line breaches flagged individually
Levee failure events	Counter	events	Failures rolled across the run, with the island named
Water exports delivered	Line graph	million acre-feet vs years	Delivery trace against the agency's stated minimum
Outcome over time	Line graph	index vs years	Biodiversity index across the full 30 years for the current plan
Commission report card	Pass-fail badge	—	Pass requires biodiversity gain, budget compliance and no red line crossed

WHAT THE STUDENT SHOULD REALISE

Students believe a real conservation problem has a right answer that a well-meaning plan can reach in full. Here every plan that satisfies the fish disappoints a stakeholder, holding every levee costs more each year while species keep declining, and a plan that passes at 0.4 m of sea level rise fails at 0.9 m. Good solutions are defensible trade-offs made under constraint. The student should be able to say: *"I cannot give everyone everything — I can choose openly, show my evidence, and say what I gave up."*